



# Automated Factor Discovery and Tradability Evaluation in US Equities

An Evaluation of RD-Agent(Q) with  
Fundamental and News Sentiment Signals

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**Disclaimer:** This report is submitted as part requirement for the MSc Computer Science degree at UCL. It is substantially the result of my own work except where explicitly indicated in the text.

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# Abstract

This project tests whether the factor-discovery process of RD-Agent(Q), a large-language-model system that generates, implements and validates quantitative factors in an iterative loop, transfers from the Chinese CSI 300 market to the US S&P 500. It asks whether the process adds predictive information beyond a price–volume baseline, whether point-in-time fundamentals and news sentiment add further information, how portfolio construction converts signals into economic value, and whether results hold on data not used in their formation.

A point-in-time S&P 500 universe is reconstructed from CRSP and Compustat. Two 20-loop RD-Factor trajectories are run under a fixed LightGBM predictor; their factors are audited at the level of hypothesis, code and numerical series, fused with fundamental and news signals, and converted into long-only TopK-Dropout portfolios. A log audit showed that the agent’s feedback loop reuses the metrics of the configured test window, so 2020–2023 is treated as a development period. Results on 2024–2025 are evaluated with moving-block bootstrap inference, with the evidence status of each specification determined by when it was formed; the final strategy is further evaluated on a 2026 holdout and in prospective IBKR paper trading.

The process transfers: both trajectories produce interpretable short-horizon price–volume factors. The predictive increment does not: it is concentrated in the development window, mainly 2020, and no reliable positive increment appears outside that window. Fundamentals carry distinct information, while news matters mainly near publication. Higher predictive metrics do not map monotonically into portfolio returns. All three portfolios evaluated on 2024–2025 trailed SPY. Over 2024 to August 2026 as a whole, the pre-frozen baseline portfolio returned 72.6% against 60.3% for SPY, although the difference is not significant. The final strategy, formed later, outperformed SPY in the 2026 holdout, with returns concentrated in a few stocks. Paper trading exposed partial fills and cash accumulation as implementation risks invisible to backtests.

# Declaration on the Use of Generative AI

Generative AI tools were used in this project in the following ways.

**As part of the experimental system.** RD-Agent(Q) uses OpenAI o4-mini, accessed through the OpenAI API (OpenAI, 2026) and LiteLLM, to generate factor hypotheses, implementation code and feedback. This use is the object of study and is described in Chapter 3.

**Code assistance.** Cursor was used to write, refactor and audit analysis and pipeline code. All code was reviewed and run by the author, and results were checked against logs and output files.

**Writing assistance.** Claude (Anthropic) was used to discuss methodology, check the consistency of arguments, and check references. All claims, numbers and citations were verified by the author against the original sources and experiment outputs.

The research design, experiments, interpretation and conclusions are the author's own.

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# Chapter 1

## Introduction

### 1.1 Motivation

Stock returns exhibit heavy tails, volatility clustering and non-stationarity, stylised facts that make them inherently difficult to predict (Cont, 2001). At the same time, the number of factors proposed in the asset-pricing literature keeps growing, and when many candidate signals are tested repeatedly, conventional significance thresholds produce false discoveries (Harvey, Liu and Zhu, 2016). The central question in factor research is therefore not only whether a predictive signal can be found, but whether it remains stable out of sample and adds information beyond what is already known.

Machine learning and automated search have moved factor research from a small number of hand-built features towards high-dimensional prediction and automated discovery. RD-Agent(Q) (R&D-Agent for Quantitative Finance; RD-Agent below) applies large language models to the research process itself, forming an iterative loop of hypothesis generation, code implementation, validation and feedback (Li et al., 2025). On the CSI 300, its factor-discovery component, RD-Factor, produced a small factor set with good predictive and strategy performance. That evidence, however, comes from the Chinese large-cap market; whether the process remains effective in the US market, with its different investor base, trading rules and information environment, has not been tested directly.

Transferring the process is not the same as copying its factors: both the returns and the construction of a given factor class can vary across markets (Fama and French, 2012; Liu, Stambaugh and Yuan, 2019). This project therefore runs the RD-Agent process itself on the S&P 500 and assesses how its hypothesis generation, implementation and evaluation behave under a different market structure.

Even if the process transfers, two questions remain. Many plausible candidates do not imply information independent of existing factors, and a higher predictive metric does not guarantee a tradable strategy once portfolio constraints and transaction costs are applied. The project therefore also asks whether fundamental and news information

adds to price–volume signals, how predictive signals translate into economic value under specific portfolio rules, and whether the results hold on later data and in paper execution.

## 1.2 Research Questions

The project evaluates whether RD-Agent’s factor-discovery process transfers to US large-cap equities, and whether the resulting signals add information, create economic value at the portfolio level and remain effective on later data and under broker execution. Four research questions are addressed:

**RQ1 – Cross-market factor discovery.**

Applied to US large-cap stocks, can RD-Agent factor discovery produce predictive information beyond a price–volume baseline, and how do the results depend on how the factor set is formed?

**RQ2 – Incremental information.**

Do fundamentals and company news sentiment add predictive information beyond price–volume and RD-Agent signals, and is this increment stable across periods and information timings?

**RQ3 – Economic value and market conditions.**

How do portfolio construction and market conditions affect the conversion of predictive signals into economic value?

**RQ4 – Later-period evaluation and implementation.**

Do the signals and portfolio rules formed during development retain their performance on data not used in their formation, and can the final strategy be executed as designed in paper trading?

## 1.3 Scope and Boundaries of the Conclusions

The project studies the RD-Factor component of RD-Agent(Q) and its downstream use; it does not examine RD-Model, the search over model architectures, or joint factor–model optimisation. The LLM backend is held fixed, and alternative backends were used only in pilot runs.

The empirical conclusions apply to the US large-cap market represented by the S&P 500 and do not extend directly to small caps, other asset classes or other countries. All portfolio results are for long-only TopK-Dropout portfolios with a limited number of holdings, equal-amount purchases and drifting weights; they do not extend directly to long–short or capitalisation-weighted portfolios. The evaluation on data not used in forming each specification covers at most about two and a half years, and paper trading covers only

eight trading days, so these results describe performance and execution in specific market environments rather than long-run stability.

## 1.4 Main Contributions

**First, cross-market transfer and factor-set formation.** The project runs RD-Factor on a reconstructed point-in-time S&P 500 universe and compares two ways of forming a factor set: online accumulation and post-hoc selection. The process generates interpretable candidates in the new market, but their predictive increment is strongly period-dependent.

**Second, auditing the search and its implementation.** Each generated factor is checked against its hypothesis, code and the numerical series it produces, which reveals implementation aliases, look-ahead fields and scale dependence. The audit also shows that RD-Factor’s default feedback loop reuses the metrics of the configured test window, turning a nominal test set into an adaptive development set.

**Third, incremental information from fundamentals and news.** Point-in-time fundamentals carry information distinct from price and volume, although their marginal value depends on the baseline and the period; the value of news is concentrated close to publication.

**Fourth, prediction versus economic value.** Comparing predictive metrics with portfolio returns at the level of search loops, models and trajectories shows that the two do not move together, and that holding breadth and replacement speed need to match the temporal structure of the signal.

**Fifth, evidence status defined by formation time.** Each result is classified by when its specification was formed relative to the data used to evaluate it. On data not used in their formation, the development-period advantages of both trajectories did not persist.

**Sixth, from backtest to broker execution.** The final strategy is evaluated on data not used in its formation and in IBKR paper trading, with holdings reconciliation and fail-closed control (suspending trading when a critical state cannot be confirmed) built into the deployment process.

## 1.5 Structure of the Report

Chapter 2 reviews the related literature. Chapter 3 describes the data, the two search trajectories, signal and portfolio construction, the time partition and the evaluation methods. Chapters 4 to 7 answer RQ1 to RQ4 in turn, covering the generated factors, fundamentals and news, portfolio construction and the final strategy, and later-period evaluation with paper execution. Chapter 8 discusses the results and threats to validity, and Chapter 9 concludes.

# Chapter 2

## Literature Review

### 2.1 Quantitative Factor Research and Evaluation

In this report, a factor is a numerical signal used to predict future returns or to rank stocks in the cross-section; it need not carry an asset-pricing interpretation. Traditional factor research began with a small number of characteristics with clear economic meaning (Fama and French, 1993, 2015; Novy-Marx, 2013; Hou, Xue and Zhang, 2015). As candidate factors multiplied, attention shifted to whether new signals are genuinely independent and persistent. When many candidates are tested repeatedly, conventional significance thresholds produce false discoveries (Harvey, Liu and Zhu, 2016), and many reported anomalies do not survive stricter replication standards (Hou, Xue and Zhang, 2020). A factor that predicts returns on its own may also become redundant once other factors are included (Fama and French, 2015). Factor research therefore needs to distinguish three properties: predictive power, incremental information and stability across periods.

### 2.2 Automated Factor Discovery and RD-Agent

Formulaic alphas express trading signals as executable expressions of market variables, operators and parameters, turning factor discovery into a search problem (Kakushadze, 2016). AlphaEvolve searches over such expressions with evolutionary methods while controlling redundancy (Cui et al., 2021); AlphaGen uses a candidate’s contribution to a factor combination as a reinforcement-learning reward (Yu et al., 2023); and AlphaForge adds time-varying factor weights (Shi et al., 2025). All of these methods search within variables, operators and rules specified in advance by the researcher.

Large language models offer a different approach. RD-Agent(Q) organises quantitative research as a loop of specification, synthesis, implementation, validation and analysis: the model proposes hypotheses, writes code, runs experiments and uses the results to pro-

pose the next round (Li et al., 2025). In RD-Factor, its factor-discovery component, an Analysis Unit compares the IC, annualised excess return and maximum drawdown of the current loop with those of the best state so far (state of the art, SOTA) and decides whether to replace it; accepted factors then form the basis of subsequent search. RD-Factor fixes LightGBM (Ke et al., 2017) as the predictor, so that differences between experiments arise mainly from the factors. Because later hypotheses depend on earlier results, the final factor set is tied to the search path and to the evaluation configuration. AlphaAgent (Tang et al., 2025) also uses LLM agents to generate formulaic factors, adding regularisation for originality, hypothesis–factor alignment and complexity to counter alpha decay, and reports results on both the CSI 500 and the S&P 500. The evidence for RD-Agent(Q), by contrast, comes mainly from the CSI 300.

## 2.3 Cross-Market Factor Transfer

Factors of the same class do not perform consistently across markets (Griffin, 2002; Fama and French, 2012). In China, directly applying classic US factor constructions does not describe stock returns adequately (Liu, Stambaugh and Yuan, 2019), and the T+1 trading rule, daily price limits and a large retail investor base affect short-horizon price relationships and the speed at which information is reflected in prices (Qiao and Dam, 2020; Chen, Gu and Ni, 2023; Tan, Zhang and Zhang, 2024). Even within US equities, many anomalies have attenuated as liquidity and trading activity have risen (Chordia, Subrahmanyam and Tong, 2014), and published predictors often weaken after academic dissemination (McLean and Pontiff, 2016). Cross-market transfer therefore concerns not only whether price–volume relationships keep their predictive power, but also whether the research environment, including the stock universe, trading rules, costs, execution timing and benchmark, is rebuilt for the target market. This project focuses on the discovery process itself: whether, in a reconstructed US market environment, RD-Factor can generate, implement and validate candidates with predictive information.

## 2.4 Factor Redundancy and Incremental Information

Even a candidate with predictive power may overlap heavily with existing signals. AlphaEvolve and RD-Agent(Q) screen for redundancy using cross-sectional correlation, RD-Agent(Q) discarding candidates whose average correlation with the existing SOTA factors reaches 0.99, while AlphaGen asks whether a new candidate improves the existing combination (Cui et al., 2021; Yu et al., 2023; Li et al., 2025). Low correlation, however, is not the same as an increment: a candidate must still improve prediction once added to the existing information set. When code is generated automatically, factors with

different names may also produce the same series. Evaluating generated factors therefore requires examining their standalone predictive power, statistical redundancy, implementation overlap, incremental contribution and cross-period stability together.

## **2.5 From Predictive Signals to Economic Value**

Predictive power becomes return only through portfolio construction. Grinold's (1989) fundamental law of active management expresses risk-adjusted return as a function of forecasting skill and the number of independent decisions; Clarke, de Silva and Thorley (2002) show that long-only, holding-count and turnover constraints weaken the transfer of predictive information into portfolio weights; and Novy-Marx and Velikov (2016) show that transaction costs substantially erode the returns of high-turnover strategies. Whether a signal is tradable therefore depends on portfolio design: the number of holdings, the speed of replacement and the holding period.

## **2.6 Backtest Validity and Information Leakage**

Automated factor discovery repeatedly uses the same historical data to generate, screen and modify candidates, and is therefore exposed to multiple testing and backtest overfitting: after many comparisons, the best historical result may be an artefact of the search (White, 2000; Harvey, Liu and Zhu, 2016; Bailey et al., 2017). In adaptive search, later candidates depend on earlier results, so post-hoc significance adjustments cannot fully remove the selection effect; once a holdout set informs successive analysis choices, it no longer gives an unbiased estimate of generalisation (Dwork et al., 2015). A second risk is information leakage: look-ahead bias and ex-post sample selection can substantially alter backtest results (Banz and Breen, 1986), so fundamental release dates, news publication times and the historical stock universe must be aligned with the information available at the time. Keeping part of the data outside the development process, and evaluating on it only after the specification is fixed, is therefore a necessary complement to statistical correction.

## **2.7 Fundamentals, News Sentiment and Market State**

Fundamental variables provide lower-frequency information tied directly to a firm's operating condition (Novy-Marx, 2013; Fama and French, 2015). Company news provides higher-frequency, unstructured information: Tetlock, Saar-Tsechansky and Macskassy (2008) find that the language of news helps predict subsequent earnings and returns. Beyond dictionary methods, pre-trained financial language models such as FinBERT (Araci,

2019) can estimate sentiment directly from text. In either case, the value of a news signal depends on whether its publication time and company attribution can be aligned accurately.

Predictive relationships may also change with market conditions. Hamilton’s (1989) Markov-switching model is the classic way to describe latent states. Momentum returns depend on the market state, with severe losses concentrated in rebounds after high-volatility declines (Cooper, Gutierrez and Hameed, 2004; Daniel and Moskowitz, 2016), and the quality premium also varies over time (Asness, Frazzini and Pedersen, 2019). Moreira and Muir (2017) find that scaling exposure by volatility improves many factor portfolios, but Cederburg et al. (2020) show that this advantage weakens considerably under stricter out-of-sample evaluation.

## 2.8 Research Gaps

The literature leaves four related gaps. First, the evidence for RD-Agent(Q) comes from the Chinese market. Other LLM-based systems such as AlphaAgent report results on US equities, but with different search and acceptance mechanisms; whether RD-Factor, which accepts factors through model-generated feedback and accumulates them across loops, can regenerate useful signals in a reconstructed US market has not been tested (RQ1). Second, less attention has been paid to whether generated factors with different names duplicate one another in implementation, to whether they add genuinely incremental information, and to what fundamentals and news add beyond price–volume signals (RQ1, RQ2). Third, there is little systematic evidence on how the predictive power of automatically generated signals translates into economic value under portfolio constraints, transaction costs and changing market states (RQ3). Fourth, the feedback loop of an automated agent can itself reuse evaluation data, yet the evaluations reviewed here do not clearly separate the data that inform the search from the data used to judge the final specification, nor follow a specification from development to later data and broker execution (RQ4).

## 2.9 Tools and Data

The empirical work is built on three software layers and several market-data sources. Factor search, model training and portfolio backtesting run in RD-Agent(Q) (Li et al., 2025) on top of Qlib, an AI-oriented quantitative investment platform that supplies the data handlers, LightGBM workflows and TopK-Dropout execution used throughout (Yang et al., 2020). The RD-Agent source code used here was obtained from the public GitHub repository (Microsoft, 2025) and adapted on the project branch `us-market-experiment`.

Large-language-model calls inside the agent use the OpenAI API (OpenAI, 2026). Point-in-time US equity prices, returns and S&P 500 membership come from CRSP (Center for Research in Security Prices, 2026) via WRDS (Wharton Research Data Services, 2026); fundamentals come from Compustat (Compustat, S&P Global Market Intelligence, 2026); company news and sentiment scores come from the Alpha Vantage News and Sentiment API (Alpha Vantage, 2026); the 2026 holdout uses Massive daily bars with approximate constituent and fundamental chains (Massive, 2026); and prospective execution uses Interactive Brokers paper trading (Interactive Brokers, 2026). Chapter 3 details how these sources are aligned to avoid look-ahead.

# **Chapter 3**

## **US Market Reconstruction, Research Design and Methods**

### **3.1 Overall Research Design**

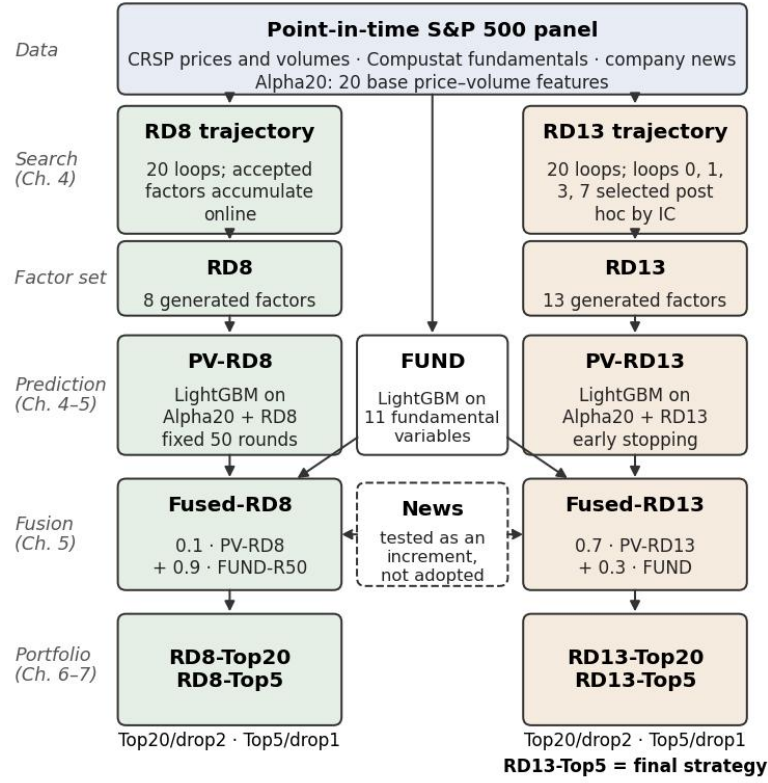
This chapter describes the data construction, the two RD-Factor search trajectories, signal and portfolio construction, the time partition and the evaluation methods. The experiments proceed along two trajectories: the RD8 trajectory forms its factor set by online accumulation, and the RD13 trajectory forms its factor set by post-hoc selection from a separate search. Each is then fused with fundamental information and converted through portfolio rules into strategies that can be evaluated (Figure 3.1). The main identifiers and their meanings are summarised in Appendix I.

### **3.2 S&P 500 Sample and Point-in-Time Data**

The S&P 500 is the main research market, and the daily investable universe is reconstructed from historical constituent relationships. A security enters the day's cross-section only during its membership spell; if it re-enters the index after leaving, this is treated as a new spell. This design avoids back-filling historical periods with current constituents and thereby reduces survivorship bias.

Long-run security identity uses the permanent identifier PERMNO of the Center for Research in Security Prices (CRSP) as the primary key, combined with ticker and company identifiers to complete the historical membership mapping. Securities whose relationship cannot be uniquely confirmed are excluded from the investable sample on the corresponding trading day.

US equity market data for 2008–2025 come from CRSP, obtained through Wharton Research Data Services (WRDS). Quarterly company financial data come from Compus-



FUND-R50: FUND trained for a fixed 50 rounds. Scores are z-scored cross-sectionally each day before fusion.

**Figure 3.1:** Research pipeline: the two RD-Factor trajectories, the price–volume and fundamental models built from them, the fused signals and the portfolios evaluated in later chapters.

tat and are mapped to PERMNO through the CRSP–Compustat linking information; company news and news sentiment come from the Alpha Vantage News and Sentiment API. The news data downloaded for this study have observations only from 2010 onwards, and coverage rises markedly over time: the share of stock–days with news is about 4% in the training period, about 11% in the validation period and about 25% in 2020–2023. There are no news observations in 2008–2009, and all stocks in that period are treated as having no news.

All information enters the prediction panel according to the time at which it actually became available. The prediction for trading day  $t$  uses only information that was public and observable by  $t$ : prices and volumes enter the feature calculation after the close of the corresponding trading day; fundamental information is mapped to the first valid trading day after the date on which the report became available; news is aligned to tradable dates according to its publication timestamp, company attribution and ticker relevance. The Compustat availability-date rule, the treatment of missing values and the field mapping are listed in Appendix A (code in Appendix J.2).

CRSP data from WRDS end in 2025, so the 2026 holdout and paper trading use daily market data from Massive, with the overlapping period 2023–2025 used to check the

consistency of the two sources’ return series and corporate-action handling. Fundamental data for 2026 come from Massive financial statements, with the availability date set by the public filing time; the stock universe is reconstructed from the constituents at the end of 2025 plus publicly announced constituent changes.

### 3.3 RD-Factor Experiment Design

The predictive model is fixed as LightGBM, so that the main variation between experiments comes from the candidate factors and the search history. RD-Factor is a stateful, feedback-driven search process: each iteration (loop) performs hypothesis generation, code implementation, validation and analysis in turn (Li et al., 2025). After a candidate hypothesis is implemented as a daily stock feature, it is fed into LightGBM together with the base price–volume feature set Alpha20 (Section 3.4); in the analysis stage, the large language model compares the current loop’s IC, excess ARR and excess MDD with those of the current SOTA state and decides whether to replace it, and accepted factors become the basis of subsequent iterations.

To control the search budget, two RD-Factor search trajectories are compared under the same length of 20 iterations. The first trajectory uses RD-Factor’s online update mechanism: accepted candidate factors enter the subsequent search state loop by loop, finally forming the RD8 factor set. The second trajectory is taken from another 20-loop experiment in which the framework’s cached baseline did not match the (correctly configured) US setting, so no candidate was accepted (Appendix E.4). The RD13 factor set was built by post-hoc selection of the loops in which the 2020–2023 development-period IC reached a new high (loops 0, 1, 3 and 7; see Section 4.3).

The two trajectories thus represent two ways of forming a factor set: online accumulation during the search, and post-hoc selection on development-period performance. How the 2020–2023 window, which supplies the search feedback, affects each of them is set out in Section 3.7.

### 3.4 Predictive Information and Signal Construction

Alpha20, a set of 20 technical features built from price and volume, is the base information set common to both trajectories (Appendix C). Each RD factor set is combined with Alpha20 and fed into LightGBM to form a price–volume model: PV-RD8 from RD8 and PV-RD13 from RD13 (Figure 3.1).

The two price–volume models are trained differently. PV-RD13 follows RD-Agent’s default setting, stopping early on the 2018–2019 validation period; all three seeds stop after the first tree, and Fused-RD13, RD13-Top20 and RD13-Top5 are built on this model.

Because early stopping yields models of very different sizes, factor-set comparisons and the RD8 trajectory use a fixed 50 boosting rounds (Appendix E.2).

The numerical scales and tail distributions of the different RD factors differ considerably, so before training a robust scaling using training-period statistics only is applied:

$$z = \frac{x - \text{median}(x)}{1.4826 \times \text{MAD}(x)},$$

where  $\text{MAD}(x) = \text{median}(|x - \text{median}(x)|)$  and the constant 1.4826 makes the MAD comparable to the standard deviation under a normal distribution. The median and MAD are estimated only from the 2008–2017 training data and held fixed in later periods; after standardisation, values with absolute value above 3 are set to  $\pm 3$  (truncation) so that a small number of extreme observations do not dominate the model.

Fundamental information is the second class of information source. It comprises 11 variables (ROE, ROA, gross profitability, sales growth, asset growth, accruals, leverage, current ratio, book-to-market, earnings yield and sales-to-price), covering profitability, growth, investment, earnings quality, capital structure, liquidity and valuation. Fundamental variables enter the prediction panel only after the relevant financial information has been made public. On each trading day they are winsorised at 1%/99% within the cross-section, then standardised cross-sectionally, and then trained separately with LightGBM to form a fundamental predictive score, FUND. Like PV-RD13, FUND stops early on the validation period and serves as the fundamental component of Fused-RD13; FUND-R50 is the same model trained for a fixed 50 rounds and is used in the RD8 trajectory. To test the increment of fundamentals, a D series of models is trained on the common sample: D0 (Alpha20), D1 (Alpha20 and fundamentals), D2 (Alpha20 and RD13, that is, PV-RD13) and D3 (Alpha20, RD13 and fundamentals).

The price–volume predictive score and the fundamental predictive score come from different models, with different ranges and dispersions; if they were simply added, the more dispersed one would dominate. On each trading day the two are therefore first converted separately into cross-sectional z-scores, placing them on the same scale, and then fused with linear weights:

$$W(i, t; s) = s \cdot \text{PV}(i, t) + (1 - s) \cdot \text{FUND}(i, t), \quad 0 \leq s \leq 1,$$

where PV is the price–volume predictive score, FUND is the fundamental predictive score, and  $s$  is the weight of price–volume information in the fused score, not a capital-allocation ratio. The fusion weights of both trajectories are determined only in the 2018–2019 validation period and held fixed thereafter, but the selection rules differ: the RD8 trajectory selects, on the grid  $s = 0, 0.1, \dots, 1$ , the weight with the highest average validation-period

Rank IC; the RD13 trajectory selects Fused-RD13 from four candidate fusion specifications on the basis of validation-period portfolio performance. The fusion weights and validation-period results of the two trajectories are reported in Chapter 5; the specific parameters and implementation details are given in Appendix C.

Company news sentiment is the third information source. Article-level sentiment is aggregated to company level with relevance weights and aligned to trading days by publication timestamp and company attribution, under two schemes: a delayed daily scheme matched to the same daily target as the price–volume models, and an immediate scheme that uses only news published before the open to predict the open-to-close return of the same day. Stock-days without news are kept as missing rather than treated as neutral sentiment.

In the RD8 trajectory, sentiment is fused linearly with Fused-RD8 as a third score; its weight is chosen on the validation period, and sentiment is adopted only if the bootstrap interval of the paired validation-period  $\Delta$ Rank IC lies entirely above zero. In the RD13 trajectory, delayed sentiment is tested under the daily target, with FinBERT scores (Araci, 2019) as a control for the Alpha Vantage scores, and the immediate scheme tests the information in pre-open news. The two trajectories use different samples and targets, so their absolute Rank ICs are not compared. Alignment and standardisation details are given in Appendices F.5–F.8.

### 3.5 Factor Evaluation

Factors are evaluated at three levels. First, statistical redundancy is checked, that is, whether the average daily cross-sectional correlation between a new candidate and the existing factors reaches 0.99 (Li et al., 2025). Second, the stated formula, underlying fields, code implementation and the series actually generated are checked for consistency, to distinguish superficial differences in formula from actual differences in input. Third, it is tested whether adding the candidate to Alpha20 or to the existing RD factors raises IC or Rank IC, and whether this effect is stable across years. For the illiquidity factor whose implementation depends on the level of adjusted volume, a sensitivity test using a dollar-volume definition is also carried out (Appendix F.10).

### 3.6 Portfolio Construction and Market-State Design

The TopK-Dropout strategy of Qlib (Yang et al., 2020) is used to convert daily stock rankings into a continuous long-only portfolio. TopK controls the target number of holdings, `n_drop` controls the maximum number of stocks replaced at a single rebalance, `hold_thresh` controls the minimum holding period in days, and `risk_degree` controls the

proportion of cash that may be deployed at each purchase; new purchases are funded in equal amounts of “cash  $\times$  risk\_degree / number of purchases”. Existing holdings drift naturally with prices and are not fully rebalanced to equal weight each day. The main experiments uniformly set hold\_thresh = 1 and risk\_degree = 0.95, and study holding breadth and replacement speed through different combinations of TopK and n\_drop.

The core portfolio rules are Top20/drop2 and Top5/drop1, with Top20/drop4 as a control for faster replacement. The correspondence between signals and portfolio rules is listed in Appendix D; the comparison and selection of specific portfolio specifications are discussed in Chapter 6.

The market-state experiments are run separately from the holding-breadth experiments, so that the researcher degrees of freedom of the two searches are not mixed in a single selection: the holding-breadth experiments fix the signal and the position size and compare only different combinations of TopK and n\_drop (Sections 6.2–6.4); the market-state experiments, on a fixed portfolio rule (Section 6.5), test whether adjusting signal weights, equity exposure or portfolio rules by state can improve performance. The two-state hidden Markov model (HMM) uses the SPY daily return and the realised volatility over the past 20 trading days; the three-state specification adds the ratio of 20-day to 60-day realised volatility. The initial HMM parameters are estimated from the 2008–2017 training data, and subsequent states are updated day by day by causal forward filtering, avoiding the use of future state information.

The state mechanisms fall into two classes: one adjusts the relative weights of the different predictive scores in the ranking according to the state posterior, and the other adjusts the target equity exposure according to the market state. They answer, respectively, “which predictive information is more useful in which market state” and “how much market risk should be taken in each market state”.

### **3.7 Time Partition and Design of the Subsequent Evaluation**

The sample is divided into training (2008–2017), validation (2018–2019), development (2020–2023) and holdout (2024–2025) periods. RD-Factor’s Analysis Unit receives the IC, portfolio return and drawdown computed on its configured test window (2020–2023) and uses them as feedback for later loops, so this window is treated as a development (search-feedback) period rather than an independent test sample.

Whether a later period counts as independent evidence depends on when each specification was formed (Table 3.1). The RD13 chain (RD13, Fused-RD13 and RD13-Top20) was fixed before 2024–2025 was opened, and no step of it used that window, so 2024–

2025 is an independent evaluation for this chain. The RD8-chain specifications were formed after the window had been opened for the RD13 chain; neither their search nor their downstream selection used it, but their 2024–2025 results are reported as a first evaluation rather than an independent test. RD13-Top5 was formed after the 2024–2025 results had been observed, so its results for that period are post-hoc diagnostics, and it is tested on the 2026 holdout.

| Period    | Original label | Main use and role in this report                                                                                                                                                                                                             |
|-----------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2008–2017 | Train          | Model training, pre-processing statistics and initial HMM estimation                                                                                                                                                                         |
| 2018–2019 | Validation     | Limited selection of specifications and weights                                                                                                                                                                                              |
| 2020–2023 | Test           | Development (search-feedback) period: RD-Factor search feedback, RD13 selection, model and portfolio research                                                                                                                                |
| 2024–2025 | Holdout        | Independent evaluation for the RD13 chain (RD13, Fused-RD13, RD13-Top20) frozen before the window was opened; first evaluation for the RD8 chain (Fused-RD8 / RD8-Top20 / RD8-Top5); post-hoc diagnosis for specifications formed afterwards |
| 2026      | –              | 2026 holdout and prospective paper trading                                                                                                                                                                                                   |

**Table 3.1:** Sample periods: original labels and role in this report

The 2020–2023 window affects the two trajectories to different degrees. In both, factors were generated and accepted under search feedback computed on this window; RD13 was in addition selected post hoc by the IC of the same window. For the RD8 chain, downstream weights and portfolio parameters were selected on 2018–2019 only, so its development-period results are less affected by selection than those of RD13, although they are still not independent test evidence.

To separate differences in signal from differences in portfolio rule, a  $2 \times 2$  matched comparison is also used for 2024–2025: the fused signals of the two trajectories are each evaluated under the Top20/drop2 and Top5/drop1 rules, with the same trading times, cost assumptions and stock sample. Each signal is computed on its own universe and the ranking is then restricted to the stocks common to both; all four combinations are reported, to avoid comparing only the combination that performed best after the fact. The main results of the RD8 trajectory are still based on its own universe.

The 2026 window serves as the holdout for RD13-Top5. Its data up to 18 August predate the formation of RD13-Top5 but were not accessed until RD13-Top5 had been fixed; the remainder of the window, and the period that follows it, accrued only after formation and is therefore forward evidence. From 31 August, RD13-Top5 was also run in prospective IBKR paper trading. Because the factors and fundamental model of the RD8 trajectory cannot be rebuilt under their frozen definitions on the 2026 data chain, only the RD13 trajectory is evaluated in 2026, against a placebo distribution from 500 sets of daily random rankings. The 2026 results use the revised price–volume data; the first replay on the earlier data is reported in Appendix H.2.

### 3.8 Evaluation Metrics and Statistical Methods

At the prediction level, IC, Rank IC, ICIR and Rank ICIR are reported. IC is the Pearson correlation between the predictive score and the future return in the daily cross-section; Rank IC uses the Spearman correlation; ICIR and Rank ICIR are the time-series mean of the corresponding daily correlation divided by its time-series standard deviation.

Incremental comparisons between models (paired  $\Delta$ IC and paired  $\Delta$ Rank IC) use only stock–days on which both models have valid predictions; the difference is computed day by day and then aggregated over time, so the result need not equal the simple difference between the two models’ full-sample means. Comparisons involving fundamentals are restricted to the common sample of stock–days on which both the price–volume and the fundamental predictions are valid, so the IC of the same price–volume model may differ between Chapters 4 and 5.

At the portfolio level, SPY (CRSP PERMNO 84398) is the market benchmark; SPY tracks the S&P 500 and corresponds directly to the research universe; factor-level IC and Rank IC do not depend on the market benchmark. The portfolio level reports cumulative return, annualised return, excess return relative to SPY, annualised volatility, Sharpe ratio, maximum drawdown, turnover and transaction costs. Development-period ARR is computed according to Qlib’s daily definition, namely  $238 \times$  the mean daily return; the US market has about 252 trading days a year, so ARR under this definition is about 5.6% lower than the value annualised over 252 days, which does not affect comparisons under the same definition. Subsequent wealth performance is additionally described by the CAGR computed from the cumulative wealth path. The two annualised metrics are reported separately and are not converted into one another.

The annualised volatility and Sharpe ratio computed in this report use a 252-trading-day conversion and do not subtract a risk-free rate. The daily excess return relative to SPY is defined as

$$r_{\text{excess},t} = r_{\text{portfolio},t} - r_{\text{SPY},t} - \text{cost}_t.$$

Excess ARR and excess MDD are both computed from this same daily excess-return series. Annual results follow the same daily definitions. Turnover is the daily ratio of two-sided (buy plus sell) traded value to account value; market exposure is described by an OLS regression of the strategy’s daily return on the SPY daily return. For shorter windows such as 2026, cumulative return, the change in account value and implementation deviation are the main descriptive quantities.

The moving-block bootstrap (Künsch, 1989) is used to handle the temporal dependence of daily series and to construct 95% confidence intervals. The main analysis uses a block length of about one trading month (20 or 21 trading days) and 2,000 replications, supplemented by Newey–West heteroskedasticity- and autocorrelation-consistent (HAC)

statistics (Newey and West, 1987); for the shorter 2026 window, a shorter block length is also used, with a 21-day block as a robustness check. The block lengths, numbers of replications and random seeds for each experiment are listed in Appendix D. Appendix D.5 summarises the threats to validity addressed by the design and the control used for each.

### 3.9 Prediction Target, Transaction Costs and Execution

The main prediction target is the one-day close-to-close return following execution at the next trading day’s close; its Qlib label is

$$y_t = \frac{\text{Close}(t+2)}{\text{Close}(t+1)} - 1.$$

The price–volume information of trading day  $t$  is formed after the close, so the temporal relationship of the historical process is: information at the close of day  $t \rightarrow$  execution at the close of day  $t+1 \rightarrow$  return from the close of  $t+1$  to the close of  $t+2$ . This design keeps feature availability, execution time and return label consistent. In the 2026 implementation phase, next-day close execution is mapped to the broker’s market-on-close (MOC) order.

Historical portfolios assume a uniform transaction cost of 1 bp each for buying and selling. In the paper-trading phase, orders, fills and commissions are saved from the broker’s records, and the theoretical target, planned orders, actual fills and broker holdings are recorded separately. Prospective execution takes the account and position state returned by the broker as the actual position; before each rebalance, the system first checks that the actual holdings, cash and contract information in the broker account agree with expectations, and if any of these cannot be confirmed (for example, if the contract for a stock cannot be identified), no orders are submitted that day. This practice is called fail-closed.

To reduce look-ahead bias, temporal order is maintained at four points: the stock universe, information availability, model estimation and trade execution. The historical universe contains only the constituents valid at the time; fundamentals and news enter the panel only after publication; model parameters and pre-processing statistics are estimated only from training-period data; and trade execution always follows the time of the information used for prediction. The candidate factors, code and prediction outputs of each loop, and the times at which models and strategies were fixed, are all saved, so that it can be confirmed that the subsequent-evaluation data did not take part in earlier selection.

## Chapter 4

# RD-Agent Factor Discovery in the US Equity Market

### 4.1 Research Question and Evaluation Framework

This chapter answers the first research question: when RD-Factor is applied to US large-cap stocks, can it generate incremental predictive information beyond Alpha20, and how do the results of the two ways of forming the factor set differ? The analysis proceeds at two levels: first, how factors were generated, accepted and implemented during the search; second, how the two factor sets compare in predictive power under an identical model setting. The period 2020–2023 took part in the search and screening of both trajectories and reflects only performance within the search window; 2018–2019 and 2024–2025 took no part in screening and are used to test whether the increment extends to other periods. As explained in Section 3.7, search feedback on this window conditions both trajectories and RD13 was also selected by its IC, so results for 2020–2023 are read as development evidence.

### 4.2 The Online Accumulation Trajectory: RD8

The first 20-loop trajectory uses RD-Factor’s online update mechanism. The trajectory contains one manual correction: Loops 0–1 were retained from an earlier run, the corrected Loop 2 factor was re-evaluated by the unchanged Analysis Unit, and Loops 3–19 were proposed and accepted by the framework without intervention (Appendix B.1). In the end only loops 1, 3 and 9 entered the SOTA state, progressively forming eight RD-generated factors. Acceptance did not follow a single ordering by IC: loop 1 had an IC close to zero but turned the portfolio excess return positive; loop 3 improved both prediction and portfolio performance; loop 9 further added adaptive momentum factors and

formed the terminal state. Loops 10–19 did not change this factor set again.

| State                  | IC       | Excess ARR | Excess MDD | Search outcome           |
|------------------------|----------|------------|------------|--------------------------|
| Alpha20 starting point | 0.004579 | −1.44%     | −26.83%    | Starting point           |
| Loop 1                 | 0.000010 | +3.95%     | −33.38%    | Accepted                 |
| Loop 3                 | 0.005944 | +16.40%    | −30.66%    | Accepted                 |
| Loop 9                 | 0.006046 | +16.61%    | −25.16%    | Accepted, terminal state |

**Table 4.1:** SOTA states in the RD8 trajectory (2020–2023; portfolio metrics are net of costs and in excess of SPY)

The terminal RD8 set contains eight factors covering short-term reversal, price range and volatility state, volume anomaly, risk-normalised momentum, return–volume co-movement and adaptive-window momentum (the full list is in Appendix C.1). These correspond to the momentum, reversal, volatility-state and volume effects documented in US equities (Jegadeesh and Titman, 1993; Campbell, Grossman and Wang, 1993; Lee and Swaminathan, 2000; Barroso and Santa-Clara, 2015; Daniel and Moskowitz, 2016).

### 4.3 The Development-Period Selection Trajectory: RD13

The second trajectory formed no cumulative factor library during its original search. Within the same budget of the first 20 loops, loops 0, 1, 3 and 7 were selected as the loops in which the 2020–2023 IC reached a new high. The IC of the four loops rises from 0.0050 to 0.0103, and their factors were merged into 13 price–volume factors, denoted RD13. RD13 refers throughout to the revised implementation (Section 4.6). Because these four loops were selected exactly on the 2020–2023 IC, the IC of RD13 in this period is inflated by “picking the best result”, and the results for this period can serve only as development-period evidence (Harvey, Liu and Zhu, 2016; Giglio, Liao and Xiu, 2021).

| Selected loop | IC       | Excess ARR (SPY) | Excess MDD (SPY) |
|---------------|----------|------------------|------------------|
| Loop 0        | 0.005037 | −1.01%           | −26.94%          |
| Loop 1        | 0.007674 | +0.24%           | −27.16%          |
| Loop 3        | 0.007972 | −0.72%           | −40.28%          |
| Loop 7        | 0.010275 | −4.41%           | −34.43%          |

**Table 4.2:** The development-period IC selection path of RD13

The loops selected by IC do not have higher portfolio returns: taking the simple average of the net-of-cost excess ARR against SPY of each loop in Table E.2, the four selected loops average about −1.47% and the remaining 16 loops about +5.13%. This trajectory formed no cumulative state because the framework reused a stale cached baseline that did not match the US configuration (Appendix E.4). An offline replay that keeps

the IC and the comparison threshold unchanged and only converts the return and draw-down in the feedback to the SPY definition shows that the replacement decision changes in some loops, and that the decision mainly follows the annualised return (Appendix E.4). A configuration item such as the return benchmark therefore affects the whole search path through the feedback.

The implementation check also found that the 13 names in RD13 initially corresponded to only four distinct numerical series. The revised RD13 rebuilds the factors according to their original economic definitions so that all 13 differ; the original four series are denoted RD4 and serve as a control for implementation duplication.

## **4.4 Comparison of the Two Trajectories**

### **4.4.1 Similar factor structures**

Although the two trajectories were formed independently, they repeatedly arrived at similar short-horizon price–volume structures: the average daily cross-sectional correlations between the 10-day ATR of RD8 and the 14-day normalised ATR of RD13, between the volatility-normalised momentum factors of the two trajectories, and between the volume-anomaly factors are about 0.96, 0.96 and 0.87 respectively (Appendix Table E.12). These correlations are below the redundancy threshold of 0.99 but close to 1, indicating that different search paths rediscovered highly similar economic structures. RD13 also contains illiquidity, rolling beta, idiosyncratic volatility and win-rate features that have no direct counterpart in RD8.

### **4.4.2 Incremental predictive power**

To compare the factor sets themselves, both groups of models were retrained under the unified setting (Section 3.4; Table 4.3). In 2018–2019, which took no part in search or screening, the increment of both factor sets relative to Alpha20 is slightly negative; in 2020–2023, the increment of RD8 is positive but not significant, while that of RD13 is larger and significant, but RD13 was selected exactly on the IC of this window; in 2024–2025 both intervals contain zero. Broken down by year, the development-period increment comes mainly from 2020 (+0.0114 for RD8 and +0.0235 for RD13, both significant), and in 2022 both are negative (Appendix Table E.13). The new price–volume information is therefore concentrated in a single year within the search window.

| Information set | 2018–2019<br>Rank IC | 2020–2023<br>Rank IC | 2024–2025<br>Rank IC | $\Delta$ Rank IC<br>2020–2023 | $\Delta$ Rank IC<br>2024–2025 |
|-----------------|----------------------|----------------------|----------------------|-------------------------------|-------------------------------|
| Alpha20         | 0.00396              | 0.00617              | −0.00202             | —                             | —                             |
| Alpha20 + RD8   | 0.00347              | 0.00945              | 0.00079              | +0.00328 ( $p = 0.098$ )      | +0.00282 ( $p = 0.167$ )      |
| Alpha20 + RD13  | 0.00309              | 0.01512              | 0.00240              | +0.00895 ( $p = 0.007$ )      | +0.00443 ( $p = 0.174$ )      |
| Alpha20 + RD4   | 0.00490              | 0.01204              | 0.00110              | +0.00587 ( $p = 0.009$ )      | +0.00312 ( $p = 0.147$ )      |

**Table 4.3:** Comparison of information sets under the unified setting (3 random seeds; paired differences use a 21-day moving-block bootstrap)

#### 4.4.3 Prediction and portfolio return

During the search phase, IC and portfolio return also lack a stable correspondence: the correlation between candidate IC and excess ARR across loops is about  $-0.05$  for RD8 and about  $-0.55$  for RD13. Because successive loops are connected, each built on the results of the previous one, they are not independent observations, and these coefficients are descriptive only. The portfolio comparison under unified predictions is more direct (Table 4.4): RD13 has the higher Rank IC, but its excess return is lower than that of RD8 under both Top20/drop2 and Top5/drop1, by about 5.3 and 4.1 percentage points respectively, neither of which is significant under the moving-block bootstrap ( $p = 0.31$  and  $0.69$ ).

| Signal         | Portfolio rule | Excess ARR | Excess MDD | Mean turnover |
|----------------|----------------|------------|------------|---------------|
| Alpha20 + RD8  | Top20/drop2    | 15.15%     | −26.63%    | 0.192         |
| Alpha20 + RD8  | Top5/drop1     | 15.07%     | −63.68%    | 0.382         |
| Alpha20 + RD13 | Top20/drop2    | 10.12%     | −48.93%    | 0.192         |
| Alpha20 + RD13 | Top5/drop1     | 11.24%     | −84.12%    | 0.382         |

**Table 4.4:**  $2 \times 2$  portfolio comparison under unified predictions (2020–2023; fixed 50 rounds)

## 4.5 Model Information Use and Tied Scores

Individual RD factors have very weak predictive power when used alone: the highest single-factor Rank IC in 2020–2023, for 5-day short-term reversal, is only 0.0091, and most factors are close to zero or negative. Within the model, however, RD factors account for about 30% (RD8) and 40% (RD13) of LightGBM split gain, indicating that they work mainly in combination with other features.

PV-RD13 contains a single tree, while the factor-set comparison in this chapter uses models trained for a fixed 50 rounds (Appendix Table E.14). A model with a single tree gives many stocks exactly the same score, and when scores are tied Qlib simply takes the first  $K$  in the original order of the data table. To measure the effect of this incidental ordering, the order of tied stocks was shuffled at random and the run repeated 100 times:

the 2020–2023 net excess return of pure PV-RD13 under Top5/drop1 is 20.6% in the original order, 22.7% on average under random orders, and 90% of the results lie between 15.2% and 30.9%. The return of a concentrated portfolio can differ by more than ten percentage points solely because of the order of tied stocks.

## 4.6 Configuration and Implementation Audit

The audit combined log inspection, configuration checks and factor-level code review (Appendix J.3); Table 4.5 lists the issues it found.

| Issue                                                                                              | Affected                                          | Handling                                                 |
|----------------------------------------------------------------------------------------------------|---------------------------------------------------|----------------------------------------------------------|
| Test-window metrics fed back to the Analysis Unit                                                  | Both trajectories                                 | 2020–2023 relabelled as development period (Section 3.7) |
| Stale framework baseline cache (experiment configuration correct)                                  | RD13 trajectory: no SOTA chain formed             | Isolated cache for RD8; Appendix E.4                     |
| Placeholder benchmark (stock P10104)                                                               | Early excess returns                              | SPY ingested; results recomputed                         |
| Generated code does not match its hypothesis (absolute price range; value scales with share price) | RD8 trajectory, Loop 2                            | Corrected and re-evaluated; search continued             |
| Price-adjustment field used as a signal                                                            | RD13 trajectory, Loops 12–13                      | Excluded from RD13                                       |
| Illiquidity factor uses split-adjusted volume                                                      | One RD13 factor                                   | Dollar-volume version as sensitivity test (Section 5.2)  |
| Autoencoder fitted on the full sample                                                              | ae_recon_error_10d (RD8, Loop 15)                 | Excluded from the main factor sets                       |
| 13 factor names map to 4 series                                                                    | Original RD13                                     | Rebuilt; original series kept as RD4                     |
| Failed backtests                                                                                   | RD8, Loops 16–17 (including autoencoder variants) | Recorded as failures; not interpreted                    |

**Table 4.5:** Configuration and implementation issues identified by the audit

Three of these issues concern the evaluation configuration rather than the factors themselves. Test-window reuse is discussed in Section 3.7. The others concern whether a factor’s natural-language hypothesis, code and numerical series agree, and two of them show how future information can enter through platform fields rather than through the factor logic. Loops 12–13 of the RD13 trajectory used Qlib’s cumulative price-adjustment field directly; because its values are set relative to later splits, they embed future events. The Amihud-type illiquidity factor in RD13 (Amihud, 2002) divides by adjusted volume,

which is scaled backwards from the last day of the sample: if a stock splits four-for-one after day  $t$ , its volume on day  $t$  is multiplied by four, so the factor value on that day reflects a future event. All 21 final factors are nevertheless invariant to a proportional scaling of prices.

## 4.7 Chapter Summary

RD-Agent can generate economically meaningful short-horizon price–volume factors in US large-cap stocks, and the two independently formed trajectories repeatedly discovered highly similar structures; the number of independent information dimensions is smaller than the number of factor names. The predictive increment of both trajectories is concentrated in 2020, within the development window, and is indistinguishable from zero in both the validation period and 2024–2025. Both trajectories also show that predictive metrics and portfolio returns have no stable one-to-one correspondence, and that ties in shallow models amplify the uncertainty of concentrated portfolios. Automated factor discovery therefore needs to be checked at several levels at once: information increment, cross-period stability, code implementation and portfolio conversion.

# Chapter 5

## Incremental Information in Fundamentals and News Sentiment

### 5.1 Research Question and Design

This chapter answers the second research question: can point-in-time fundamentals and company news sentiment provide incremental predictive information beyond price–volume signals? Both trajectories are used: the RD13 trajectory tests how fundamentals enter Fused-RD13, the signal actually used; the RD8 trajectory compares price–volume and fundamentals under a fixed number of training rounds, so that model sizes are identical and the increment is easier to interpret. The chapter concentrates on the increment at the prediction level, with portfolio returns as supporting description only.

### 5.2 The RD13 Trajectory: The D Series and Fused-RD13

The RD13 trajectory changes the information set step by step on the same common sample (Table 5.1; D2 is the price–volume model PV-RD13) and compares the models with paired tests (Table 5.2).

| Spec. | Information set               | 2018–2019<br>Rank IC | 2020–2023<br>Rank IC | 2020–2023 net<br>excess ARR |
|-------|-------------------------------|----------------------|----------------------|-----------------------------|
| D0    | Alpha20                       | $\approx 0.00136$    | $\approx 0.00668$    | +3.65%                      |
| D1    | Alpha20 + Fundamentals        | 0.01121              | 0.00873              | +8.72%                      |
| D2    | Alpha20 + RD13                | 0.00296              | 0.00729              | +11.18%                     |
| D3    | Alpha20 + RD13 + Fundamentals | 0.01090              | 0.01425              | +8.28%                      |

**Table 5.1:** The D series of the RD13 trajectory (the Rank IC of D0 is inferred from the paired difference)

The effect of adding fundamentals depends on the baseline. On the model with Alpha20 alone, fundamentals raise Rank IC significantly in the validation period but not in

| Comparison                              | 2018–2019<br>$\Delta$ Rank IC | 95% CI               | 2020–2023<br>$\Delta$ Rank IC | 95% CI               |
|-----------------------------------------|-------------------------------|----------------------|-------------------------------|----------------------|
| D1 – D0 (fundamentals added to Alpha20) | +0.00985                      | [+0.00215, +0.01844] | +0.00205                      | [–0.00578, +0.00787] |
| D3 – D2 (fundamentals added to RD13)    | +0.00868                      | [–0.00093, +0.01714] | +0.00696                      | [+0.00116, +0.01228] |
| D3_R50 – D2_R50 (fixed 50 rounds)       | +0.00609                      | [–0.00078, +0.01219] | +0.00467                      | [+0.00007, +0.00890] |
| D3 – D1 (RD13 added to fundamentals)    | –0.00026                      | [–0.00738, +0.00651] | +0.00558                      | [–0.00279, +0.01442] |
| Fused-RD13 – PV-RD13                    | +0.00581                      | [+0.00209, +0.00906] | +0.00329                      | [+0.00077, +0.00606] |
| Fused-RD13 – FUND                       | –0.00579                      | [–0.01283, +0.00275] | +0.00275                      | [–0.00415, +0.00992] |

**Table 5.2:** Paired  $\Delta$ Rank IC of the RD13 trajectory (common sample; 21-day moving-block bootstrap)

the development period; on the model containing RD13 the pattern is reversed, with a significant increment in the development period. Because D2 has only one tree, this difference may partly reflect model size; when both are fixed at 50 rounds, the development-period increment remains positive but becomes clearly smaller (+0.0047, with the lower bound of the interval just above zero). Although the Rank IC of D3 is higher than that of D2, its development-period portfolio return is lower; the reason is discussed in Section 6.1.

Fused-RD13 is formed by standardising PV-RD13 and FUND separately and fusing them 0.7/0.3, the weight being chosen from four candidates on the basis of validation-period portfolio performance. The paired tests show that the Rank IC of Fused-RD13 is significantly higher than that of PV-RD13 in both the validation and the development period (+0.0058 and +0.0033), and not significantly different from that of pure FUND, suggesting that the advantage of Fused-RD13 is driven largely by its fundamental component. Changing the illiquidity factor in RD13 to the dollar-volume definition leaves the ranking and Rank IC of Fused-RD13 essentially unchanged (daily rank correlation about 0.96; Appendix F.10).

### 5.3 The RD8 Trajectory: Fixed-Round Fundamentals

In the RD8 trajectory, PV-RD8 and FUND-R50 are both trained for a fixed 50 rounds and evaluated on their common stock–day sample (Table 5.3; because the sample differs, the values for PV-RD8 differ slightly from those in Chapter 4). The relative strength of the two reverses clearly between the two periods: in the validation period the Rank IC of the fundamental model is 0.0123, nearly nine times that of the price–volume model (0.0014); in 2020–2023 the price–volume model overtakes it (0.0081 against 0.0054). The daily correlation between the two prediction streams is close to zero, so the ranking

information they contain is almost entirely different.

| Signal    | 2018–2019 IC | 2018–2019 Rank IC | 2020–2023 IC | 2020–2023 Rank IC |
|-----------|--------------|-------------------|--------------|-------------------|
| PV-RD8    | 0.00028      | 0.00138           | 0.00721      | 0.00807           |
| FUND-R50  | 0.01135      | 0.01230           | 0.00513      | 0.00539           |
| Fused-RD8 | 0.01132      | 0.01247           | 0.00585      | 0.00637           |

**Table 5.3:** Price–volume, fundamental and fused signals under the fixed-50-round setting

The fusion weight was chosen by average validation-period Rank IC, which is highest at a price–volume weight of 0.1, denoted Fused-RD8; in bootstrap resampling, combinations with a large fundamental weight win most often, so the more stable finding is that the fundamental weight is high, rather than any exact ratio (Appendix Figure F.1). The paired tests (Table 5.4) show that the increment of Fused-RD8 over PV-RD8 in the validation period is +0.0111, but the interval contains zero; in 2020–2023 it turns to  $-0.0017$ . At the same time, it has a small but significant increment over pure FUND-R50 in 2020–2023 (+0.0010), indicating that the relative value of the two classes of information varies by period.

| Period    | Candidate – baseline | $\Delta$ Rank IC | 95% CI                  | NW $p$ |
|-----------|----------------------|------------------|-------------------------|--------|
| 2018–2019 | Fused-RD8 – PV-RD8   | +0.01109         | [ $-0.00132, 0.01999$ ] | 0.0505 |
| 2018–2019 | FUND-R50 – PV-RD8    | +0.01092         | [ $-0.00191, 0.02055$ ] | —      |
| 2018–2019 | Fused-RD8 – FUND-R50 | +0.00017         | [ $-0.00066, 0.00105$ ] | 0.6792 |
| 2020–2023 | Fused-RD8 – PV-RD8   | $-0.00170$       | [ $-0.00847, 0.00504$ ] | 0.5971 |
| 2020–2023 | FUND-R50 – PV-RD8    | $-0.00269$       | [ $-0.00940, 0.00387$ ] | —      |
| 2020–2023 | Fused-RD8 – FUND-R50 | +0.00099         | [ $0.00035, 0.00166$ ]  | 0.0027 |

**Table 5.4:** Paired  $\Delta$ Rank IC of the RD8 trajectory

## 5.4 Synthesis of the Two Trajectories

The conclusions of the two trajectories agree: fundamentals provide information different from price and volume, but their increment depends on the strength of the price–volume baseline and on the period. In the RD13 trajectory, adding fundamentals to the single-tree PV-RD13 raises development-period Rank IC by +0.0070; when both models are fixed at 50 rounds the increment shrinks to +0.0047, with the lower bound of the interval only just above zero (Table 5.2). In the RD8 trajectory, where fundamentals enter through score fusion rather than as model features, Fused-RD8 shows no development-period increment over PV-RD8 ( $-0.0017$ , interval containing zero; Table 5.4). This pattern is consistent with the increment narrowing as the price–volume baseline becomes stronger, although the RD8 comparison also differs in how fundamentals enter the model. The 70/30 weighting of Fused-RD13 and the 10/90 weighting of Fused-RD8 arise from different underly-

ing signals and selection criteria and should be understood as selection outcomes tied to specification and sample.

## 5.5 The Increment from News Sentiment

The predictive role of news depends more heavily on publication time and return window (see the design in Section 3.4). The two trajectories test different questions: the RD13 trajectory focuses on event stocks with pre-open news and the open-to-close return, supplemented by a delayed daily test; the RD8 trajectory targets the daily return over the full cross-section.

### 5.5.1 RD13 trajectory: event information

In the delayed daily setting, neither the Alpha Vantage sentiment nor the FinBERT scores produced repeatable evidence: the former has a Rank IC of 0.0033 in the validation period and 0.0012 in 2020–2023, and the latter turns from 0.0040 to  $-0.0005$ . A shorter event window gives a different picture: using only information available before the open, the baseline Rank IC without news is  $-0.0003$ , rising to 0.0105 once pre-open sentiment is added (0.0060 for FinBERT), indicating that pre-open news contains immediate cross-sectional information among event stocks with news. In the full model already containing Alpha20, RD13 and fundamentals, however, the main effect of sentiment does not improve prediction stably (Table 5.5); the version that keeps only the interaction terms between sentiment and other features has a significant increment in the validation period ( $+0.0231$ ), suggesting that the meaning of news is conditional.

| Spec. | Information added relative to I0                    | 2018–2019<br>Rank IC | 2020–2023<br>Rank IC | Intraday<br>net ARR |
|-------|-----------------------------------------------------|----------------------|----------------------|---------------------|
| I0    | No raw Alpha Vantage sentiment                      | $-0.00897$           | 0.00722              | $-0.27\%$           |
| I1    | + 6 raw Alpha Vantage features                      | 0.01049              | 0.00283              | $-0.68\%$           |
| I2    | + Alpha Vantage + 9 interaction terms               | 0.00098              | 0.00632              | $-3.16\%$           |
| I3    | Interaction terms only, no raw Alpha Vantage levels | 0.01417              | 0.00833              | $-1.68\%$           |

**Table 5.5:** Full-control comparison of the I series in the RD13 trajectory

### 5.5.2 RD8 trajectory: daily sentiment fusion

In the RD8 trajectory, sentiment is added as a third predictive score to the fixed Fused-RD8, with candidate weights from 0 to 0.3 (Table 5.6). Under both time alignments the validation-period point estimates are positive, but both intervals straddle zero, the pre-set inclusion gate is not passed, and the final sentiment weight is 0; 2020–2023 likewise provides no

interval support. The rise in news coverage from about 11% in the validation period to about 25% in 2020–2023 also limits cross-period comparison.

| Timing    | $s^*$ (2018–2019) | 2018–2019<br>$\Delta\text{Rank IC}$ | 2018–2019 95% CI                 | 2020–2023<br>$\Delta\text{Rank IC}$ | 2020–2023 95% CI                |
|-----------|-------------------|-------------------------------------|----------------------------------|-------------------------------------|---------------------------------|
| Delayed   | 0.1               | $+0.32 \times 10^{-4}$              | $[-2.22, +3.00] \times 10^{-4}$  | $+1.02 \times 10^{-4}$              | $[-1.60, +3.62] \times 10^{-4}$ |
| Immediate | 0.3               | $+3.33 \times 10^{-4}$              | $[-4.38, +10.61] \times 10^{-4}$ | $+1.44 \times 10^{-4}$              | $[-6.41, +9.70] \times 10^{-4}$ |

**Table 5.6:** Incremental Rank IC of news sentiment in the RD8 trajectory ( $s^*$  is the validation-period candidate weight; because the intervals straddle zero, the final weight is 0)

## 5.6 Chapter Summary

Fundamentals provide information different from price–volume signals, but their relative value varies with period and model setting, and the validation-period tilt towards fundamentals did not carry through stably to 2020–2023. The value of news is concentrated close to publication: pre-open event news improves the immediate ranking of event stocks, but under daily fusion and full controls no stable increment appears. Later-period economic performance of the fused portfolios is reported in Chapter 7; the next chapter turns to portfolio construction.

## Chapter 6

# Portfolio Construction and the Final Strategy

### 6.1 Research Question

This chapter answers the third research question: how do portfolio construction and market conditions affect the conversion of predictive signals into economic value? It also explains how the final deployment strategy RD13-Top5 was formed. Chapters 4 and 5 have already shown that a signal with a higher predictive metric does not necessarily bring a higher portfolio return: under the unified setting RD13 has a higher Rank IC than RD8 but a lower portfolio return (Table 4.4); D3 has a higher Rank IC than D2, and again a lower return. The reason is that Rank IC measures the ranking of the whole cross-section of about 500 stocks, whereas a long-only portfolio buys only the 5 to 20 highest-ranked stocks, and its return also depends on how long they are held, on transaction costs and on market exposure.

This chapter fixes the signal, first examines replacement speed and holding breadth on the fused signals of the two trajectories, and then tests market-state adjustments. The ranking of Fused-RD13 changes a great deal from day to day, while that of Fused-RD8 is very stable, so the two form an informative contrast. The results in this chapter are confined to the validation and development periods; performance on later data is reported in Chapter 7.

### 6.2 A Signal with Rapidly Changing Rankings: Fused-RD13

The research baseline for Fused-RD13 is RD13-Top20, which uses Top20/drop2, replacing at most one tenth of the holdings each day; this proportion was carried over from RD-

Agent's default setting (Top50/drop5) and was not selected on returns. The development-period net excess return of RD13-Top20 is 9.1%, with an interval containing zero; its market beta is 1.27, meaning that for every 1% move in SPY the portfolio moves about 1.27% on average, and the annualised alpha after controlling for this exposure is 6.7% ( $t = 0.86$ ). Because this alpha is not significant, the development-period excess return cannot be distinguished from the return to the higher market exposure.

Merely raising the maximum number of stocks replaced each day from 2 to 4 increases the net excess return by 3.6 percentage points (Table 6.1): the share of holdings that have already fallen out of the top 20 declines, and a stock is usually sold within one trading day of its rank slipping. Turnover doubles as a result, however, and the advantage disappears at a one-way cost of about 8.7 bp; in the sharp fall and rebound of the first quarter of 2020, the slower replacement was actually ahead by about 35 percentage points.

| Metric                                       | At most 2 replaced per day | At most 4 replaced per day |
|----------------------------------------------|----------------------------|----------------------------|
| Net excess ARR (one-way cost 1 bp)           | 9.06%                      | 12.68%                     |
| Net excess ARR (one-way cost 5 bp)           | 7.13%                      | 8.88%                      |
| Net excess ARR (one-way cost 10 bp)          | 4.72%                      | 4.13%                      |
| Mean daily turnover                          | 0.202                      | 0.399                      |
| Share of holdings already outside the top 20 | 76.7%                      | 72.2%                      |
| Exit delay after rank slips (median)         | 2 trading days             | 1 trading day              |

**Table 6.1:** Replacement-speed contrast for Fused-RD13 under Top20 (2020–2023)

Under all three breadths, Top20, Top10 and Top5, replacing 20% beats replacing about 10%, and the more concentrated the holdings, the larger the gap (Table 6.2); Top5 with faster replacement (RD13-Top5) has the highest development-period net excess return, 18.8%. The high return of a concentrated portfolio, however, depends on picking the right few stocks: removing the five largest contributors reduces the return of RD13-Top5 from about 17.8% to 6.9% (on a daily-contribution basis), and merely changing the illiquidity factor to the dollar-volume definition reduces its development-period net excess ARR from 18.8% to 13.7%, whereas the change for Top20 is under 2 percentage points. The market beta of RD13-Top5 is about 1.49, and its alpha after controlling for market exposure is also not significant ( $t = 1.23$ ).

| Holding breadth | Replace about 10%                  | Replace 20%        |
|-----------------|------------------------------------|--------------------|
| Top20           | 9.06% (RD13-Top20)                 | 12.68%             |
| Top10           | 12.53%                             | 13.26%             |
| Top5            | 7.96% (1 replaced every other day) | 18.78% (RD13-Top5) |

**Table 6.2:** Development-period net excess ARR of Fused-RD13 under different holding breadths and replacement proportions (2020–2023)

### 6.3 A Signal with Stable Rankings: Fused-RD8

For the stable-ranking signal Fused-RD8, 16 static portfolios across five holding breadths and several replacement proportions were compared, together with rules that switch by market state. It took place after the 2024–2025 results of the RD13 trajectory had been observed and after RD13-Top5 had been formed, but before Fused-RD8 itself had been evaluated on 2024–2025. Two rules were carried forward in parallel, with neither designated as primary: the inherited Top20/drop2 (RD8-Top20) and Top5/drop1 (RD8-Top5), the best validation performer. RD8-Top5 was ahead by 11.3 percentage points in the validation period, but a test against Top20/drop2 with correction for multiple comparisons gives  $p = 0.39$ , so this advantage is not statistically established, mainly because a two-year validation period has low power. Both rules were fixed before their 2024–2025 results were run. The validation-period advantage of RD8-Top5 almost disappears in the development period, and its risk-adjusted performance is worse (Table 6.3).

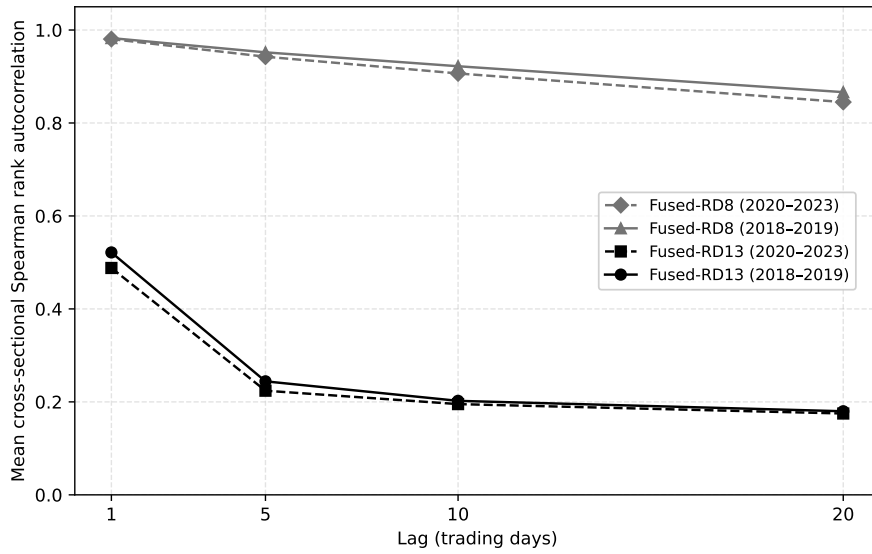
| Portfolio   | Period    | Net excess ARR | Information ratio | Max drawdown | Mean daily turnover |
|-------------|-----------|----------------|-------------------|--------------|---------------------|
| Top20/drop2 | 2018–2019 | 7.16%          | 0.97              | −21.2%       | 0.169               |
| Top5/drop1  | 2018–2019 | 18.45%         | 1.37              | −17.3%       | 0.245               |
| Top20/drop2 | 2020–2023 | 10.07%         | 0.62              | −52.5%       | 0.174               |
| Top5/drop1  | 2020–2023 | 11.09%         | 0.43              | −67.9%       | 0.232               |

**Table 6.3:** Performance of diversified and concentrated portfolios on Fused-RD8 (maximum drawdown in the Qlib definition)

Further analysis (Appendix G) shows that with five holdings, replacing only one stock a day is best, and that performance is stable for breadths between 4 and 6. The difference in next-day return between ranks 1–5 and ranks 6–20 is very small and negligible relative to daily volatility; but when the ordering within the top 20 is shuffled at random, the true ordering ranks at the 98th percentile in the validation period, with lower turnover. The advantage is therefore more likely to come from the stability of the ranking, that is, from holding well-performing stocks for longer, than from picking more accurately; and this effect had already disappeared by the development period. The return of the concentrated portfolio also depends heavily on a few stocks (the top five contribute about two thirds) and on 2018, and its beta (1.22 to 1.35) is higher than that of the diversified portfolio.

### 6.4 Two Signals Compared: Replacement Speed and Ranking Stability

The two signals have opposite preferences over replacement speed, and ranking stability explains the difference (Figure 6.1): the rank correlation of Fused-RD13 is about 0.5 at a



**Figure 6.1:** Ranking stability of the two fused signals: mean daily cross-sectional Spearman correlation between rankings separated by the given lag, over common stocks

one-day lag and about 0.2 at five days, whereas that of Fused-RD8 is still about 0.85 after twenty days. Under the same Top5/drop1 rule, the mean daily turnover of Fused-RD13 is about 0.40 and that of Fused-RD8 only about 0.23.

Re-running both signals on the same 16 portfolio rules and raising the replacement proportion from 10% to 20% at each breadth: Fused-RD13 benefits under all four breadths in the validation period and under three breadths in the development period, by 2.0 to 3.8 percentage points; Fused-RD8 is slightly worse under most breadths (Table 6.4). Fused-RD13 always uses its full replacement capacity, whereas Fused-RD8 replaces only about 10% even when allowed 20%. The individual differences were not tested for significance one by one, but the direction is consistent across both periods and most breadths. The replacement rule sets only an upper bound; how much is actually replaced depends on how quickly the ranking changes. The portfolio rule therefore needs to match the update speed of the signal, which can be seen as an extension into the time dimension of the signal-transfer problem discussed by Clarke, de Silva and Thorley (2002).

| Holding breadth                        | Fused-RD13 |            | Fused-RD8       |                 |
|----------------------------------------|------------|------------|-----------------|-----------------|
|                                        | 2018–2019  | 2020–2023  | 2018–2019       | 2020–2023       |
| Top10                                  | +1.42 pp   | −0.19 pp   | −0.31 pp        | −0.56 pp        |
| Top20                                  | +0.39 pp   | +3.82 pp   | +0.23 pp        | −1.98 pp        |
| Top30                                  | +0.93 pp   | +2.01 pp   | −0.93 pp        | −0.73 pp        |
| Top50                                  | +2.32 pp   | +2.75 pp   | −0.33 pp        | +0.34 pp        |
| Actual replacement rate at 20% nominal | about 0.20 | about 0.20 | about 0.10–0.11 | about 0.10–0.11 |

**Table 6.4:** Change in net excess ARR when the replacement proportion is raised from 10% to 20% (same grid of 16 portfolio rules)

## 6.5 Market-State Adjustment

Market states are identified with the two-state HMM of Section 3.6, and three adjustments are tested (Table 6.5). Adjusting the signal composition by state (more weight on price–volume in the low-risk state and on fundamentals in the high-risk state) lags fixed weights in the validation period and is ahead by about 1.2 percentage points in the development period, but the interval is  $[-5.7, +7.8]$ ; reducing exposure by state lowers the maximum drawdown from  $-45\%$  to about  $-27\%$ , but the net excess return falls from  $9.1\%$  to about  $-2\%$ ; the RD8 trajectory (Fused-RD8) searched portfolio rules separately in each state, and both states selected Top20/drop2 again.

| Adjustment                                   | Validation net excess ARR | Development net excess ARR | Development max drawdown |
|----------------------------------------------|---------------------------|----------------------------|--------------------------|
| No adjustment (Fused-RD13, exposure 0.95)    | +1.16%                    | +9.06%                     | −45.1%                   |
| Signal composition adjusted by state         | −0.37%                    | +10.27%                    | −44.7%                   |
| Exposure adjusted by state                   | −5.92%                    | −2.17%                     | −26.8%                   |
| Composition and exposure both adjusted       | Not evaluated             | −1.58%                     | −21.9%                   |
| Portfolio rule adjusted by state (Fused-RD8) | Same as default rule      | Same as default rule       | —                        |

**Table 6.5:** Results of adjusting the portfolio using market states (maximum drawdown in the Qlib definition)

For both signals, the development-period excess return of the portfolios comes mainly from the 404 days the model classes as high-risk (about  $+26\%$  for the Top20 portfolio on Fused-RD13, against about  $-2\%$  over the 602 low-risk days; the corresponding figures for Fused-RD8 are about  $+20\%$  and  $+3\%$ ; Appendix Table G.20), and the exposure-reduction rule cuts exposure to about 0.31 on exactly those days. The lag in state identification makes the problem worse: the model did not confirm the switch from the high-risk to the low-risk state until December 2020, about half a year after the actual rebound. Portfolios on both signals show this pattern, indicating that it is more probably related to the common portfolio structure and the higher market exposure. The HMM is therefore used as a tool for describing the market environment.

## 6.6 The Final Strategy

The final strategy (RD13-Top5) is adopted as the deployment strategy. It was formed after RD13-Top20 had been evaluated on 2024–2025. RD13-Top20 replaces at most 10% of its holdings each day, a proportion inherited from RD-Agent’s default setting; because rankings of US stocks appeared to change faster, the replacement proportion was raised to 20% and the Top5/drop1 rule was re-evaluated over the full history. The mechanism ev-

idence in this chapter supports this direction: the ranking of Fused-RD13 updates quickly, and a higher replacement capacity reduces stale holdings (Sections 6.2 and 6.4); concentrated holdings earn more in the development period; and adjusting by market state brings no stable improvement, so the final strategy keeps a fixed exposure. Because RD13-Top5 was chosen after the 2024–2025 results had been observed (the rule pre-registered for that round favoured Top10, and the validation-period net excess return of RD13-Top5 is only 2.5%), its results up to 2025 carry a selection effect. RD13-Top20 is retained as the pre-frozen research baseline, and RD13-Top5 is tested on the 2026 holdout, which was kept closed until RD13-Top5 had been fixed (Section 7.3).

## **6.7 Chapter Summary**

The chapter has three conclusions. First, a signal with a higher predictive metric does not necessarily bring a higher portfolio return; the portfolio rule is an independent link between the two. Second, the effect of replacement speed matches the ranking stability of the signal: Fused-RD13 usually benefits from a higher replacement capacity, Fused-RD8 mostly does not; faster replacement is at the same time constrained by transaction costs and the market period. Third, concentrated holdings deliver high returns only when the right few stocks are chosen, so their results depend heavily on a few stocks, a few years and market exposure; adjusting the signal, exposure or rule by market state brings no stable improvement.

# Chapter 7

## Later-Period Evaluation and Paper Execution

### 7.1 Research Question and Evidence Status

This chapter answers the fourth research question: does performance persist on later data, and can the final strategy be executed as designed? The status of each specification in the later periods is given in Table 7.1. RD13-Top20 was fixed before 2024–2025 was opened and is treated as a pre-frozen research baseline with independent evaluation in that window. For the RD8 chain, RD8-Top20 and RD8-Top5 are reported as first evaluations on 2024–2025 (both were fixed before Fused-RD8 was run on that window, but after the 2024–2025 results of the RD13 trajectory had been observed). The 2024–2025 result of RD13-Top5 is post-hoc diagnosis only; its main test is the 2026 holdout and prospective paper trading. Returns are reported mainly as CAGR and its difference from SPY.

| Specification | 2024–2025                                             | 2026                                                                                 |
|---------------|-------------------------------------------------------|--------------------------------------------------------------------------------------|
| RD13-Top20    | Pre-frozen research baseline (independent evaluation) | Control (regenerated under the same rule)                                            |
| RD8-Top20     | First evaluation                                      | Not evaluated (data chain cannot be extended)                                        |
| RD8-Top5      | First evaluation                                      | Not evaluated (data chain cannot be extended)                                        |
| RD13-Top5     | Post-hoc diagnosis                                    | Main test (2026 holdout, including a forward segment, and prospective paper trading) |

**Table 7.1:** Evidence status of each specification in the later periods

## 7.2 2024–2025

### 7.2.1 Predictive power

The IC of both signals in 2024–2025 is indistinguishable from zero, positive in 2024 and negative in 2025 (Table 7.2); the higher development-period predictive power of Fused-RD13 did not persist, and the two signals cannot be distinguished in this period either.

| Signal                       | 2024   | 2025    | 2024–2025 (95% interval) |
|------------------------------|--------|---------|--------------------------|
| Fused-RD8 (RD8 trajectory)   | 0.0022 | −0.0013 | 0.0005 [−0.0054, 0.0057] |
| Fused-RD13 (RD13 trajectory) | 0.0049 | −0.0029 | 0.0010 [−0.0062, 0.0067] |

**Table 7.2:** IC of the two signals in 2024–2025 (common universe; intervals from a 21-day moving-block bootstrap)

### 7.2.2 Portfolio results under different evidence statuses

All three portfolios trailed SPY in 2024–2025 (Table 7.3; evidence status in Table 7.1): SPY annualised 19.9%, RD13-Top20 14.4%, and RD8-Top20 and RD8-Top5 12.2% and 8.5%. None of these shortfalls is statistically significant (the  $t$ -statistics of the daily excess returns are about  $-0.7$  to  $-0.9$ , and none of the intercepts is significant), and they are mainly associated with betas below 1 (0.6–0.8): over two years in which SPY rose about 20% a year, the shortfall is consistent with lower market exposure. All portfolios fell behind by a large margin in 2024; in 2025 RD13-Top20 and RD8-Top20 were slightly ahead of SPY.

| Portfolio  | CAGR   | vs SPY    | Max drawdown | Sharpe ratio | Market beta |
|------------|--------|-----------|--------------|--------------|-------------|
| RD13-Top20 | 14.39% | −5.47 pp  | −14.43%      | 0.92         | 0.77        |
| RD8-Top20  | 12.20% | −7.66 pp  | −16.31%      | 0.79         | 0.80        |
| RD8-Top5   | 8.50%  | −11.37 pp | −20.73%      | 0.51         | 0.62        |
| SPY        | 19.86% | —         | —            | —            | 1.00        |

**Table 7.3:** Portfolio performance in 2024–2025 under different evidence statuses (native universe; annualised over 252 trading days)

### 7.2.3 The two trajectories under the same rules

Placing the two signals under the same two rules on the common universe (Appendix Table H.3): under Top20/drop2 the two are almost identical, differing by only 0.3 percentage points; under Top5/drop1 Fused-RD13 is ahead by 19.3 percentage points, but the interval is  $[-34.9, +0.02]$ : the point estimate is large, yet the interval just contains zero, so this difference is still not significant at the 5% level.

## 7.2.4 RD13-Top5 and the market environment

The annualised return of RD13-Top5 in this period is 28.2%, about 8 percentage points above SPY, but the advantage comes almost entirely from 2025 (11.3% in 2024, again behind SPY), and merely changing the illiquidity factor to the dollar-volume definition turns its excess return from positive to negative (Appendix Table H.4).

These two years were dominated by a steady rise: 433 of the 502 trading days were in the low-risk state. All four portfolios earned positive excess returns on the few high-risk days and fell behind or roughly matched SPY on low-risk days (Appendix Table H.6); over the same period the daily equal-weighted portfolio of the universe also annualised 8.2 percentage points below SPY. This is consistent with a rise driven by a few very large-cap stocks: in such an environment, a portfolio that is not capitalisation-weighted and has limited holdings tends to fall behind. But the market environment explains only part of the result; the predictive power of both signals weakened as well.

## 7.3 2026 Holdout of the Final Strategy

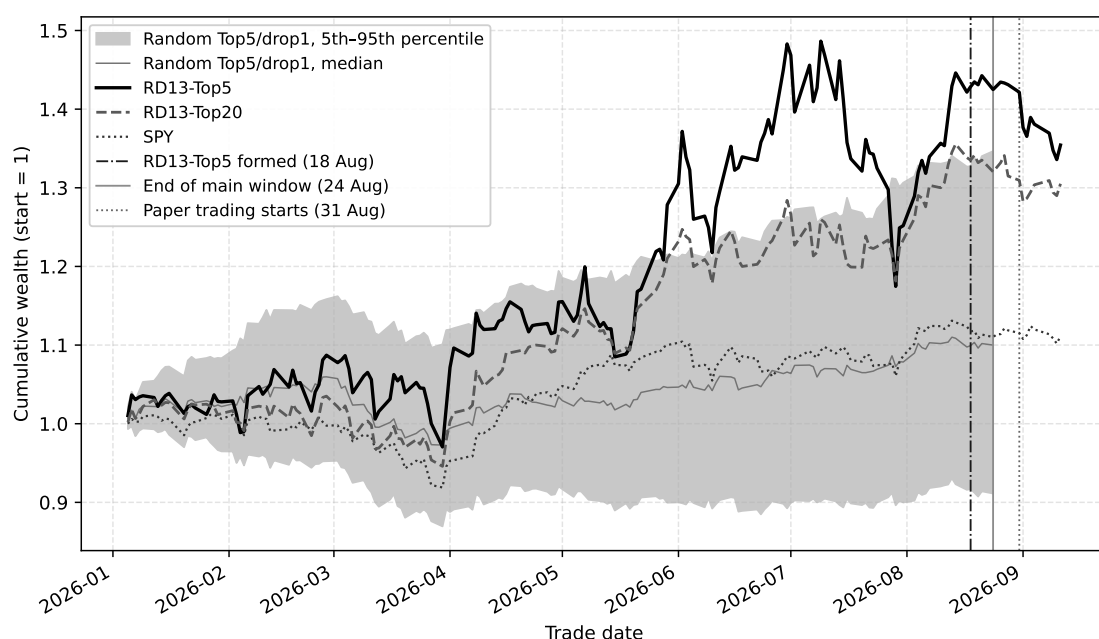
Because WRDS coverage ends in 2025, the 2026 holdout uses Massive data with approximate constituent and fundamental chains (Section 3.2). Only the RD13 trajectory (RD13-Top5 and RD13-Top20) is evaluated in 2026, because the factors and the fundamental model of the RD8 trajectory cannot be rebuilt on the 2026 data chain under their frozen definitions. In the main window from 5 January to 24 August 2026, the cumulative return of RD13-Top5 is 42.5%, against 11.8% for SPY over the same period (Table 7.4); RD13-Top20 returns 32.0%. Among 500 simulations that rank stocks at random each day and trade under the same rules, RD13-Top5 sits at the 97th percentile and RD13-Top20 at the 99th.

| Portfolio                         | Cumulative return | vs SPY (geometric) | Percentile in random-portfolio distribution |
|-----------------------------------|-------------------|--------------------|---------------------------------------------|
| RD13-Top5                         | 42.49%            | +27.5 pp           | 97th percentile                             |
| RD13-Top20                        | 32.04%            | +18.2 pp           | 99th percentile                             |
| PV-RD13 × Top5/drop1 (diagnostic) | 15.62%            | +3.5 pp            | —                                           |
| FUND × Top5/drop1 (diagnostic)    | 25.00%            | +11.9 pp           | 80th percentile                             |
| FUND × Top20/drop2 (diagnostic)   | 21.22%            | +8.5 pp            | 89th percentile                             |
| SPY                               | 11.75%            | —                  | —                                           |

**Table 7.4:** Performance in the 2026 main window (5 January to 24 August 2026)

Diagnostic rows apply the same rules to one component of Fused-RD13 and were not used in any selection.

Figure 7.1 shows the path: RD13-Top5 rose above the band of random portfolios in June and stayed near or above its upper edge until the end of the main window, then



**Figure 7.1:** Cumulative wealth in 2026 of RD13-Top5, RD13-Top20 and SPY against the distribution of 500 random portfolios under the Top5/drop1 rule; the vertical lines mark the formation of RD13-Top5, the end of the main window and the start of paper trading

fell back after its formation. Neither component alone matches the fused signal: under the same Top5/drop1 rule the price–volume score returns 15.6%, close to the random average, and the fundamental score 25.0% (80th percentile), against 42.5% for Fused-RD13 (Table 7.4). The 2026 result therefore depends on combining the two sources rather than on either of them alone.

The limits of this result are also clear. The Rank IC of Fused-RD13 is 0.0065 (Pearson IC 0.0096), with a Rank IC interval  $[-0.012, +0.023]$  containing zero, so it cannot be established that the signal ranked stocks correctly across the whole cross-section. Returns are highly concentrated: the top five stocks (CIEN, NTAP, ON, COHR and MU, all related to AI hardware and semiconductors) contributed 75%, and after removing them only 7.8% to 11.2% remains. The evidence therefore cannot distinguish stable stock-selection skill from a single successful thematic exposure. In the 12 trading days after the main window, which together with the paper-trading period are the only data that accrued after RD13-Top5 was formed, RD13-Top5 fell 5.0% and SPY fell 0.2%.

RD13-Top20 is the only specification whose whole later period, from January 2024 to the end of the 2026 main window, was unseen when it was fixed. Over that period it returned 72.6% against 60.3% for SPY, behind in 2024–2025 and ahead in 2026, a relative gain of 7.7% that is not statistically significant (21-day moving-block bootstrap 95% interval  $-23.6\%$  to  $+53.0\%$ ). The two segments use different price sources (CRSP and Massive) under the same cost rule.

## 7.4 Prospective IBKR Paper Trading

RD13-Top5 ran in an IBKR paper-trading account for eight trading days from 31 August to 10 September 2026. The account is denominated in sterling and all orders were market-on-close (MOC). Of 18 orders, 15 filled in full and 3 partially; on the evening of 7 September the contracts for two stocks could not be identified, the system placed no orders that evening under the fail-closed rule, and the rebalance was completed on the next trading day (Appendix J.6). Fill prices were on average about 4 bp below the close, and total commissions were about £39.

| Item                                                       | Result                                    |
|------------------------------------------------------------|-------------------------------------------|
| Number of orders (fully / partially filled)                | 18 (15 / 3)                               |
| Fail-closed triggers                                       | 1 (7 September, contract mapping failure) |
| Fill price relative to close (mean)                        | about -4 bp (buys -5.3 bp, sells -3.4 bp) |
| Total commissions                                          | about £39                                 |
| Overlap between target and actual holdings (daily mean)    | 0.96                                      |
| Closing cash as share of net asset value (design about 5%) | about 40%                                 |
| Cumulative account return / SPY (in sterling)              | -6.47% / -0.91%                           |
| Theoretical replay under frozen rules, same period         | -7.64%                                    |

**Table 7.5:** Execution summary of IBKR paper trading (31 August to 10 September 2026)

The largest deviation concerns exposure. Sells almost always filled, but buys were partially filled or placed in too small a quantity; for example, on 10 September the planned purchase of 1,307 shares of KLAC filled only 36 shares. The proceeds of sales were not fully redeployed into purchases, so cash rose from about 6% to about 40% of net asset value. In this falling window, the lower equity exposure meant the account lost about 1.2 percentage points less than the theoretical replay; the account fell 6.5% over eight trading days while SPY in sterling fell 0.9%, and the loss came mainly from the stocks held.

## 7.5 Chapter Summary

The 2026 holdout is the main test of the final strategy: RD13-Top5 outperformed SPY and ranked near the top of the random-portfolio distribution, and neither of its components alone matched the fused result (Section 7.3). The pre-frozen RD13-Top20 finished 2024 to August 2026 ahead of SPY, but not significantly. In 2024–2025 all three portfolios trailed SPY, largely consistent with betas below 1 in a narrow, capitalisation-led rally, although predictive power also weakened. Paper trading confirmed that the end-to-end pipeline and fail-closed control work, and that incomplete fills can leave cash undeployed.

# Chapter 8

## Discussion and Threats to Validity

### 8.1 Cross-Market Transfer: The Process Transfers, the Increment Does Not

The results for RQ1 show that RD-Agent’s research process transfers to a new market. On the S&P 500 it still generates interpretable, implementable price–volume factors, and two independent search trajectories repeatedly converged on similar economic structures. LLM-driven search therefore does not depend on particular stocks or formulas from the source market; it rediscovers more general short-horizon relationships among price, volume and risk.

The additional predictive power of these factors beyond Alpha20 is concentrated in the development window, mainly 2020, and cannot be distinguished from zero in the other periods. Plausible explanations include selection on the window that supplied the search feedback, an advantage specific to the 2020 crash and rebound, and a limited search budget of 20 loops, a single run and a small LLM backend; the design cannot separate them, and Section 9.6 proposes how to do so. The increment is also small because the starting point is low: Alpha20 has an IC of about 0.005 on the S&P 500 in 2020–2023 (Table 4.1).

This gap is consistent with the market differences discussed in Section 2.3. Short-horizon price–volume effects are stronger where retail investors dominate and trading is constrained by T+1 settlement and price limits, whereas US large caps are liquid, institutionally traded and closely followed, so information in past prices and volumes is likely to be priced quickly. A transferred process therefore starts from less exploitable structure. This interpretation fits the evidence but was not tested directly.

## 8.2 Adaptive Search and Evidence Status

Results in this report differ in evidence status. RD13 was formed by post-hoc selection on development-period IC; RD13-Top20 was fixed before 2024–2025 was opened and is evaluated independently there; the RD8-chain portfolios receive a first evaluation on that window; and RD13-Top5, formed after it had been observed, is tested on the 2026 holdout. Once a period has helped to form a specification, it can no longer evaluate that specification independently, and the evaluation window must move forward.

The same problem arises inside the search. RD-Factor feeds the metrics of its configured test window back to the Analysis Unit, so an automated agent can turn a nominal test set into an adaptive development set without the researcher choosing to do so, a risk discussed in the literature on adaptive data reuse (Dwork et al., 2015; Harvey, Liu and Zhu, 2016). More generally, automation does not remove researcher degrees of freedom but moves some of them into system configuration and the feedback mechanism: a baseline cache that is not keyed by market can silently change the acceptance reference (Appendix E.4). Reproducible automated research therefore has to record the search state, benchmark, acceptance rule, data version and the time each strategy was formed, and should keep a window that the agent never observes closed until the specification is fixed.

## 8.3 The Implementation Layer between Prediction and Return

Chapters 4 to 6 repeatedly show that better predictive metrics do not translate monotonically into higher portfolio returns. The link between the two, called here the implementation layer, has three dimensions. The first is the temporal structure of the signal: the fast-updating Fused-RD13 and the stable-ranking Fused-RD8 produce different turnover under the same rule, so the rule must match the signal. The second is concentration: Fused-RD13 and Fused-RD13a have almost identical overall rankings, yet their Top5 results differ markedly, because a concentrated portfolio depends on a few top stocks; return concentration should therefore be reported alongside returns. The third is transaction cost: faster replacement turns more predictive information into trades but also gives up more return to costs (Novy-Marx and Velikov, 2016).

## 8.4 Additional Information and Market State

Fundamental and price–volume predictions are almost uncorrelated, so fundamentals bring genuinely different information, although their marginal value depends on the price–

volume baseline and the period.

News carries immediate information for stocks with pre-open announcements, consistent with Tetlock, Saar-Tsechansky and Macskassy (2008), but it did not become a stable daily signal for the full cross-section, and the data explain much of this. News coverage is sparse in the early years and rises steeply over the sample, so the models learn mostly from stock-days without news and are then evaluated on periods in which news is far more common; a weight chosen in one period is applied to a signal whose composition has changed in the next. Because most stock-days have no news, sentiment also behaves more like an event indicator than a continuous score, and its information is concentrated around announcements, which a daily cross-sectional model with a next-day target largely misses. The negative result therefore concerns this data source and fusion method rather than the value of news itself. News offers many more dimensions than a vendor sentiment score, including the event type, the novelty of the information and the market's reaction around publication, but the sparse early coverage limited this study to testing whether a daily sentiment score adds to the existing signals.

Market states can be identified, but adjusting the signal, exposure or portfolio rule by state brought no stable improvement, in line with the finding of Cederburg et al. (2020) that the benefits of volatility management weaken out of sample.

## **8.5 Later-Period Evidence and Paper Execution**

The 2024–2025 results show how quickly a development-period advantage can fade once the evaluation window no longer feeds the search. Because the signals carried little ranking information in this period, portfolio returns were driven mainly by market exposure, and the shortfall against SPY reflects lower market beta in a strong market rather than evidence that the signals ranked stocks in the wrong direction. This is consistent with the literature on data reuse: results formed on a window that has been used repeatedly tend to overstate what can be expected on new data (Harvey, Liu and Zhu, 2016).

The 2026 results are favourable in direction and support the complementarity between fundamental and price–volume information. They cannot yet be read as evidence of stable stock-selection skill, because the gain is concentrated in one theme and the cross-sectional ranking is not significant; a concentrated, fast-updating rule may have captured a thematic exposure rather than a repeatable edge. The evidence status of the two RD13 specifications matters here: RD13-Top20, fixed before any later data were seen, provides the cleaner test, whereas RD13-Top5 was formed with knowledge of 2024–2025. Separating skill from exposure requires a longer prospective record, which is what the paper-trading framework is designed to accumulate.

Paper trading confirmed end-to-end broker execution and fail-closed control. It also

showed why execution must treat the broker’s actual holdings, not the theoretical target, as the reference state when fills are incomplete.

## 8.6 Threats to Validity

The design controls listed in Appendix D.5 address the main threats; those that remain are as follows.

**Internal validity.** Development data were used repeatedly for research decisions. The moving-block bootstrap and multiple-comparison corrections quantify uncertainty for a given candidate set but cannot remove the selection effect of adaptive search, which is why the later windows carry the main evidential weight.

**Measurement.** The original illiquidity factor is affected by later splits; its dollar-volume version is a sensitivity check added after the specifications were fixed. The 2026 holdout uses Massive data with approximate constituent and fundamental chains, and rising news coverage limits cross-period comparison of the sentiment results.

**Model and statistics.** Fixing LightGBM improves comparability but restricts the conclusions to this model, and the two-year validation period has low statistical power.

**LLM backend.** Both trajectories used o4-mini, a small reasoning model chosen within the project’s API budget. Some audit findings, such as implementation aliases, look-ahead fields, nominal-price dependence, coding failures and a drift towards momentum variants, may reflect model capability and might be reduced by a larger model at higher cost; this was not tested. Configuration issues, such as test-window reuse, the stale baseline cache and the placeholder benchmark, would not be. The results therefore describe RD-Agent with this backend, not its upper bound.

**Portfolio and benchmark.** The backtests assume a one-way cost of 1 bp, execution at the close and zero return on cash, and all portfolios are long-only, concentrated and not capitalisation-weighted, unlike SPY.

**External validity.** The conclusions apply mainly to US large caps. The 2026 holdout covers less than a year and, because of the project timeline, paper trading lasted eight trading days; capacity, market impact and live returns require a longer run with real capital.

## 8.7 Implications for Automated Quantitative Research

Taken together, RD-Agent is better understood as an automated research system rather than an automatic alpha machine. A reliable automated process needs five layers of evidence: an economically meaningful hypothesis; code and numerical series that implement it; a repeatable predictive increment; a portfolio rule that turns the increment into net value; and a specification that holds on new data and in execution. This project

checks each layer explicitly. Many candidates pass the first two, several add information within the development window, and the final strategy outperformed SPY on its holdout, although no candidate yet passes all five layers robustly.

Evaluating automated research platforms therefore requires more than a single IC or ARR: immutable data and experiment versions, recorded formation times, factor checks from name to numerical series, separate evaluation of prediction and portfolio, and the broker's actual state recorded as part of the research. Automation first raises research throughput; whether it also raises research credibility depends on whether validation and auditing are automated at the same pace.

# Chapter 9

## Conclusions and Future Work

This chapter answers the four research questions, evaluates the contributions of the project and sets out future work.

### 9.1 RQ1: Incremental Prediction across Markets

RD-Agent’s factor-discovery process transfers to the S&P 500: it generates interpretable, implementable price–volume factors, and two independent trajectories rediscovered similar structures. The additional predictive power of these factors beyond Alpha20 is concentrated in the development window, mainly 2020, and cannot be distinguished from zero in the other periods. The US price–volume baseline itself is weak, which leaves little room for an increment.

### 9.2 RQ2: Extra Information from Fundamentals and News

Fundamentals carry information almost uncorrelated with price–volume signals and complement them: in 2026 the fused signal outperformed either component used alone. The marginal value of fundamentals nevertheless depends on the baseline and the period. News contains immediate information for stocks with pre-open announcements but did not become a stable daily component.

### 9.3 RQ3: Portfolio Construction, Market Conditions and Economic Value

Between prediction and return lies a distinct implementation layer. Holding breadth, replacement speed, ranking stability and transaction costs jointly determine the economic outcome: a fast-updating signal benefits from faster replacement, a stable one does not,

and a higher Rank IC does not guarantee a higher return. Market-state adjustments brought no stable improvement.

## **9.4 RQ4: Later-Period Generalisation and Implementation**

The three portfolios evaluated on 2024–2025 trailed SPY, mainly through lower market exposure. On the 2026 holdout the final strategy outperformed SPY, the random portfolios and each of its components used alone, but its return was concentrated in a few AI-hardware stocks and the cross-sectional ranking was not significant. The pre-frozen RD13-Top20 finished 2024 to August 2026 ahead of SPY without a statistically significant edge. Paper trading confirmed that the pipeline executes at a broker and that cash can accumulate when fills are incomplete.

## **9.5 Main Contributions of the Project and Their Evaluation**

Of the six contributions set out in the introduction, the first, third and fourth are addressed by the answers to the research questions above. The other three go beyond them. The audit of the search and its implementation showed, in addition to implementation aliases and look-ahead fields, that RD-Factor’s feedback loop reuses its configured test window, so that an automated agent can turn a nominal test set into a development set; keeping 2024–2025 closed preserved an independent evaluation for the RD13 chain. The classification of evidence by formation time made the status of each later-period result explicit, although with a single run per trajectory it cannot measure how much of the development-period advantage came from selection. The extension to broker execution demonstrated that the pipeline operates end to end, but eight trading days test its mechanics rather than the strategy.

From a computer science perspective, the main technical output is an end-to-end pipeline covering point-in-time data construction, automated factor search, model training, portfolio backtesting and broker execution, in which implementation checks, evidence status and execution state are recorded as part of the pipeline itself; the code is available in the repository described in Appendix B.2.

## 9.6 Future Work

Five extensions have priority. First, rerun the search with several random seeds, more loops and different starting periods, with evaluation rules fixed in advance, to estimate the variance of automated search and test whether more robust factors emerge. Second, repeat it with larger reasoning models under the same data, budget and rules, to separate the contribution of the framework from that of the model. Third, study why the development-period increment is concentrated in 2020, and strengthen the data chain with a dollar-volume illiquidity measure and a maintained fundamental source for 2026 onwards. Fourth, with a news source of denser historical coverage, move from a vendor sentiment score to the text itself: classify events by type, measure how new the information is, let a language model read the articles directly, and test the signal in event windows rather than only in the daily cross-section. Fifth, run paper trading over a longer period, manage broker holdings, cash and partial fills as one state, and evaluate exposure errors as well as holdings overlap.

## 9.7 Final Conclusion

RD-Agent does not simply succeed or fail in the US market. It transfers as a research process, generates economically meaningful factors and, combined with fundamentals and a suitable portfolio rule, produced a strategy that outperformed SPY on a holdout it had not seen, although with concentrated returns. Its advantages depend on the search path, implementation details, market environment and portfolio rule, and a stable predictive increment across periods has not yet been established. A broader lesson concerns evaluation: automation raises research throughput, but it becomes credible research only when data lineage, formation times, implementation and execution are recorded and tested as carefully as the results themselves.

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# Appendix A

## Data Construction and Point-in-Time Rules

### A.1 Compustat Availability-Date Rule

Compustat quarterly observations are mapped to PERMNO through the CRSP–Compustat link. When `rdqe` is available, the availability date is  $\max(\text{rdqe}, \text{datadate})$ ; when `rdqe` is missing, it falls back to `datadate`. Constructions such as rolling sums, year-on-year changes and the conversion of year-to-date figures to quarterly values use only historical observations that were public at the time, and later revised values are not back-filled into the past.

### A.2 News Time Alignment

News is aligned to company–trading-day pairs by article publication timestamp, company attribution and ticker relevance. The delayed daily scheme uses the more conservative mapping to the next tradable day; the immediate event-time scheme uses only articles observable before the target execution window. The raw panel keeps no-news observations as missing; in the joint models they are filled with 0 according to a pre-specified rule and standardised cross-sectionally each day.

# Appendix B

## RD-Factor Experimental Configuration

### B.1 The Two Search Trajectories

| Item                                    | Trajectory A: RD8                                                             | Trajectory B: RD13                                                 |
|-----------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Search length                           | 20 loops                                                                      | First 20 loops of an earlier experiment                            |
| Universe                                | Historical S&P 500                                                            | Historical S&P 500                                                 |
| Training / validation / development     | 2008–2017 / 2018–2019 / 2020–2023                                             | 2008–2017 / 2018–2019 / 2020–2023                                  |
| Predictor                               | Fixed LightGBM                                                                | Fixed LightGBM                                                     |
| LLM backend                             | o4-mini (OpenAI API, routed via LiteLLM)                                      | o4-mini (OpenAI API, routed via LiteLLM)                           |
| Code base                               | RD-Agent, branch <code>us-market-experiment</code>                            | RD-Agent, branch <code>us-market-experiment</code>                 |
| Run period                              | 2026-09-16 – 2026-09-18                                                       | 2026-07-07 – 2026-07-08                                            |
| Loops without a valid result            | Loop 6 (framework duplicate check); Loops 16–17 (dependency/interface errors) | Loops 12–13 recorded but excluded (\$factor field, Appendix E.1.2) |
| Factor-set formation                    | Candidates accepted in online feedback accumulate loop by loop                | Selected after the run by successive new highs in 2020–2023 IC     |
| Downstream price–volume information set | RD8 → PV-RD8                                                                  | RD13 → PV-RD13                                                     |
| Portfolio benchmark in the report       | SPY                                                                           | SPY                                                                |

**Table B.1:** Experimental configuration of the two search trajectories

Trajectory A was not produced in a single uninterrupted run. In the original run, the Loop 2 factor (an ATR–volume interaction) used the absolute price range, so its value scaled with the nominal share price (Section 4.6), and that loop had been accepted. The trajectory was therefore regenerated: Loops 0–1 were retained from the matched prior run (forked checkpoint; not re-searched); the corrected Loop 2 factor, with the range divided by the previous close, was injected as the Loop 2 experiment and judged by the unchanged Analysis Unit, which rejected it (IC 0.002091, excess ARR +2.88%); Loops 3–19 were

then proposed, evaluated and accepted by the framework’s native mechanism without further intervention, in two consecutive sessions that continued the same search state (Loops 0–9, then Loops 10–19). Because the correction changed the state from which all later loops were proposed, the RD8 set is the terminal state of the regenerated trajectory, not of the original run. This is the only manual intervention in either trajectory.

The accepted loops, specific factors and search content of the two trajectories are reported in Chapter 4; portfolio comparisons in the report uniformly use SPY as the market benchmark.

## B.2 Code Repository and Reproduction Notes

The code, configuration files and result tables that support this report are archived at:

[https://github.com/Tiechuiifeifei/COMP0073\\_2026\\_Yufei\\_Hu](https://github.com/Tiechuiifeifei/COMP0073_2026_Yufei_Hu)

The repository allows the reported tables and figures to be checked against the shipped result files (`results/manifest.csv`). The underlying data are not redistributed. CRSP daily stock data, the CRSP–Compustat link table and Compustat quarterly fundamentals were obtained through an institutional WRDS subscription; company news and sentiment through the Alpha Vantage News & Sentiment API; and 2026 prices and financial statements through the Massive API, each of which requires its own account or key. The exact datasets, fields, sample periods and extraction steps are listed in `docs/DATA_SOURCES.md`. RD-Factor search is not bit-reproducible, because the language model generates different hypotheses on each run; `docs/REPRODUCIBILITY.md` describes what can be rerun and what each step requires. A brief reproduction path is as follows. (1) Obtain WRDS/CRSP and Compustat access and rebuild the historical S&P 500 Qlib universe as described in Chapter 3 and Appendix A. (2) Checkout the RD-Agent branch `us-market-experiment` and run the two RD-Factor trajectories under the fixed LightGBM predictor with the period split in Table B.1. (3) Rebuild downstream panels and portfolios from the frozen factor sets ( $\text{RD8} \rightarrow \text{PV-RD8}$ ;  $\text{RD13} \rightarrow \text{PV-RD13}$ ), then regenerate the Chapter 7 evaluation tables under the fixed specifications. Detailed manifests and path pointers are given in the chapter appendices and in the repository README.

# Appendix C

## Signal and Factor Definitions

### C.1 RD8 and PV-RD8

| RD8 factor                   | Main economic meaning                               |
|------------------------------|-----------------------------------------------------|
| reversal_1d                  | Very short-term price reversal                      |
| atr_10d                      | Recent price range and volatility state             |
| volume_ratio_10d             | Relative volume anomaly                             |
| vol_norm_price_momentum_10d  | Risk-normalised price trend                         |
| vol_norm_volume_momentum_20d | Risk-normalised volume trend                        |
| rv_corr_20d                  | Co-movement of returns and volume changes           |
| dynamic_price_momentum       | Price momentum conditional on the volatility state  |
| dynamic_volume_momentum      | Volume momentum conditional on the volatility state |

**Table C.1:** RD8 factors and their main economic meaning

### C.2 RD13 and PV-RD13

RD13 comprises the 13 RD-generated price–volume factors selected from the second trajectory by the rule of successive new highs in 2020–2023 development-period IC; the revised implementation is denoted RD13. PV-RD13 is the predictive score obtained by feeding Alpha20 and RD13 jointly into LightGBM, trained with validation-period early stopping.

| RD13 factor                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10-day momentum                                                                                                                                                                            |
| 20-day realised volatility                                                                                                                                                                 |
| 5-day volume deviation                                                                                                                                                                     |
| 10-day price-range ratio                                                                                                                                                                   |
| Volatility-adjusted 10-day momentum                                                                                                                                                        |
| 5-day short-term reversal                                                                                                                                                                  |
| Amihud-type illiquidity (actually implemented as $ \text{daily return} /\text{adjusted volume}$ ; for the alternative with dollar volume as denominator see Section 5.2 and Appendix F.10) |
| 20-day rolling beta                                                                                                                                                                        |
| 10-day idiosyncratic volatility                                                                                                                                                            |
| 5-day cross-sectional momentum rank                                                                                                                                                        |
| 5-day win rate                                                                                                                                                                             |
| 5-day up-day volume ratio                                                                                                                                                                  |
| 14-day ATR normalised by close                                                                                                                                                             |

**Table C.2:** RD13 factors

### C.3 FUND Fundamental Variables

| Variable            | Economic dimension  |
|---------------------|---------------------|
| ROE                 | Profitability       |
| ROA                 | Profitability       |
| Gross profitability | Profitability       |
| Sales growth YoY    | Growth              |
| Asset growth YoY    | Investment / Growth |
| Accruals            | Earnings quality    |
| Leverage            | Capital structure   |
| Current ratio       | Liquidity           |
| Book-to-market      | Valuation           |
| Earnings yield      | Valuation           |
| Sales-to-price      | Valuation           |

**Table C.3:** Fundamental variables and their economic dimensions

## C.4 Fusion of Predictive Scores and the Joint Sentiment Model

| Signal                | Components              | Final implementation note                                                                                                                           |
|-----------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Fused-RD13            | PV-RD13 + FUND          | PV-RD13 and FUND are winsorised at 1/99 and then standardised; direction set by the sign of the validation-period IC; price–volume weight $s = 0.7$ |
| Fused-RD8             | PV-RD8 + FUND-R50       | Each leg standardised cross-sectionally each day; price–volume weight $s = 0.1$                                                                     |
| Joint sentiment model | Fused-RD8 + sentiment_z | Sentiment missing $\rightarrow 0$ , then daily cross-sectional z-score; $s \in \{0, 0.05, 0.1, 0.2, 0.3\}$ , selected in the validation period only |

**Table C.4:** Implementation of the predictive-score fusion and the joint sentiment model

# Appendix D

## Portfolios, Metrics and Statistical Methods

### D.1 Portfolio Specifications

| Spec.            | Signal     | Portfolio rule | Use                                                                               |
|------------------|------------|----------------|-----------------------------------------------------------------------------------|
| RD13-Top20       | Fused-RD13 | Top20/drop2    | Pre-frozen research baseline of the RD13 trajectory                               |
| RD8-Top20        | Fused-RD8  | Top20/drop2    | Broad-portfolio control of the RD8 trajectory                                     |
| RD8-Top5         | Fused-RD8  | Top5/drop1     | Concentrated strategy of the RD8 trajectory, frozen by the validation-period rule |
| RD13-Top5        | Fused-RD13 | Top5/drop1     | Final deployment strategy of the RD13 trajectory                                  |
| RD13-Top20/drop4 | Fused-RD13 | Top20/drop4    | Replacement-speed control                                                         |

**Table D.1:** Portfolio specifications

### D.2 Market-State Specifications

| Spec.            | Input                                                  | Update method                                      |
|------------------|--------------------------------------------------------|----------------------------------------------------|
| 2-state HMM      | SPY daily return + 20-day realised volatility          | Initial fit on 2008–2017; causal forward filtering |
| 3-state HMM      | The above + 20-day/60-day volatility ratio             | Causal forward filtering with frozen parameters    |
| Dynamic weights  | State posterior $\rightarrow$ predictive-score weights | Evaluated separately from dynamic exposure         |
| Dynamic exposure | State posterior $\rightarrow$ target equity exposure   | Evaluated separately from signal weights           |

**Table D.2:** Market-state specifications

## D.3 Return and Risk Definitions

| Metric                | Definition / convention                                                                                    |
|-----------------------|------------------------------------------------------------------------------------------------------------|
| Qlib ARR              | $238 \times$ mean daily return; used to preserve the convention of the historical Qlib/RD-Agent logs       |
| CAGR                  | Geometric annualisation based on the cumulative wealth path; used for subsequent wealth performance        |
| Cumulative return     | $\prod(1 + r_t) - 1$                                                                                       |
| Daily excess return   | Portfolio return – SPY return – transaction cost                                                           |
| Annualised volatility | Standard deviation of daily returns $\times \sqrt{252}$                                                    |
| Sharpe ratio          | Mean daily return / standard deviation of daily returns $\times \sqrt{252}$ ; no risk-free rate subtracted |
| Maximum drawdown      | Largest peak-to-trough fall of the corresponding return or excess-return path                              |
| Turnover              | Two-sided traded value / account value (daily mean)                                                        |

**Table D.3:** Conventions for return and risk metrics

## D.4 Statistical Inference

| Experiment family                                     | Protocol                                                                           |
|-------------------------------------------------------|------------------------------------------------------------------------------------|
| Main paired $\Delta\text{RankIC}$ / $\Delta\text{IC}$ | Moving-block bootstrap; block 21 days; $B = 2,000$ ; 95% percentile CI             |
| Main portfolio differences                            | As above; common evaluation dates and consistent cost convention                   |
| 2026 short window                                     | Shorter block as the main setting, 21-day block as a sensitivity check             |
| Joint sentiment model                                 | Validation paired $\Delta\text{RankIC}$ ; $B = 2,000$ ; block 20 days; 95% CI gate |
| 2026 placebo                                          | Daily random ranking; Top5/drop1 and Top20/drop2 each repeated 500 times           |

**Table D.4:** Statistical inference protocol for each experiment family

## D.5 Threats and Controls

Table D.5–D.6 list the main threats to validity addressed by the design and the control used for each; the threats that remain are discussed in Section 8.6.

| Threat                               | Control                                                                                     | Where          |
|--------------------------------------|---------------------------------------------------------------------------------------------|----------------|
| <i>Data</i>                          |                                                                                             |                |
| Survivorship bias                    | Historical S&P 500 membership reconstructed from CRSP                                       | 3.2            |
| Identifier mismatch                  | Compustat mapped to PERMNO through the CRSP–Compustat link                                  | 3.2, A.1       |
| Fundamental look-ahead               | Availability date $\max(\text{rdqe}, \text{datadate})$ ; residual records disclosed         | A.1, F.1       |
| News look-ahead                      | Publication-time alignment; immediate scheme uses pre-open news only                        | A.2, F.5       |
| Label errors                         | LABEL0 cross-checked against CRSP returns and adjustment factors                            | G.8            |
| Change of data source in 2026        | Massive data checked against CRSP over 2023–2025                                            | 3.2            |
| Wrong benchmark                      | SPY (PERMNO 84398) replaces the placeholder benchmark                                       | 3.8, 4.6       |
| <i>Search and factors</i>            |                                                                                             |                |
| Test-window reuse in search feedback | 2020–2023 treated as development; later windows kept closed                                 | 3.7            |
| Stale baseline cache                 | RD8 trajectory rerun with an isolated cache                                                 | 4.6, E.4       |
| Search-path dependence               | Two independent trajectories; feedback-sensitivity replay (partial: one run per trajectory) | 4.2–4.4, E.4   |
| Redundant and aliased factors        | Correlation threshold; series-level check; RD4 control                                      | 4.3, 4.6, E.3  |
| Look-ahead inside factor code        | Code review; adjustment-field and volume checks                                             | 4.6, E.5, F.10 |
| Nominal-price dependence             | Scale-invariance test ( $\text{OHLC} \times 2, \times 4$ )                                  | E.6            |

**Table D.5:** Threats to validity and the controls used for each (data; search and factors)

| Threat                                | Control                                                        | Where          |
|---------------------------------------|----------------------------------------------------------------|----------------|
| <i>Models and statistics</i>          |                                                                |                |
| Training-length confound              | Unified fixed-50-round comparison                              | 4.5, E.2       |
| Model randomness                      | Three LightGBM seeds                                           | 4.4, Table 4.3 |
| Order of tied scores                  | Random tie-break test (100 runs)                               | 4.5, E.7       |
| Serial dependence                     | Moving-block bootstrap; Newey–West HAC statistics              | 3.8, D.4       |
| Multiple comparisons                  | Holm correction in the portfolio searches                      | 6.3, G.2       |
| Selection by luck                     | Random-portfolio placebos (500 in 2026; 300 within the top 20) | 7.3, G.4, H.2  |
| <i>Portfolio and later evaluation</i> |                                                                |                |
| Tuning on later data                  | Pre-commitment; evidence status set by formation time          | 3.7, Table 7.1 |
| Market exposure                       | Beta regressions; equal-weight diagnostic                      | 7.2, G.6       |
| Return concentration                  | Contribution analysis excluding top contributors               | 6.2, 7.3, G.5  |
| Cost sensitivity                      | One-way costs of 1, 5 and 10 bp                                | 6.2, Table 6.1 |
| Inconsistent metrics                  | Unified return and drawdown conventions                        | D.3, G.7       |
| <i>Execution</i>                      |                                                                |                |
| Execution mismatch                    | Broker reconciliation; fail-closed control                     | 3.9, 7.4, H.3  |

**Table D.6:** Threats to validity and the controls used for each (models; portfolio; execution)

# Appendix E

## Supplementary Results and Implementation Checks for Chapter 4

### E.1 Loop-by-Loop Records of the Two Search Trajectories

#### E.1.1 RD8 online accumulation trajectory

Table E.1 lists every loop of the RD8 trajectory.

#### E.1.2 RD13 candidate trajectory

Note to Table E.2: the `$factor` field used in loops 12–13 does not meet the strict point-in-time requirement; the corresponding IC, ARR and MDD are retained only for a complete record of the search trajectory, are not given an economic interpretation, and neither loop entered the final RD13 selection set.

### E.2 Sensitivity to Training Length

The unified comparison was repeated at 10, 20 and 50 boosting rounds. The  $\Delta\text{RankIC}$  of RD8 relative to Alpha20 is positive under all three settings but reaches 5% significance in none; RD13 is significant at 20 and 50 rounds and not at 10. This shows that the development-period ordering is not produced entirely by a single training length, but the advantage of RD13 still has to be interpreted together with its development-period selection process.

| Loop | Search theme                                                 | IC       | Excess ARR | Excess MDD | Outcome             |
|------|--------------------------------------------------------------|----------|------------|------------|---------------------|
| 0    | Simple momentum, volatility and volume                       | 0.005222 | −1.87%     | −37.53%    | Replace=no          |
| 1    | Reversal, range volatility and volume ratio                  | 0.000010 | +3.95%     | −33.38%    | Replace=yes         |
| 2    | Corrected ATR–volume interaction                             | 0.002091 | +2.88%     | −35.30%    | Replace=no          |
| 3    | Volatility-normalised momentum and return–volume correlation | 0.005944 | +16.40%    | −30.66%    | Replace=yes         |
| 4    | Downside semi-variance normalisation                         | 0.002605 | −1.23%     | −19.30%    | Replace=no          |
| 5    | EWMA downside risk and idiosyncratic momentum                | 0.000710 | +0.85%     | −24.93%    | Replace=no          |
| 6    | Asymmetric semi-variance (skipped as duplicate)              | —        | —          | —          | Skipped (duplicate) |
| 7    | Conventional technical indicators                            | 0.005620 | −4.09%     | −34.82%    | Replace=no          |
| 8    | Volume-weighted momentum, OBV and breakout                   | 0.005676 | +1.22%     | −21.86%    | Replace=no          |
| 9    | State-adaptive momentum and return–volume correlation        | 0.006046 | +16.61%    | −25.16%    | Replace=yes         |
| 10   | EWMA adaptive window and ensemble                            | 0.005508 | +7.30%     | −11.11%    | Replace=no          |
| 11   | Adaptive momentum ensemble                                   | 0.004492 | +3.87%     | −14.69%    | Replace=no          |
| 12   | Dynamic momentum and return skewness                         | 0.004515 | −3.50%     | −21.20%    | Replace=no          |
| 13   | ADX and volume z-score composite                             | 0.008445 | −8.10%     | −39.08%    | Replace=no          |
| 14   | Continuous ADX and volume-rank composite                     | 0.005465 | +2.50%     | −25.73%    | Replace=no          |
| 15   | LGBM, autoencoder and PCA hybrid                             | 0.005397 | −3.17%     | −42.78%    | Replace=no          |
| 16   | Autoencoder and LGBM hybrid                                  | —        | —          | —          | Invalid             |
| 17   | Log-return autoencoder and LGBM hybrid                       | —        | —          | —          | Invalid             |
| 18   | Random forest, k-means and AR(1)                             | 0.006038 | +0.60%     | −24.23%    | Replace=no          |
| 19   | Volatility-normalised dynamic momentum                       | 0.005482 | −0.18%     | −28.24%    | Replace=no          |

**Table E.1:** Loop-by-loop summary of the RD8 trajectory (2020–2023)

| Loop | Search theme                                 | IC       | Excess ARR (SPY) | Excess MDD (SPY) | Selection |
|------|----------------------------------------------|----------|------------------|------------------|-----------|
| 0    | Momentum, volatility and price–volume        | 0.005037 | −1.01%           | −26.94%          | Selected  |
| 1    | Volatility, momentum and reversal            | 0.007674 | 0.24%            | −27.16%          | Selected  |
| 2    | Overnight and intraday skewness              | 0.002531 | 4.51%            | −29.93%          | —         |
| 3    | Beta and idiosyncratic volatility            | 0.007972 | −0.72%           | −40.28%          | Selected  |
| 4    | Multi-horizon volatility and momentum        | 0.005219 | 0.61%            | −51.13%          | —         |
| 5    | EWMA volatility rank                         | 0.001916 | −0.22%           | −23.11%          | —         |
| 6    | Breakout, RSI and downside                   | 0.004968 | 5.61%            | −36.42%          | —         |
| 7    | Win rate, volume and ATR                     | 0.010275 | −4.41%           | −34.43%          | Selected  |
| 8    | Ten-day win rate and VWAP                    | 0.005790 | −0.87%           | −41.05%          | —         |
| 9    | Momentum and volume acceleration             | 0.003408 | 4.47%            | −24.55%          | —         |
| 10   | Dollar volume and liquidity                  | 0.001050 | 4.70%            | −16.16%          | —         |
| 11   | Return PCA exposures                         | 0.002428 | 4.36%            | −25.87%          | —         |
| 12   | Transformations of the raw \$factor field    | 0.003302 | 4.71%            | −20.13%          | —         |
| 13   | \$factor neutralisation                      | 0.003633 | 10.87%           | −27.07%          | —         |
| 14   | Stock-level PCA latent factors               | 0.003905 | 10.67%           | −30.79%          | —         |
| 15   | AE + LGBM hybrid                             | 0.004200 | 8.54%            | −24.56%          | —         |
| 16   | Three-dimensional AE + LGBM                  | 0.007254 | 4.70%            | −26.74%          | —         |
| 17   | LSTM + network centrality                    | 0.004827 | 9.06%            | −24.97%          | —         |
| 18   | Attention graph network                      | 0.001989 | 11.71%           | −29.30%          | —         |
| 19   | Hurst exponent, entropy and fractal measures | 0.005574 | −1.42%           | −29.31%          | —         |

**Table E.2:** Loop-by-loop summary of the RD13 candidate trajectory (2020–2023; selection criterion: successive new highs in IC)

| Boosting rounds | RD8 – Alpha20            | RD13 – Alpha20           | RD4 – Alpha20            |
|-----------------|--------------------------|--------------------------|--------------------------|
| 10              | +0.00242 ( $p = 0.381$ ) | +0.00644 ( $p = 0.136$ ) | +0.00535 ( $p = 0.062$ ) |
| 20              | +0.00438 ( $p = 0.080$ ) | +0.00929 ( $p = 0.009$ ) | +0.00694 ( $p = 0.003$ ) |
| 50              | +0.00328 ( $p = 0.098$ ) | +0.00895 ( $p = 0.007$ ) | +0.00587 ( $p = 0.009$ ) |

**Table E.3:** Paired  $\Delta$ RankIC under different training lengths, 2020–2023

### E.3 Factor Redundancy and Model Use

There are several highly correlated correspondences between RD8 and RD13, but none reaches the 0.99 redundancy threshold. The maximum absolute average daily cross-sectional correlation with Alpha20 is likewise below 0.99. The split gain of the joint models shows that RD factors are used frequently in the multivariate trees.

| Item                                                         | RD8                                           | RD13                                              |
|--------------------------------------------------------------|-----------------------------------------------|---------------------------------------------------|
| Representative highest correlation with the other trajectory | ATR 0.960; volatility-adjusted momentum 0.963 | ATR 0.960; volatility-adjusted momentum 0.963     |
| Highest absolute correlation with Alpha20                    | reversal_1d vs resi5: 0.764                   | 10-day price-range ratio vs klen: about 0.66–0.72 |
| Share of split gain from RD factors                          | 30.2%                                         | 40.0%                                             |
| RD feature with highest gain                                 | atr_10d: 6.43%                                | rolling_beta_20d: 5.51%                           |

**Table E.4:** Summary of redundancy and model use

### E.4 Baseline Cache and Feedback Sensitivity of the Replacement Decision

The experiment configuration of the RD13 trajectory was correct (S&P 500 universe, 2008–2017 training, 2020–2023 test window). RD-Agent, however, caches the baseline result under a key derived only from the sub-task text, which does not include the market or the evaluation window, so the run silently reused an earlier CSI 300 Alpha20 baseline ( $IC \approx 0.031$ , 2017–2020) instead of computing the US baseline ( $IC \approx 0.0046$ ; Table 4.1). No candidate was accepted against this reference. The RD8 trajectory was run with an isolated cache (Listing J.1).

A supplementary analysis fixes the historical prompt, IC and comparator of each loop, replaces only the current loop’s portfolio ARR/MDD, and repeats each of the 20 loops 10 times. Under the original feedback, all 200 decisions were “do not replace”; under portfolio feedback in the unified SPY convention, the replacement proportion rises in some loops. This result shows only that the single-loop replacement decision is sensitive to

the portfolio evaluation information; it does not allow the complete counterfactual search path under a different evaluation configuration to be inferred.

| Loop | Replacement proportion under unified SPY feedback |
|------|---------------------------------------------------|
| 6    | 30%                                               |
| 9    | 40%                                               |
| 10   | 50%                                               |
| 11   | 10%                                               |
| 12   | 60%                                               |
| 13   | 40%                                               |
| 14   | 20%                                               |
| 15   | 70%                                               |
| 16   | 70%                                               |
| 17   | 90%                                               |
| 18   | 60%                                               |

**Table E.5:** Loops in which replacement occurred in the feedback-sensitivity analysis (10 repetitions per loop)

## E.5 Point-in-Time Risk of the Cumulative Adjustment-Factor Field

The Qlib `$factor` field used in RD13 loops 12–13 comes from the CRSP `DlyCumFacPr` cumulative price-adjustment factor. In a full data dump, the level of `$factor` on a given day embeds the effect of future splits or reverse splits on the cumulative adjustment factor, so it is unsuitable as a strictly point-in-time raw predictive feature. Around the 4:1 split of AAPL on 31 August 2020, the field jumps from about 0.0359 to 0.1437. Because loops 12–13 did not enter the RD13 selection path, the problem does not affect the final RD13, but their results are not used for economic interpretation.

## E.6 Nominal Price-Scale Test of the Final Factor Sets

For the 8 factors of RD8 and the 13 factors of RD13, the OHLC of a single stock was multiplied proportionally by 2 and by 4 with volume held fixed. All 21 factors passed the scale-invariance test with `max_abs_diff = 0`, so no factor reconstruction or retraining of the unified R50 models was triggered. The test concerns nominal price scaling and is not equivalent to a full validation of volume units or point-in-time data semantics.

| Family | Factor                           | Formula / definition summary                     | OHLC $\times \{2, 4\}$ |
|--------|----------------------------------|--------------------------------------------------|------------------------|
| RD8    | reversal_1d                      | $-(C_t/C_{t-1} - 1)$                             | Pass                   |
| RD8    | atr_10d                          | $\text{mean}((H - L)/C, 10)$                     | Pass                   |
| RD8    | volume_ratio_10d                 | $V/\text{mean}(V_{t-10:t-1}) - 1$                | Pass                   |
| RD8    | vol_norm_volume_momentum_20d     | Volume momentum / return volatility              | Pass                   |
| RD8    | vol_norm_price_momentum_10d      | 10-day price momentum / return volatility        | Pass                   |
| RD8    | rv_corr_20d                      | $\text{corr}(\Delta C/C, \Delta V/V, 20)$        | Pass                   |
| RD8    | dynamic_price_momentum           | Adaptive-window price momentum                   | Pass                   |
| RD8    | dynamic_volume_momentum          | Adaptive-window volume momentum                  | Pass                   |
| RD13   | 10_day_momentum                  | $C/C.\text{shift}(10) - 1$                       | Pass                   |
| RD13   | 20_day_realized_volatility       | $\text{std}(\text{pct\_change}(C), 20)$          | Pass                   |
| RD13   | 5_day_volume_deviation           | $V/\text{mean}(V, 5) - 1$                        | Pass                   |
| RD13   | 10_day_price_range_ratio         | $(H - L)/\text{MA}_{10}(H - L)$                  | Pass                   |
| RD13   | volatility_adjusted_10d_momentum | $\text{mom}_{10} / \text{std}(\log\_ret, 20)$    | Pass                   |
| RD13   | 5d_short_term_reversal           | $-(C/C.\text{shift}(5) - 1)$                     | Pass                   |
| RD13   | daily_amihud_illiquidity         | $ r /V$ (non-classical $P \times V$ denominator) | Pass                   |
| RD13   | rolling_beta_20d                 | 20-day rolling beta                              | Pass                   |
| RD13   | idiosyncratic_volatility_10d     | 10-day idiosyncratic volatility                  | Pass                   |
| RD13   | cs_momentum_rank_5d              | Cross-sectional rank of 5-day momentum           | Pass                   |
| RD13   | win_rate_5d                      | 5-day proportion of up days                      | Pass                   |
| RD13   | upday_volume_ratio_5d            | 5-day share of volume on up days                 | Pass                   |
| RD13   | atr_norm_close_14d               | $\text{ATR}_{14} / \text{MA}_{14}(C)$            | Pass                   |

**Table E.6:** Scale-invariance test of the final 21 RD factors

## E.7 Tie Diagnostics for the Original PV-RD13

The original PV-RD13 has only one tree per seed. Averaged over the three seeds, the mean number of distinct scores per day is 264.6 (median 269.5, range 58–345). Ties still occur frequently at the Top5 and Top20 ranking boundaries; Qlib’s TopK-Dropout has no explicit secondary tie-break key, so the order of tied stocks depends on the input ordering.

| Metric                                         | Top5 | Top20 |
|------------------------------------------------|------|-------|
| Stocks at the boundary score (daily mean)      | 5.22 | 5.04  |
| Stocks with score $\geq$ boundary (daily mean) | 8.06 | 21.83 |
| Share of trading days exceeding $K$            | 52%  | 49%   |

**Table E.7:** TopK boundary ties for PV-RD13 (2020–2023)

| Pure PV-RD13 $\times$ Top5/drop1            | Net excess ARR        |
|---------------------------------------------|-----------------------|
| Original ordering                           | 20.61%                |
| Random tie-break, 100 runs: mean $\pm$ s.d. | 22.66% $\pm$ 5.07 pp  |
| p5 / p50 / p95                              | 15.2% / 22.2% / 30.9% |
| Minimum / maximum                           | 10.4% / 35.7%         |

**Table E.8:** Portfolio sensitivity to random tie-breaking

This sensitivity analysis uses pure PV-RD13 predictions and is not equivalent to the final Fused-RD13 or RD13-Top5; its purpose is to identify the implementation uncertainty that arises when the shallow PV-RD13 is converted from predictive scores into a concentrated portfolio.

## E.8 Single-Factor Results and Summary of Feature Importance

| Factor                                 | 2020–2023 Rank IC |
|----------------------------------------|-------------------|
| RD13: 5d_short_term_reversal           | 0.0091            |
| RD8: reversal_1d                       | 0.0053            |
| RD13: daily_amihud_illiquidity         | 0.0052            |
| RD13: volatility_adjusted_10d_momentum | 0.0009            |
| RD8: vol_norm_price_momentum_10d       | 0.0008            |

**Table E.9:** RD factors with higher single-factor Rank IC in 2020–2023

| Rank | Alpha20 + RD8 | Alpha20 + RD13           |
|------|---------------|--------------------------|
| 1    | atr_10d 6.43% | rolling_beta_20d 5.51%   |
| 2    | rsqr5 4.84%   | atr_norm_close_14d 4.46% |
| 3    | rsqr60 4.38%  | resi5 4.31%              |
| 4    | rsqr10 4.21%  | rsqr5 4.18%              |
| 5    | rsqr20 4.16%  | rsqr20 3.83%             |

**Table E.10:** Top five features by split gain in the R50 models (mean over 3 seeds). Alpha20 feature names follow the Qlib Alpha158 convention: RSQR is the  $R^2$  of a linear regression of price on time, RESI is the residual of that regression, KLEN and KLOW are the candlestick length and lower shadow, WVMA is the volume-weighted return volatility, CORR and CORD are price–volume correlations, VSTD is volume volatility and ROC is the rate of price change; the number is the look-back window in days.

## E.9 Evidence Index for RQ1

| Question                                                         | Evidence in this chapter                                                                                                                                                                         | Boundary of the conclusion                                                             |
|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Can economically meaningful price–volume factors be generated?   | RD8 and RD13 cover reversal, volatility, volume and risk-adjusted momentum; some cross-trajectory correlations about 0.96                                                                        | Supports interpretability and rediscovery; not equivalent to independent alpha         |
| Is there an increment over Alpha20?                              | R50: RD8 $\Delta\text{RankIC} +0.00328$ ; RD13 $+0.00895$                                                                                                                                        | Present in the development period; RD13 carries a selection effect                     |
| Is the increment stable?                                         | Paired differences in 2018–2019 and 2024–2025 both non-significant; increment most concentrated in 2020                                                                                          | No stable cross-period advantage established                                           |
| Does improved prediction convert directly into portfolio return? | No stable loop-level IC–ARR correspondence; RD13 not better than RD8 in the $2 \times 2$ portfolios                                                                                              | Portfolio construction needs separate evaluation                                       |
| Do implementation details matter?                                | Semantic scale, implementation aliases, look-ahead in the cumulative adjustment factor, volume adjustment in the illiquidity factor, PV-RD13 ties and run failures all change the interpretation | Hypothesis, code, actual series and portfolio implementation must be validated jointly |

**Table E.11:** Evidence index of Chapter 4 for RQ1

**Independence of the 2024–2025 window for the RD13 chain.** RD13 was rebuilt before the 2024–2025 window was first accessed under the formal holdout precommit; the audit flag `holdout_2024_accessed=0` holds for all search and selection steps of the RD13 path. Fused-RD13 was fixed by validation-period weight selection (scheme recorded in `selected_factor_weights.json`); RD13-Top20 was frozen in the same holdout precommit that opened the formal evaluation of that window. RD13-Top5 was defined after the window had been opened, and is therefore classified as post hoc for 2024–2025. Evidence: `RD13_v2_manifest.json`; `final_holdout_precommit.json`; `selected_factor_weights.json`; `topk_breadth/precommit.json`; and `reports/dissertation_ch3_method_audit.md`.

| RD8 factor                  | Closest RD13 factor              | Mean daily cross-sectional correlation           |
|-----------------------------|----------------------------------|--------------------------------------------------|
| atr_10d                     | atr_norm_close_14d               | 0.960                                            |
| vol_norm_price_momentum_10d | volatility_adjusted_10d_momentum | 0.963                                            |
| volume_ratio_10d            | 5_day_volume_deviation           | 0.866                                            |
| dynamic_price_momentum      | 10_day_momentum                  | 0.889                                            |
| rv_corr_20d                 | 10_day_momentum                  | 0.327                                            |
| —                           | daily_amihud_illiquidity         | Maximum absolute correlation with RD8 only 0.033 |

**Table E.12:** Correspondence between representative factors of the two trajectories

| Period               | RD8 – Alpha20           | RD13 – Alpha20          |
|----------------------|-------------------------|-------------------------|
| 2018–2019 validation | −0.0005 ( $p = 1.000$ ) | −0.0009 ( $p = 0.979$ ) |
| 2020                 | +0.0114*                | +0.0235*                |
| 2021                 | +0.0021                 | +0.0068                 |
| 2022                 | −0.0031                 | −0.0038                 |
| 2023                 | +0.0027                 | +0.0091                 |
| 2024                 | +0.0044                 | +0.0059                 |
| 2025                 | +0.0012                 | +0.0030                 |

**Table E.13:** Annual  $\Delta\text{RankIC}$  under the unified setting (\* indicates two-sided  $p < 0.05$  from a 21-day moving-block bootstrap)

| Version                                     | Training                                  | 2018–2019 Rank IC | 2020–2023 Rank IC | 2024–2025 Rank IC |
|---------------------------------------------|-------------------------------------------|-------------------|-------------------|-------------------|
| Original PV-RD13 (actually used downstream) | Validation early stopping; in fact 1 tree | 0.00297           | 0.00729           | 0.00290           |
| Unified comparison version                  | Fixed 50 rounds; no early stopping        | 0.00309           | 0.01512           | 0.00240           |

**Table E.14:** Original training of PV-RD13 and the unified setting

# Appendix F

## Supplementary Results and Comparability Notes for Chapter 5

This appendix retains the model capacity, PIT data, full weight grids, news time definitions and standardisation sensitivities that lie outside the main text. Core results already reported in the main text are not repeated here.

### F.1 FUND Fundamental Variables and PIT Treatment

The 11 fundamental variables and their economic dimensions are given in Appendix C.3; this section adds only the details of the point-in-time treatment.

Fundamental variables are winsorised at 1%/99% and z-standardised cross-sectionally on each trading day; invalid denominators are kept as missing. The initial candidate set had 12 variables; operating profitability was removed in a pre-specified collinearity ablation, and the final FUND was determined among F1A–F1D by the 2018–2019 validation-period Rank IC rule.

The PIT audit found 24 quarterly records with missing `rdqe`; the availability rule in the implementation is `max(rdqe, datadate)`, so these records fall back to `datadate`. The affected records that enter the main daily fundamental sample number 237 rows (about 0.011%): 14 rows in 2020–2023 and 0 rows in 2024–2025. The available material cannot confirm the true announcement dates for these companies, so this is disclosed as a residual measurement risk rather than described as confirmed pre-announcement look-ahead leakage.

## F.2 Model Capacity and Implementation Boundaries of the RD13 Trajectory

| Model           | Training cap / early stopping | Actual trees (3 seeds) | Use in main text                       |
|-----------------|-------------------------------|------------------------|----------------------------------------|
| D2              | 1000 / patience 50            | 1 / 1 / 1              | RD13 price–volume/RD prediction stream |
| Historical FUND | 1000 / patience 50            | 14 / 4 / 11            | Fundamental leg of Fused-RD13          |
| D1              | 1000 / patience 50            | 2 / 8 / 5              | Alpha20 + fundamentals                 |
| D3              | 1000 / patience 50            | 3 / 6 / 2              | Alpha20 + RD13 + fundamentals          |

**Table F.1:** Verified capacity of the early-stopped models

The 1/1/1 of D2 and the 3/6/2 of D3 are capacity boundaries that must be kept in mind when interpreting D3–D2, which is why the main text also reports D3\_R50–D2\_R50 with both fixed at 50 rounds. The Rank IC of D0 is inferred from the paired difference D1–D0 on the same common sample.

The illiquidity factor of RD13 is implemented as  $|\text{return}|/\text{volume}$ , where volume is adjusted with the end of the sample as the reference and therefore does not meet the strict point-in-time requirement. The alternative definition with dollar volume as the denominator and its results are given in Section 5.2 and Appendix F.10.

## F.3 RD8 Fixed-50-Round Path: Full Weight Results

| $w$ (price–volume) | 2018–2019 Rank IC | 2018–2019 95% CI    | 2020–2023 Rank IC | 2020–2023 95% CI   |
|--------------------|-------------------|---------------------|-------------------|--------------------|
| 0.0                | 0.012300          | [0.00113, 0.01890]  | 0.005386          | [0.00010, 0.01015] |
| 0.1                | 0.012472          | [0.00143, 0.01909]  | 0.006372          | [0.00097, 0.01121] |
| 0.2                | 0.012365          | [0.00149, 0.01884]  | 0.007415          | [0.00180, 0.01244] |
| 0.3                | 0.012079          | [0.00177, 0.01812]  | 0.008476          | [0.00271, 0.01380] |
| 0.4                | 0.011352          | [0.00195, 0.01734]  | 0.009412          | [0.00371, 0.01481] |
| 0.5                | 0.010219          | [0.00146, 0.01576]  | 0.010024          | [0.00431, 0.01576] |
| 0.6                | 0.008683          | [0.00033, 0.01408]  | 0.010177          | [0.00451, 0.01608] |
| 0.7                | 0.006811          | [−0.00107, 0.01197] | 0.010059          | [0.00447, 0.01601] |
| 0.8                | 0.004865          | [−0.00260, 0.01003] | 0.009533          | [0.00414, 0.01523] |
| 0.9                | 0.002967          | [−0.00426, 0.00834] | 0.008793          | [0.00343, 0.01448] |
| 1.0                | 0.001380          | [−0.00543, 0.00697] | 0.008074          | [0.00287, 0.01361] |

**Table F.2:** Full weight grid for Fused-RD8 (PV-RD8 and FUND-R50)

| $w$       | 0.0    | 0.1    | 0.2   | 0.3    | 0.4   | 0.5   | 0.6   | 0.7   | 0.8   | 1.0   |
|-----------|--------|--------|-------|--------|-------|-------|-------|-------|-------|-------|
| Frequency | 34.70% | 21.65% | 7.95% | 15.35% | 8.35% | 6.45% | 3.45% | 1.50% | 0.20% | 0.40% |

**Table F.3:** Frequency with which each weight achieves the highest Rank IC in the validation-period bootstrap ( $B = 2,000$ , block = 20)

The absolute confidence intervals in Table F.2 describe the Rank IC of each signal on its own; only the paired intervals in Table 5.4 of the main text answer whether the candidate raises Rank IC relative to the baseline. The two kinds of interval cannot be interpreted interchangeably.

## F.4 Correlation of the RD8 Prediction Streams and Model Identity

| Period    | Pearson  | Spearman | 2018–2019 trading days |
|-----------|----------|----------|------------------------|
| 2018–2019 | 0.003331 | 0.004970 | 503                    |
| 2020–2023 | 0.004305 | 0.003420 | 1006                   |

**Table F.4:** Mean daily cross-sectional correlation between PV-RD8 and FUND-R50

A correlation close to zero shows that the cross-sectional rankings of the two prediction streams differ clearly, but this only describes a difference in information and is not equivalent to a stable increment. PV-RD8 in Chapter 5 refers to the re-evaluation on the common sample aligned with FUND-R50; PV-RD8 in Chapter 4 refers to the broader price–volume evaluation sample, so the absolute Rank ICs of the two chapters should not be checked digit by digit against one another.

## F.5 News Data Coverage and Time Semantics

The upstream Alpha Vantage sample contains about 307,075 article–ticker records. The RD13 event experiments and the RD8 daily sentiment post-fusion use different stock sets and return targets, so their absolute Rank ICs cannot be treated as a single leaderboard.

| Dimension                   | RD13 event / I series                                                                  | RD8 revised daily post-fusion                                             |
|-----------------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Main sample                 | Event stock-days with tradable pre-open news / comparable event sample                 | Broader daily stock cross-section                                         |
| Main target                 | Event-day open→close                                                                   | LABEL0: $t + 1 \rightarrow t + 2$<br>close-to-close                       |
| Price–volume/RD information | Strictly information from $t - 1$ and earlier                                          | Normal availability time of the daily baseline                            |
| News information            | Only news published before the open                                                    | Delayed or Immediate publication-time mapping                             |
| Main question               | Does news provide immediate event information; does an increment survive full controls | Can news become a stable third predictive component of the daily baseline |

**Table F.5:** Time semantics of the two classes of news experiment

| Timing    | Train | 2018–2019 | 2020–2023 | Mean articles per stock-day<br>with news (2018–2019 / 2020–2023) |
|-----------|-------|-----------|-----------|------------------------------------------------------------------|
| Delayed   | 4.39% | 11.34%    | 25.34%    | 1.30 / 1.64                                                      |
| Immediate | 3.43% | 10.03%    | 19.61%    | 1.27 / 1.50                                                      |

**Table F.6:** Coverage of the RD8 revised sentiment specification

## F.6 Supplementary News Results for RD13

| Spec.                           | 2018–2019 Rank IC | 2020–2023 / event<br>development-period Rank IC | Interpretation                                  |
|---------------------------------|-------------------|-------------------------------------------------|-------------------------------------------------|
| Alpha Vantage delayed           | 0.00330           | 0.00119                                         | Weak; no stable daily increment                 |
| FinBERT delayed                 | 0.00396           | −0.00046                                        | Turns negative in the later period              |
| E0: $t - 1$ PV, no news         | —                 | −0.00034                                        | Event baseline                                  |
| E1: E0 + pre-open Alpha Vantage | —                 | 0.01045                                         | Immediate information in pre-open Alpha Vantage |
| FinBERT E2                      | —                 | 0.00600                                         | Same direction but weaker                       |

**Table F.7:** Delayed daily and pre-open event results

The full-control results of the I series are reported in Table 5.5 of the main text and are not repeated here. The validation-period paired Rank IC increment of I3 over I0 is +0.02314,

with a 21-day moving-block bootstrap 95% CI of [0.00646, 0.03787].

## F.7 Full $s$ -Grid of the Main RD8 Sentiment Specification

| Timing    | $s$   | 2018–2019 $\Delta$ | 2018–2019 95% CI | 2020–2023 $\Delta$ | 2020–2023 95% CI |
|-----------|-------|--------------------|------------------|--------------------|------------------|
| Delayed   | 0.05  | +0.17              | [−1.13, +1.41]   | +0.50              | [−0.71, +1.72]   |
| Delayed   | 0.10* | +0.32              | [−2.22, +3.00]   | +1.02              | [−1.60, +3.62]   |
| Delayed   | 0.20  | +0.10              | [−5.43, +5.74]   | +2.31              | [−3.13, +7.58]   |
| Delayed   | 0.30  | −0.00              | [−8.47, +9.49]   | +2.66              | [−5.75, +10.65]  |
| Immediate | 0.05  | +0.69              | [−0.37, +1.69]   | +0.36              | [−0.74, +1.42]   |
| Immediate | 0.10  | +1.23              | [−0.94, +3.32]   | +0.66              | [−1.64, +2.85]   |
| Immediate | 0.20  | +2.23              | [−2.67, +6.50]   | +1.62              | [−3.40, +6.32]   |
| Immediate | 0.30* | +3.33              | [−4.38, +10.61]  | +1.44              | [−6.41, +9.70]   |

**Table F.8:** Main specification: standardise within stocks with news, then set the contribution of stocks without news to 0 ( $\Delta\text{Rank IC} \times 10^{-4}$ )

\* indicates the  $s$  selected in the validation period by average Rank IC; the asterisk does not denote statistical significance. The development period is used for evaluation only and takes no part in weight selection.

## F.8 Sensitivity of Fill-Zero-then-Standardise

| Timing    | $s$   | 2018–2019 $\Delta$ | 2018–2019 95% CI | 2020–2023 $\Delta$ | 2020–2023 95% CI |
|-----------|-------|--------------------|------------------|--------------------|------------------|
| Delayed   | 0.05  | +0.51              | [−2.85, +3.30]   | +0.03              | [−2.23, +2.43]   |
| Delayed   | 0.10* | +0.89              | [−5.76, +6.91]   | +0.03              | [−4.61, +5.04]   |
| Delayed   | 0.20  | −1.11              | [−14.57, +11.47] | −2.17              | [−11.51, +7.70]  |
| Delayed   | 0.30  | −5.73              | [−24.65, +12.36] | −5.93              | [−19.88, +8.93]  |
| Immediate | 0.05  | +2.07              | [−0.71, +4.51]   | −0.10              | [−2.21, +2.29]   |
| Immediate | 0.10* | +3.67              | [−2.08, +8.94]   | −0.48              | [−5.22, +4.36]   |
| Immediate | 0.20  | +3.44              | [−8.71, +14.25]  | −2.62              | [−12.21, +6.57]  |
| Immediate | 0.30  | −0.34              | [−17.09, +16.76] | −5.83              | [−20.04, +8.83]  |

**Table F.9:** Full  $s$ -grid of the fill0→z sensitivity specification ( $\Delta\text{Rank IC} \times 10^{-4}$ )

This specification first fills stocks without news with 0 and then z-scores the full cross-section, so stocks without news receive a systematically negative exposure. Under this sensitivity, both Delayed and Immediate select  $s = 0.1$  in the validation period, but all paired confidence intervals still straddle zero; the standardisation semantics therefore change the point estimates but not the decision not to include a sentiment component.

## F.9 Statistical Protocol and Comparability Boundaries

The paired  $\Delta$ Rank IC of the RD8 fixed-50-round path uses a 20-day moving-block bootstrap with  $B = 2,000$ , supplemented by Newey–West HAC statistics. The RD8 revised sentiment experiment uses the same paired bootstrap with  $B = 2,000$  and block = 20; the interaction mechanism of the RD13 I series uses a 21-day block. Candidate weights are selected only in the 2018–2019 validation period; 2020–2023 takes no part in selection.

In cross-trajectory comparisons, a distinction is drawn between “the descriptive performance of the final signal systems” and “attribution to the underlying factor sets”. Fused-RD13 and Fused-RD8 are both linear fused scores after daily standardisation and can be compared on a unified label and the common intersection of stock–days; absolute Rank ICs computed on different samples remain mainly for within-trajectory interpretation, and the difference between the two cannot be attributed only to the RD13 and RD8 factors.

## F.10 Sensitivity to the Definition of the Illiquidity Factor

The sensitivity version RD13a changes only the illiquidity factor from  $|r|/V$  to  $|r|/(P \cdot V)$ ; the other 12 factors, the validation-period early-stopping training of PV-RD13 (the best iteration for all three seeds remains 1) and the 0.7/0.3 fusion rule of Fused-RD13 are unchanged, giving PV-RD13a and Fused-RD13a. The suffix a denotes the dollar-volume definition of the illiquidity factor. This analysis was carried out after the original signals were frozen and takes no part in any model or specification selection; the corresponding 2024–2025 results are reported in Chapter 7.

| Signal      | Rank IC (2018–2019) | Rank IC (2020–2023) |
|-------------|---------------------|---------------------|
| PV-RD13     | 0.00296             | 0.00729             |
| PV-RD13a    | 0.00253             | 0.00815             |
| Fused-RD13  | 0.00878             | 0.01058             |
| Fused-RD13a | 0.00835             | 0.01129             |

**Table F.10:** Predictive performance under the two illiquidity definitions

| Period    | Daily rank correlation of Fused-RD13 and Fused-RD13a | Top5 overlap |
|-----------|------------------------------------------------------|--------------|
| 2018–2019 | 0.959                                                | 80.6%        |
| 2020–2023 | 0.958                                                | 85.9%        |

**Table F.11:** Similarity of Fused-RD13 and Fused-RD13a

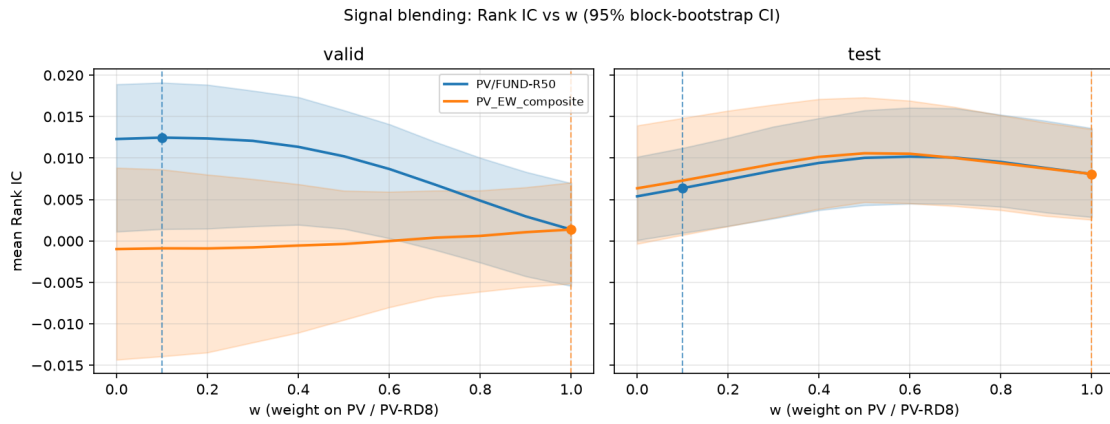
| Portfolio                        | Net excess ARR (2020–2023) | Excess CAGR (2020–2023) |
|----------------------------------|----------------------------|-------------------------|
| Fused-RD13 $\times$ Top20/drop2  | +8.96%                     | +7.63%                  |
| Fused-RD13a $\times$ Top20/drop2 | +10.56%                    | +9.35%                  |
| Fused-RD13 $\times$ Top5/drop1   | +18.72%                    | +16.27%                 |
| Fused-RD13a $\times$ Top5/drop1  | +13.65%                    | +9.42%                  |

**Table F.12:** Development-period portfolio performance under the two illiquidity definitions (one-way 1 bp, SPY benchmark)

The rankings under the two definitions are almost identical, but about 0.7–1 stock in the daily Top5 differs, which is already enough to move the development-period return of the concentrated portfolio by about 5 percentage points; the diversified Top20 portfolio changes much less.

| Signal                 | 2018–2019 IC | 2018–2019 Rank IC | 2020–2023 IC | 2020–2023 Rank IC |
|------------------------|--------------|-------------------|--------------|-------------------|
| PV-RD13                | 0.00328      | 0.00296           | 0.00948      | 0.00729           |
| FUND (historical)      | 0.01118      | 0.01456           | 0.00715      | 0.00784           |
| Fused-RD13 (0.7 / 0.3) | 0.00725      | 0.00878           | 0.01145      | 0.01058           |

**Table F.13:** Predictive performance of PV-RD13, FUND and Fused-RD13



**Figure F.1:** Price–volume versus fundamental weight and Rank IC under fixed training rounds

# Appendix G

## Supplementary Results for Chapter 6

Sections G.1–G.8 give the complete results of the RD8-trajectory portfolio search and the RD8-Top5 diagnostics; G.9 gives the multi-horizon Rank IC of the two signals; the breadth ablation, sub-periods and cost grid of the RD13 trajectory are in G.10.

### G.1 Backtest Engine and Search Settings

| Item            | Setting                                                                                           |
|-----------------|---------------------------------------------------------------------------------------------------|
| Signal          | Fused-RD8 = $0.1 \cdot z(\text{PV-RD8}) + 0.9 \cdot z(\text{FUND-R50})$ ; sentiment $s = 0$       |
| Strategy        | Qlib TopkDropoutStrategy; method_sell = bottom; method_buy = top; hold_thresh = 1                 |
| Execution       | Score on day $t \rightarrow$ fill at the close of day $t + 1$ (deal_price = close)                |
| Cost / exposure | 1 bp; risk_degree = 0.95; zero return on cash                                                     |
| Benchmark       | SPY (CRSP PERMNO 84398)                                                                           |
| Periods         | 2018–2019: 2018-01-02 to 2019-12-31 (503 days); development 2020-01-02 to 2023-12-29 (1,006 days) |
| Holdout         | 2024–2025: not accessed in these experiments; all series assert maximum date $\leq$ 2023-12-31    |

**Table G.1:** Fixed backtest settings

| Stage         | Content                                                                                                                          | Decision rule                                                                                                                  | Result                                                                                                                  |
|---------------|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| S1            | $K \in \{5, 10, 20, 30, 50\} \times$<br>about 10%/20% nominal<br>replacement (9<br>configurations)                               | Significance test against<br>RD8-Top20:<br>Holm-corrected one-sided<br>block-bootstrap $p < 0.05$<br>( $B = 2,000$ , block 21) | None significant; best<br>RD8-Top5, $\Delta\text{ARR} +0.113$ ,<br>Holm $p = 0.39$                                      |
| S2            | About 5–7% and 30%<br>nominal replacement (7<br>configurations)                                                                  | As S1                                                                                                                          | None significant                                                                                                        |
| H1            | HMM controller,<br>LOW/HIGH_RISK split by<br>training-period volatility;<br>state on day $t \rightarrow$ trade on<br>day $t + 1$ | As S1                                                                                                                          | Both states learned<br>RD8-Top20; closest to<br>passing: high-risk<br>RD8-Top5 (raw $p = 0.006$ ,<br>Holm $p = 0.090$ ) |
| Re-evaluation | 16 static configurations<br>ranked by validation period                                                                          | Highest validation-period<br>net excess ARR                                                                                    | RD8-Top5 first on ARR, IR<br>and MDD; carried forward<br>in parallel with RD8-Top20                                     |

**Table G.2:** Search stages

## G.2 Neighbourhood of RD8-Top5

| Config.    | Nominal | 2018–2019<br>ARR | 2018–2019<br>IR | 2018–2019<br>turnover | 2018–2019<br>repl. rate | 2020–2023<br>ARR | 2020–2023<br>IR | 2020–2023<br>turnover | 2020–2023<br>repl. rate |
|------------|---------|------------------|-----------------|-----------------------|-------------------------|------------------|-----------------|-----------------------|-------------------------|
| RD8-Top5   | 20%     | 0.1845           | 1.371           | 0.245                 | 0.127                   | 0.1109           | 0.430           | 0.232                 | 0.116                   |
| Top5/drop2 | 40%     | 0.1631           | 1.231           | 0.292                 | 0.147                   | 0.0800           | 0.316           | 0.280                 | 0.138                   |
| Top5/drop3 | 60%     | 0.1620           | 1.225           | 0.294                 | 0.149                   | 0.0792           | 0.310           | 0.285                 | 0.140                   |
| Top5/drop4 | 80%     | 0.1620           | 1.225           | 0.294                 | 0.149                   | 0.0747           | 0.294           | 0.285                 | 0.140                   |
| Top5/drop5 | 100%    | 0.1620           | 1.225           | 0.294                 | 0.149                   | 0.0747           | 0.294           | 0.285                 | 0.140                   |

**Table G.3:** Replacement-intensity neighbourhood ( $K = 5$ )

ARR is net excess ARR; turnover is mean daily turnover; the replacement rate is the actual daily replacement rate. Top5/drop5 denotes a nominal replacement capacity of 100%, not full replacement every day. Development-period values are taken from the neighbourhood backtest CSV (reports/k5d1\_posthoc\_diagnosis\_20260920\_110303/).

| Config.    | 2018–2019 ARR | 2018–2019 IR | 2018–2019 turnover | 2020–2023 ARR | 2020–2023 IR |
|------------|---------------|--------------|--------------------|---------------|--------------|
| Top3/drop1 | 0.1497        | 0.926        | 0.315              | 0.0501        | 0.176        |
| Top4/drop1 | 0.2021        | 1.428        | 0.252              | 0.0799        | 0.297        |
| RD8-Top5   | 0.1845        | 1.371        | 0.245              | 0.1109        | 0.430        |
| Top6/drop1 | 0.1641        | 1.354        | 0.208              | 0.0824        | 0.354        |
| Top7/drop1 | 0.1192        | 1.051        | 0.196              | 0.0541        | 0.224        |

**Table G.4:** Holding-breadth neighbourhood ( $n_{\text{drop}} = 1$ )

The development-period net excess ARR / IR of Top4/drop1 and Top6/drop1 are 0.0799 / 0.297 and 0.0824 / 0.354 respectively (differences relative to RD8-Top5:  $\Delta\text{ARR}$

−0.0310 and −0.0285;  $\Delta$ IR −0.133 and −0.076). Top3/drop1 and Top7/drop1 are 0.0501 / 0.176 and 0.0541 / 0.224 in the development period. All values come from the same neighbourhood backtest CSV; approximations read from charts are no longer used.

Interpretation under the frozen rule: on the replacement axis, “concentrated holding of the top five” appears to matter more than the specific replacement intensity; on the breadth axis, concentration shows local stability across periods. Neither dimension was used to re-select the configuration.

### G.3 Information at the Top of the Ranking

$D_{\text{top}}$  is defined as the equal-weighted mean LABEL0 of daily ranks 1–5 minus the equal-weighted mean LABEL0 of ranks 6–20, where LABEL0 =  $\text{Ref}(\$close, -2) / \text{Ref}(\$close, -1) - 1$ , taken on the scoring day (consistent with  $t \rightarrow t + 1$  close execution). Confidence intervals come from a moving-block bootstrap with  $B = 2,000$  and block length 20.

| Period    | Mean      | 95% CI                | One-sided $p$ |
|-----------|-----------|-----------------------|---------------|
| 2018–2019 | 0.000561  | [−0.000263, 0.001480] | 0.096         |
| 2020–2023 | −0.000051 | [−0.000852, 0.000770] | 0.553         |

**Table G.5:** Top-of-ranking difference  $D_{\text{top}}$  (daily)

Frozen interpretation: ranks 1–5 cannot be statistically distinguished from ranks 6–20.

| Period    | 1–5  | 6–10 | 11–20 | 21–50 | 51–100 | Rest |
|-----------|------|------|-------|-------|--------|------|
| 2018–2019 | 11.3 | 2.5  | 7.3   | 5.6   | 3.5    | 2.6  |
| 2020–2023 | 8.7  | 6.3  | 10.6  | 7.2   | 5.0    | 4.0  |

**Table G.6:** Equal-weighted mean daily LABEL0 by ranking bucket (bp)

The validation-period values are obtained by weighting the 2018 and 2019 annual results by trading days and are consistent with  $D_{\text{top}}$  (1–5 minus 6–20  $\approx$  5.6 bp). The bucket chart originally labelled “valid” in the diagnostic report mistakenly used the 2008–2019 window (including the training period; about 44 bp for bucket 1–5) and should not be used; the corresponding bucket chart needs to be redrawn for 2018–2019.

### G.4 Top-20 Placebo Test

For each random seed (0–299), the daily top-20 membership was held fixed, the scores within it were shuffled at random, and the shuffled scores were fed into a continuously running RD8-Top5 TopkDropout backtest. The table below reports the percentile of the true RD8-Top5 net excess ARR among the 300 placebo results. Frozen rule: a percentile

of at least 95% indicates that the ordering within the top 20 contributes substantively to the RD8-Top5 result.

| Period    | True ARR | Percentile | $p$   | Interpretation                                            |
|-----------|----------|------------|-------|-----------------------------------------------------------|
| 2018–2019 | 0.185    | 98.3%      | 0.017 | Ordering within the top 20 contributes substantively      |
| 2020–2023 | 0.111    | 66.7%      | 0.333 | Indistinguishable from random selection within the top 20 |

**Table G.7:** Placebo test results

The turnover of the true validation-period path is 0.151 lower than the placebo mean, suggesting a holding-path / trading-intensity channel.

## G.5 Contribution Concentration and Time Concentration

Individual stock contributions are computed as “weight at the close of  $t - 1 \times$  stock return on  $t$ ”, with zero return on cash, and reconciled against the gross portfolio return. The maximum reconciliation error across the four “configuration  $\times$  period” combinations is below  $5 \times 10^{-8}$  (stopping threshold  $10^{-6}$ ). The results after removing the largest contributors are contribution-removal approximations, not rebalanced backtests.

| Config.   | Period    | Top 1 | Top 3 | Top 5 | ARR excl.<br>top 1 | ARR excl.<br>top 2 |
|-----------|-----------|-------|-------|-------|--------------------|--------------------|
| RD8-Top5  | 2018–2019 | 0.162 | 0.447 | 0.669 | 0.138              | 0.095              |
| RD8-Top5  | 2020–2023 | 0.198 | 0.508 | 0.681 | 0.065              | 0.028              |
| RD8-Top20 | 2018–2019 | 0.125 | 0.300 | 0.458 | —                  | —                  |
| RD8-Top20 | 2020–2023 | 0.081 | 0.206 | 0.302 | —                  | —                  |

**Table G.8:** Contribution concentration (share of total contribution P&L)

The largest individual contributors: for RD8-Top5, Netflix (NFLX, PERMNO 89393, held 65 days) and Edwards Lifesciences (EW, PERMNO 87657) in the validation period, and Cardinal Health (CAH, PERMNO 21371) and DXC Technology (DXC, PERMNO 40125) in the development period; Netflix is also the largest contributor to RD8-Top20 in the validation period.

| Year              | RD8-Top5 | RD8-Top20 | Difference | RD8-Top5 absolute<br>return | RD8-Top20 absolute<br>return |
|-------------------|----------|-----------|------------|-----------------------------|------------------------------|
| 2018 (validation) | 32.6%    | 11.9%     | +20.7      | 30.6%                       | 5.8%                         |
| 2019 (validation) | 4.4%     | 2.5%      | +1.9       | 34.1%                       | 32.1%                        |
| 2020              | 18.4%    | 13.1%     | +5.3       | 23.3%                       | 25.1%                        |
| 2021              | 25.4%    | 14.5%     | +10.8      | 63.6%                       | 47.3%                        |
| 2022              | 1.2%     | 14.4%     | −13.2      | −20.8%                      | −6.9%                        |
| 2023              | −0.8%    | −1.8%     | +1.0       | 22.6%                       | 21.9%                        |

**Table G.9:** Net excess ARR by year

| Config.   | Period    | Best month (share) | Share of best 10 days | ARR excl. best 10 days | Label                         |
|-----------|-----------|--------------------|-----------------------|------------------------|-------------------------------|
| RD8-Top5  | 2018–2019 | 2018-02 (0.203)    | 0.589                 | 0.078                  | Time-concentrated             |
| RD8-Top5  | 2020–2023 | 2020-04 (0.429)    | 1.689                 | −0.077                 | Time-concentrated (over 100%) |
| RD8-Top20 | 2018–2019 | 2018-03 (0.236)    | 0.896                 | 0.008                  | Time-concentrated             |
| RD8-Top20 | 2020–2023 | 2020-11 (0.338)    | 1.076                 | −0.008                 | Time-concentrated (over 100%) |

**Table G.10:** Time concentration of net excess return

Removing the best trading days is a path-concentration diagnostic, not a tradable strategy. The labelling rule originally produced no label when the share of the best 10 days exceeded 1; this was corrected as a bug (Appendix G.9).

## G.6 Market Exposure

| Config.   | Period    | Beta  | Daily alpha | Annualised approx. | $R^2$ | vol20 vs universe | 12-1 momentum vs universe |
|-----------|-----------|-------|-------------|--------------------|-------|-------------------|---------------------------|
| RD8-Top5  | 2018–2019 | 1.224 | 0.000706    | 17.8%              | 0.652 | +0.0005           | +0.093                    |
| RD8-Top5  | 2020–2023 | 1.349 | 0.000318    | 8.0%               | 0.594 | +0.0025           | −0.034                    |
| RD8-Top20 | 2018–2019 | 1.154 | 0.000253    | 6.4%               | 0.852 | +0.0000           | +0.109                    |
| RD8-Top20 | 2020–2023 | 1.193 | 0.000346    | 8.7%               | 0.737 | +0.0026           | −0.003                    |

**Table G.11:** Daily regression on SPY and holding characteristics

Annualised alpha = daily alpha  $\times$  252. The local data contain no Fama–French, momentum or size exposures, and none were downloaded separately for this analysis.

Table G.12 extends the same regression to the Fused-RD13 portfolios. The convention is as above: an OLS regression of daily returns on SPY daily returns, no risk-free rate subtracted; Newey–West standard errors (5 lags); annualised alpha over 252 days. All portfolios have betas above 1 and no annualised alpha is significant; the annual results in Table G.13 show that every portfolio’s beta was highest in 2020.

| Portfolio        | Development beta | Development annualised alpha ( $t$ ) | 2018–2019 beta |
|------------------|------------------|--------------------------------------|----------------|
| RD13-Top20       | 1.27             | 6.7% (0.86)                          | 1.11           |
| RD13-Top20/drop4 | 1.28             | 11.0% (1.26)                         | —              |
| RD13-Top5        | 1.49             | 14.8% (1.23)                         | 1.24           |
| RD8-Top20        | 1.19             | 8.7% (1.23)                          | 1.15           |
| RD8-Top5         | 1.35             | 8.0% (0.70)                          | 1.22           |

**Table G.12:** Regression of each portfolio on SPY in the development and validation periods

| Portfolio        | 2020 | 2021 | 2022 | 2023 |
|------------------|------|------|------|------|
| RD13-Top20       | 1.32 | 1.06 | 1.31 | 1.05 |
| RD13-Top20/drop4 | 1.34 | 1.11 | 1.28 | 1.05 |
| RD13-Top5        | 1.53 | 1.39 | 1.52 | 1.29 |
| RD8-Top20        | 1.28 | 1.08 | 1.10 | 1.10 |
| RD8-Top5         | 1.49 | 1.18 | 1.24 | 1.00 |

**Table G.13:** Market beta by year (2020–2023)

## G.7 Drawdown Definitions

Both drawdown definitions are computed from the same daily return series. The main definition used throughout is the default of Qlib `risk_analysis` (`mode = "sum"`):  $MDD = \min(\text{cumsum}(r) - \text{running max of cumsum}(r))$ . The supplementary definition uses the compounded wealth path  $W = \text{cumprod}(1 + r)$ :  $MDD = \min(W / \text{running max of } W - 1)$ . The two share the same return source, window, costs, benchmark and calendar, and their peak and trough dates coincide; the difference comes only from the formula. Because the additive drawdown overstates the percentage loss in a deep decline, the gap between the two is largest during the 2020 crash.

| Config.   | Period    | Absolute (Qlib) | Absolute (compound) | Peak → trough           | Excess (Qlib) | Excess (compound) |
|-----------|-----------|-----------------|---------------------|-------------------------|---------------|-------------------|
| RD8-Top5  | 2018–2019 | −0.173          | −0.169              | 2018-09-28 → 2018-12-24 | −0.108        | −0.103            |
| RD8-Top5  | 2020–2023 | −0.679          | −0.522              | 2020-02-13 → 2020-03-23 | −0.325        | −0.292            |
| RD8-Top20 | 2018–2019 | −0.212          | −0.199              | 2018-10-01 → 2018-12-24 | −0.053        | −0.053            |
| RD8-Top20 | 2020–2023 | −0.525          | −0.432              | 2020-02-19 → 2020-03-23 | −0.202        | −0.192            |

**Table G.14:** Maximum drawdown under the two definitions

The peak / trough dates refer to the absolute drawdown. The excess-drawdown intervals are: RD8-Top5 validation 2018-09-28 → 2018-10-10; RD8-Top20 validation 2018-06-18 → 2018-10-10; both in the development period 2020-06-08 → 2020-10-01.

## G.8 Label Data Audit

Before the configuration was frozen, the LABEL0 series used for the diagnostics was cross-checked against CRSP. Among the records ranked 1–20 in each period, the 30 with the largest absolute LABEL0 were selected and compared with the CRSP daily return over the same  $t + 1 \rightarrow t + 2$  window, and the CRSP price-adjustment factor (`cfacpr`) within the surrounding 5 trading days was checked. No record differed by more than 0.05 (maximum absolute difference  $5.4 \times 10^{-7}$ ), no record had an absolute value above 0.5, and the adjustment factors of all 60 records were stable. The audit status is `CLEAN_METHOD_DIFFERENCE`:

the only problem found was the date-window error in the validation-period bucket chart described in Appendix G.3.

## G.9 Multi-Horizon Rank IC

| Signal     | Period    | 1-day lag | 5 days | 10 days | 20 days |
|------------|-----------|-----------|--------|---------|---------|
| Fused-RD13 | 2018–2019 | 0.52      | 0.24   | 0.20    | 0.18    |
| Fused-RD13 | 2020–2023 | 0.49      | 0.22   | 0.20    | 0.18    |
| Fused-RD8  | 2018–2019 | 0.98      | 0.95   | 0.92    | 0.87    |
| Fused-RD8  | 2020–2023 | 0.98      | 0.94   | 0.91    | 0.85    |

**Table G.15:** Ranking stability of the two fused signals (daily cross-sectional Spearman correlation over common stocks)

Table G.16 reports the mean daily Rank IC of Fused-RD13 and Fused-RD8 against future returns at horizons  $h = 1, 5, 10, 20$  days (the return window is consistent with LABEL0, starting from the close of  $t + 1$ ; the sample is the stocks tradable in common in each period). Both signals contain ranking information beyond one day; Fused-RD8 has a higher Rank IC at every horizon in the validation period and rises more smoothly with  $h$ , consistent with its high rank autocorrelation in Figure 6.1 of the main text and Table G.15. Source: reports/ch5\_signal\_persistence\_decay\_20260923/rank\_ic\_horizons.csv.

| Signal     | Period    | $h = 1$ | $h = 5$ | $h = 10$ | $h = 20$ | Days |
|------------|-----------|---------|---------|----------|----------|------|
| Fused-RD13 | 2018–2019 | 0.0088  | 0.0128  | 0.0182   | 0.0230   | 503  |
| Fused-RD13 | 2020–2023 | 0.0106  | 0.0203  | 0.0168   | 0.0197   | 1006 |
| Fused-RD8  | 2018–2019 | 0.0125  | 0.0217  | 0.0263   | 0.0285   | 503  |
| Fused-RD8  | 2020–2023 | 0.0064  | 0.0096  | 0.0142   | 0.0211   | 1006 |

**Table G.16:** Multi-horizon Rank IC of the two fused signals

## G.10 Supplementary Grids for the RD13 Trajectory

This section summarises the supplementary grids of the RD13 trajectory (the main text reports only the conclusions). (1) Holding-breadth study (Fused-RD13, 2020–2023): RD13-Top20 net excess +9.06% (turnover 0.202); K10 +12.53% (0.204); K5 with alternating replacement +7.96% (0.202); P10 +13.26% (0.399); RD13-Top5 +18.78% (0.399); the mechanism control RD13-Top20/drop4 +12.68% (0.399). The pre-written breadth conclusion supported Top10; RD13-Top5 is a post-hoc deployment specification. (2) RD13-Top20 vs RD13-Top20/drop4 costs: at one-way 1/5/10 bp the net excess difference is +3.61 / +1.75 / −0.59 percentage points (see Table 6.1); the denser cost grid and the annual/quarterly daily tables are in the mechanism-experiment archive. (3) HMM state

shares: validation low-risk 360 days, high-risk 143 days; development period about 602 / 404. W2 (dynamic weights) has a validation excess of  $-0.37\%$  (Fused-RD13:  $+1.16\%$ ) and is  $+1.21$  pp relative to T0 in the development period but unstable; Soft-2A has a validation excess of  $-5.92\%$  (not selected), and T2/T3 in the development period are  $-2.17\%$  /  $-1.58\%$ . The full T0–T3 results are in Appendix G.11. (4) The early Alpha20 TopK / n\_drop / hold\_thresh ablation (including sub-periods) is kept as a methodological exploration archive and does not alter the frozen status of this chapter; details are in the phase3\_portfolio\_ablation archive and the signal-code registry of Appendix I.

## G.11 Supplementary Results of the Market-State Experiments

The HMM uses the SPY daily return and 20-day realised volatility as observed variables, random seed 42, and 2008–2017 as the training period. Two-state and three-state models were compared in the validation period; the two-state model scored higher and was frozen. Inference uses forward filtering only, not full-sample smoothing; the state on day  $t$  is used for trading on day  $t + 1$ . Transition audits and leakage checks confirmed that no future observations were used. The RD13 trajectory names the two states bull and bear by the within-state SPY return; the RD8 trajectory names them low-risk and high-risk by the training-period mean 20-day volatility (about 0.0070 against 0.0193); the two correspond to the same state partition: 360 and 143 days in the validation period, and about 602 and 404 days in the development period. The main text refers to them as low-risk and high-risk. The specification W2, which adjusts the signal composition by state, weights PV-RD13 and FUND 0.8/0.2, 0.6/0.4 and 0.4/0.6 in the bull, neutral and bear states respectively; its daily holding overlap with Fused-RD13 in the development period is about 66%. The specification that adjusts exposure by state (Soft-2A) sets total equity exposure to  $\min(0.95 \cdot P(\text{bull}) + 0.30 \cdot P(\text{bear}), 0.95)$ , with a mean development-period exposure of about 0.69. When adjusting the portfolio rule by state, the RD8 trajectory searched the 16 static portfolios of Section 6.4 separately in each state under the same Holm gate; the closest to passing was Top5/drop1 in the high-risk state (raw one-sided  $p = 0.006$ , Holm-corrected  $p = 0.090$ ), and both states finally selected Top20/drop2 again.

| Arm | Signal weights | Exposure   | 2018–2019 net<br>excess ARR | Development net<br>excess ARR | Development MDD<br>(Qlib / compound) |
|-----|----------------|------------|-----------------------------|-------------------------------|--------------------------------------|
| T0  | Fused-RD13     | Fixed 0.95 | +1.16%                      | +9.06%                        | −45.1% / −39.3%                      |
| T1  | W2             | Fixed 0.95 | −0.37%                      | +10.27%                       | −44.7% / −38.4%                      |
| T2  | Fused-RD13     | Soft-2A    | −5.92%                      | −2.17%                        | −26.8% / −24.8%                      |
| T3  | W2             | Soft-2A    | Not evaluated               | −1.58%                        | −21.9% / −21.1%                      |

**Table G.17:** Results for T0–T3 (T3 was not evaluated in the validation period; in the development period the daily returns of T0 and RD13-Top20 coincide day by day)

The development-period difference of T1 relative to T0 is +1.21 percentage points, with a 21-day moving-block bootstrap (2,000 replications) 95% interval of [−5.70, +7.78] percentage points. Tables G.18 and G.19 give the annual results and the split by HMM state respectively, and Table G.20 compares the main portfolios of the two signals across states.

| Year | T0     | T1     | T2     | T3     | T2 mean exposure |
|------|--------|--------|--------|--------|------------------|
| 2020 | +34.2% | +22.0% | −3.1%  | −11.6% | 0.54             |
| 2021 | +3.4%  | +12.2% | +2.4%  | +9.3%  | 0.93             |
| 2022 | +7.4%  | +12.9% | +5.1%  | +6.5%  | 0.38             |
| 2023 | −9.0%  | −6.3%  | −13.1% | −10.7% | 0.90             |

**Table G.18:** Net excess ARR of T0–T3 by year

| State     | Days | T0 conditional excess | T1 − T0 | T2 − T0  |
|-----------|------|-----------------------|---------|----------|
| High-risk | 404  | +25.9%                | +2.0 pp | −30.8 pp |
| Low-risk  | 602  | −2.3%                 | +0.7 pp | +1.9 pp  |

**Table G.19:** Development-period results split by HMM state (conditional annualisation, approximated over 238 days; the state is the one-day-lagged label used for next-day trading)

| Portfolio                          | High-risk (404 days) | Low-risk (602 days) |
|------------------------------------|----------------------|---------------------|
| RD13-Top20                         | +25.9%               | −2.3%               |
| RD13-Top5                          | +44.5%               | +1.5%               |
| RD8-Top20 (Fused-RD8, Top20/drop2) | +20.4%               | +3.2%               |
| RD8-Top5 (Fused-RD8, Top5/drop1)   | +24.5%               | +2.1%               |

**Table G.20:** Development-period conditional excess return of the main portfolios of the two signals, split by HMM state (2020–2023; annualised approximately over 238 days)

# Appendix H

## Supplementary Results for Chapter 7

### H.1 Supplementary Results for 2024–2025

Table H.1 gives the 2024–2025 results of each research arm in the same TopK-breadth study (Qlib net excess ARR convention). All results in Table H.1 are post-hoc diagnostics. Table H.2 gives the annual performance of the portfolios reported in Section 7.2.2.

| Research arm (Fused-RD13)       | 2024    | 2025    | 2024–2025 |
|---------------------------------|---------|---------|-----------|
| K10 (Top10/drop1)               | −5.55%  | +3.45%  | −1.09%    |
| K5 (1 replaced every other day) | −7.05%  | +27.46% | +10.07%   |
| P10 (Top10/drop2)               | −10.05% | +5.57%  | −2.30%    |
| RD13-Top5 (Top5/drop1)          | —       | —       | +7.71%    |

**Table H.1:** Net excess ARR of each research arm of the TopK-breadth study in 2024–2025 (Qlib convention)

| Portfolio  | 2024 CAGR | 2025 CAGR |
|------------|-----------|-----------|
| RD13-Top20 | 9.86%     | 19.15%    |
| RD8-Top20  | 2.75%     | 22.61%    |
| RD8-Top5   | 4.55%     | 12.63%    |
| RD13-Top5  | 11.29%    | 47.86%    |
| SPY        | 23.30%    | 16.49%    |

**Table H.2:** Annual performance (native universe)

| Signal                              | Top20/drop2              | Top5/drop1                |
|-------------------------------------|--------------------------|---------------------------|
| Fused-RD8                           | 14.06%                   | 8.92%                     |
| Fused-RD13                          | 14.39%                   | 28.20%                    |
| Difference (Fused-RD8 − Fused-RD13) | −0.33 pp ( $t = -0.05$ ) | −19.28 pp ( $t = -1.15$ ) |

**Table H.3:** CAGR of the two signals under the same portfolio rules (2024–2025, common universe)

| Portfolio                       | Original def.<br>net excess ARR | Dollar-volume def.<br>net excess ARR | Original def.<br>excess CAGR | Dollar-volume def.<br>excess CAGR |
|---------------------------------|---------------------------------|--------------------------------------|------------------------------|-----------------------------------|
| Fused-RD13 $\times$ Top20/drop2 | −4.86%                          | −5.37%                               | −5.99%                       | −6.61%                            |
| Fused-RD13 $\times$ Top5/drop1  | +8.06%                          | −0.74%                               | +9.09%                       | −2.70%                            |

**Table H.4:** Effect of the illiquidity-factor definition on 2024–2025 portfolio results (post-hoc sensitivity, not used in any selection; both definitions come from the same rerun, so the original-definition values differ slightly from the frozen archive in Table H.1)

## H.2 Supplementary Results for 2026

Before the price–volume data were revised, the first replay gave RD13-Top5 a cumulative return of 46.1% (+30.7 pp against SPY, 97th percentile of the random portfolios). The first replay and the regenerated results differ in holdings on 5 decision days (27–29 April and 23–24 July 2026), with a maximum single-day return difference of about 1.9 percentage points. The random-signal placebo under the Top5 rule has a mean of 13.8%, a median of 13.0% and a 5%–95% range of −8.2% to 37.7%; under the Top20 rule the figures are 12.5%, 12.2% and 1.7% to 24.4%. The 2026 Rank IC of Fused-RD13 is 0.0065 (Pearson IC 0.0096), with a Rank IC 95% interval of [−0.012, +0.023]. Split by HMM state, 145 days of the 2026 main window are in the low-risk state and 15 in the high-risk state; the number of high-risk days is too small to interpret. In the 12 trading days after the main window, RD13-Top5 returned −4.95%, Top20 −1.15% and SPY −0.21%.

## H.3 Day-by-Day Record of Paper Trading

An earlier deployment on 28 August used intraday market orders instead of market-on-close orders; it is recorded as a test only and is not included in the results.

| Fill date | Theoretical holdings        | Anomaly                                                      | NAV (£) | Cash share |
|-----------|-----------------------------|--------------------------------------------------------------|---------|------------|
| 08-31     | AAPL, AXON, CNC, INTU, MCHP | —                                                            | 993,256 | 6.0%       |
| 09-01     | AAPL, AXON, CNC, INTU, IT   | MCHP sell partially filled, residual 555 shares              | 974,890 | 3.0%       |
| 09-02     | AAPL, AXON, CSGP, INTU, IT  | —                                                            | 973,889 | 7.7%       |
| 09-03     | AXON, CSGP, INTU, IT, PANW  | —                                                            | 997,011 | 8.1%       |
| 09-04     | AXON, CAH, CSGP, INTU, IT   | CAH order quantity too small                                 | 971,304 | 22.3%      |
| 09-08     | AXON, CAH, CIEN, CSGP, IT   | Fail-closed the previous evening; INTU sell partially filled | 943,095 | 13.8%      |
| 09-09     | AXON, CAH, CIEN, CSGP, TTWO | —                                                            | 927,988 | 19.3%      |
| 09-10     | AXON, CAH, CSGP, KLAC, TTWO | KLAC buy filled only 36/1,307 shares                         | 929,016 | 39.7%      |

**Table H.5:** Day-by-day record of IBKR paper trading (2026; theoretical holdings are the targets computed under the frozen rules)

Data notes: there are no trade-by-trade fill records for 1–3 September; fill quantities were reconstructed from differences between position snapshots, and the fill prices of those days are excluded from the price-deviation statistics. The theoretical replay uses an approximate daily return path, so the difference between account and theory cannot be decomposed exactly into each execution factor; there is a difference of about £2,000 between the account statement and the NAV differences, which was not forcibly reconciled.

| Portfolio  | High-risk state (69 days) | Low-risk state (433 days) |
|------------|---------------------------|---------------------------|
| RD13-Top20 | +3.7%                     | −5.8%                     |
| RD13-Top5  | +31.3%                    | +3.8%                     |
| RD8-Top20  | +12.1%                    | −9.2%                     |
| RD8-Top5   | +6.5%                     | −11.2%                    |

**Table H.6:** Conditional annualised excess return in 2024–2025 split by HMM state (approximated over 238 days)

# **Appendix I**

## **Main Signal and Specification Identifiers**

Table I.1 maps the names used in this report to the identifiers in the code repository.

| Name in this report | Identifier in the code repository | Description                                                                    |
|---------------------|-----------------------------------|--------------------------------------------------------------------------------|
| Alpha20             | Alpha20                           | Base set of 20 price–volume technical features (a subset of Qlib Alpha158).    |
| RD8                 | RD8                               | Eight RD factors formed by online accumulation in the RD8 trajectory.          |
| RD13                | RD13_v2                           | Thirteen RD factors from the RD13 trajectory in the revised implementation.    |
| RD13a               | RD13_v2a                          | RD13 with the illiquidity factor redefined on dollar volume.                   |
| RD4                 | RD4                               | Four distinct numerical series in the initial RD13 implementation (control).   |
| D0                  | D0                                | Alpha20-only model in the RD13 D-series grid.                                  |
| D1                  | D1                                | Alpha20 plus fundamentals in the D-series grid.                                |
| D2                  | D2                                | D-series label for the price–volume model PV-RD13.                             |
| D3                  | D3                                | Alpha20, RD13 and fundamentals in the D-series grid.                           |
| D2_R50 / D3_R50     | D2_R50 / D3_R50                   | D2 and D3 retrained for a fixed 50 rounds.                                     |
| PV-RD13             | D2                                | Price–volume LightGBM on Alpha20 + RD13 (validation early stopping).           |
| PV-RD13a            | D2a                               | PV-RD13 with the dollar-volume illiquidity definition.                         |
| PV-RD8              | B1_R50 / B1-R50                   | Price–volume LightGBM on Alpha20 + RD8 (fixed 50 rounds).                      |
| FUND                | F1C                               | Fundamentals-only LightGBM with validation early stopping.                     |
| FUND-R50            | F1C_R50 / F1C-R50                 | Same fundamental model trained for a fixed 50 rounds.                          |
| Fused-RD13          | W1                                | Linear fusion<br>$0.7 \cdot z(\text{PV-RD13}) + 0.3 \cdot z(\text{FUND})$ .    |
| Fused-RD13a         | W1a                               | Fused-RD13 under the dollar-volume illiquidity definition.                     |
| Fused-RD8           | F_BASE_W010                       | Linear fusion<br>$0.1 \cdot z(\text{PV-RD8}) + 0.9 \cdot z(\text{FUND-R50})$ . |
| RD13-Top20          | H0 / P20D2                        | Fused-RD13 under Top20/drop2 (pre-frozen research baseline).                   |
| RD13-Top5           | P5                                | Fused-RD13 under Top5/drop1 (final deployment strategy).                       |
| RD13-Top20/drop4    | P20D4                             | Fused-RD13 under Top20/drop4 (replacement-speed control).                      |
| RD8-Top20           | K20_D2                            | Fused-RD8 under Top20/drop2.                                                   |
| RD8-Top5            | K5D1_WD                           | Fused-RD8 under Top5/drop1.                                                    |

**Table I.1:** Name mapping between this report and the code repository

# Appendix J

## Code Listing

This appendix lists the parts of the code written for this project that are referred to in the main chapters. It covers the adaptation of RD-Agent to the US setting (J.1), the point-in-time data construction (J.2), the factor audit (J.3), signal fusion and statistical inference (J.4), portfolio evaluation (J.5) and broker execution and reconciliation (J.6). Only the core functions are shown; argument parsing, logging and error handling are omitted. The source code of RD-Agent and Qlib, the factor code generated by the LLM (except the Loop 2 comparison in J.3), one-off analysis scripts, one-off summary scripts, the Holm-adjustment routine and plotting code are not listed. The complete code is available in the repository given in Appendix B.2.

Identifiers in the code follow the repository naming; Table I.1 maps them to the names used in this report (for example, W1 is Fused-RD13 and P5 is RD13-Top5).

### J.1 RD-Agent Adaptation to the US Market

The excerpts below show the isolated cache and CoSTEER data directories for the RD8 trajectory, and the corrected Loop 2 injection after forking Loop 0–1 (Chapters 3 and 4).

Isolates the pickle cache and the CoSTEER data directories for the RD8 trajectory; the TEST variables set the window that this report calls the development period (Sections 3.7 and 4.6).

**Listing J.1:** Isolated RD8 cache and data directories

```
1 export PICKLE_CACHE_FOLDER_PATH_STR="${ROOT}/pickle_cache"
2 export FACTOR_CoSTEER_DATA_FOLDER="${ROOT}/source_data_clean"
3 export FACTOR_CoSTEER_DATA_FOLDER_DEBUG="${ROOT}/source_data_clean_debug"
4 export QLIB_FACTOR_TRAIN_START=2008-01-01
5 export QLIB_FACTOR_TRAIN_END=2017-12-31
6 export QLIB_FACTOR_VALID_START=2018-01-01
```

```

7 export QLIB_FACTOR_VALID_END=2019-12-31
8 export QLIB_FACTOR_TEST_START=2020-01-01
9 export QLIB_FACTOR_TEST_END=2023-12-31
10 export QLIB_FACTOR_MARKET=sp500
11 export QLIB_FACTOR_TOPK=20
12 export QLIB_FACTOR_N_DROP=2

```

After forking Loop 0–1, injects the corrected Loop 2 experiment once on the native FactorRDLoop path (Section 4.6).

**Listing J.2:** Corrected Loop 2 injection after Loop 0–1 fork

```

1 class CorrectedSemanticFactorRDLoop(FactorRDLoop):
2     corrected_loop2_injected: bool = False
3     allow_corrected_injection: bool = True
4     async def direct_exp_gen(self, prev_out: dict[str, Any]):
5         if (
6             self.allow_corrected_injection
7             and (not getattr(self, "corrected_loop2_injected", False))
8             and len(self.trace.hist) == 2
9         ):
10             hypo, exp = build_corrected_loop2_experiment(self.trace)
11             return {"propose": hypo, "exp_gen": exp}
12         return await super().direct_exp_gen(prev_out)
13     def coding(self, prev_out: dict[str, Any]):
14         exp = prev_out["direct_exp_gen"]["exp_gen"]
15         if getattr(exp, "_is_corrected_loop2", False):
16             self.corrected_loop2_injected = True
17             return exp
18         return super().coding(prev_out)

```

## J.2 Point-in-Time Data Construction

These excerpts reconstruct historical S&P 500 membership, adjust OHLC by CRSP cumulative factors, set Compustat availability dates, expand fundamentals to trading days, and align news under the delayed and pre-open schemes (Section 3.2).

Builds point-in-time membership intervals from CRSP MbrStartDt/MbrEndDt and writes instruments/sp500.txt (Section 3.2).

**Listing J.3:** Historical S&P 500 membership spells

```
1 def build_membership_spells(source_csv: Path) -> tuple[pd.DataFrame,  
    ↳ MembershipStats]:  
2     ...  
3     for chunk in pd.read_csv(source_csv, usecols=MEMBERSHIP_READ_COLUMNS,  
    ↳ chunksize=CHUNK_SIZE, low_memory=False):  
4         ...  
5         for row in valid.itertuples(index=False):  
6             key = (int(row.PERMNO), row.MbrStartDt.normalize(), row.MbrEndDt.  
    ↳ normalize())  
7             trade_date = row.DlyCalDt.normalize()  
8             bucket = spell_dates[key]  
9             ...  
10        ...  
11        for (permno, mbr_start, mbr_end), bounds in spell_dates.items():  
12            start = max(mbr_start, bounds["min_trade"])  
13            end = min(mbr_end, bounds["max_trade"])  
14            rows.append({"symbol": permno_to_symbol(permno), "start_date": start.  
    ↳ strftime("%Y-%m-%d"), "end_date": end.strftime("%Y-%m-%d"), "permno":  
    ↳ permno})  
15        ...  
16 def write_sp500_txt(spells_df: pd.DataFrame) -> None:  
17     spells_df[["symbol", "start_date", "end_date"]].to_csv(SP500_PATH, sep="\n",  
    ↳ header=False, index=False)
```

Adjusts OHLC by DlyCumFacPr, stores the cumulative factor, then normalises each series by its first close (Section 3.2).

**Listing J.4:** Split-adjusted OHLC and factor

```
1 def clean_and_adjust(chunk: pd.DataFrame, stats: CleanStats) -> pd.DataFrame:  
2     ...  
3     factor = 1.0 / work["DlyCumFacPr"]  
4     out = pd.DataFrame(  
5         {  
6             "date": work["DlyCalDt"].dt.strftime("%Y-%m-%d"),  
7             "open": work["DlyOpen"] / work["DlyCumFacPr"],  
8             "high": work["DlyHigh"] / work["DlyCumFacPr"],  
9             "low": work["DlyLow"] / work["DlyCumFacPr"],
```

```

10         "close": work["DlyClose"] / work["DlyCumFacPr"],
11         "volume": work["DlyVol"] / factor,
12         "factor": factor,
13         "_raw_close": work["DlyClose"],
14         "_permno": work["PERMNO"].astype(int),
15     }
16 )
17 ...
18     return out
19 ...
20 def normalize_instrument(df: pd.DataFrame) -> pd.DataFrame:
21     out = df.sort_values("date").copy()
22     first_close = out["close"].iloc[0]
23 ...
24     for col in ("open", "high", "low", "close", "factor"):
25         out[col] = out[col] / first_close
26     out["volume"] = out["volume"] * first_close
27     return out

```

Sets `accounting_available_date` to `max(rdqe, datadate)` (Section 3.2; Appendix A.1).

#### Listing J.5: Compustat availability date

```

1 def attach_membership_and_availability(df: pd.DataFrame, membership: pd.
    ↪ DataFrame) -> pd.DataFrame:
2     out = df.copy()
3     ...
4     out["accounting_available_date"] = out[["rdqe", "datadate"]].max(axis=1)
5     ...
6     return out

```

Backward as-of join of quarterly signal features onto each PERMNO trading day (Section 3.2; Appendix A.1).

#### Listing J.6: Expand fundamentals to trading days

```

1 def merge_daily_to_signals(daily: pd.DataFrame, signal_payload: pd.DataFrame)
    ↪ -> pd.DataFrame:
2     parts: list[pd.DataFrame] = []
3     signal_groups = {p: g for p, g in signal_payload.groupby("permno", sort=
    ↪ False)}

```

```

4 ...
5     for permno, dgrp in daily.groupby("permno", sort=False):
6         sgrp = signal_groups.get(int(permno))
7     ...
8         left = dgrp.sort_values("date").reset_index(drop=True)
9         right = (
10             sgrp.sort_values("signal_start_date")
11             .drop(columns=[c for c in drop_from_right if c in sgrp.columns])
12             .reset_index(drop=True)
13         )
14         parts.append(
15             pd.merge_asof(
16                 left,
17                 right,
18                 left_on="date",
19                 right_on="signal_start_date",
20                 direction="backward",
21                 allow_exact_matches=True,
22             )
23         )
24     return pd.concat(parts, ignore_index=True)

```

Delayed scheme maps to the next session; the immediate scheme keeps only pre-open eligible articles (Section 3.2; Appendices A.2 and F.5).

#### Listing J.7: Delayed and pre-open news alignment

```

1 def eligible_signal_date_conservative(pub_ts_et: pd.Timestamp, trading_dates:
    ↪ list[date]) -> date | None:
2     if pub_ts_et is pd.NaT or not trading_dates:
3         return None
4     pub_date = pub_ts_et.date()
5     for d in trading_dates:
6         if d > pub_date:
7             return d
8     return None
9 ...
10 def filter_eligible_articles(df: pd.DataFrame) -> pd.DataFrame:
11     out = df[df["tradable_intraday_v2"] == 1].copy()

```

```

12     out = out[out["event_trade_date_v2"].notna()]
13     out = out[out["publication_bucket_v2"] != "timestamp_ambiguous"]
14     out = out[out["publication_bucket_v2"] != "same_day_regular_hours"]
15     pub_d = pd.to_datetime(out["publication_date_et"])
16     evt_d = pd.to_datetime(out["event_trade_date_v2"])
17     same_day_am = (out["publication_bucket_v2"] == "same_day_aftermarket") &
    ↪ (pub_d == evt_d)
18     out = out[~same_day_am]
19     if "publication_ts_et" in out.columns:
20         ts = pd.to_datetime(out["publication_ts_et"], utc=True, errors="
    ↪ coerce").dt.tz_convert("America/New_York")
21         pre = out["publication_bucket_v2"] == "same_day_premarket"
22         bad_pre = pre & ts.notna() & (ts.dt.time >= MARKET_OPEN)
23         out = out[~bad_pre]
24     return out

```

## J.3 Factor Audit

These excerpts check series-level aliases, scale invariance under OHLC scaling, and the Loop 2 absolute-versus-relative ATR correction (Section 4.6).

Hashes factor value series and marks exact aliases (the RD13→RD4 finding) (Sections 4.3 and 4.6; Appendix E.3).

**Listing J.8:** Series-level exact aliases

```

1 def sha256_series(s: pd.Series) -> str:
2     arr = np.ascontiguousarray(s.to_numpy(dtype=np.float64, na_value=np.nan))
3     return hashlib.sha256(arr.tobytes()).hexdigest()
4 ...
5 hash_to_feats: dict[str, list[str]] = {}
6 for out, h in value_hashes.items():
7     hash_to_feats.setdefault(h, []).append(out)
8 exact_groups = [{ "sha256": h, "features": feats} for h, feats in
    ↪ hash_to_feats.items() if len(feats) > 1]
9 alias_of: dict[str, str | None] = {}
10 for _, row in prov.iterrows():
11     out = row["output_feature_name"]
12     if row["is_workspace_canonical_factor"]:

```

```

13         alias_of[out] = None
14     else:
15         canonical = [f for f in hash_to_feats.get(row["sha256_v1"], [])
↪ if f in WORKSPACE_CANONICAL]
16         alias_of[out] = canonical[0] if canonical else hash_to_feats[row[
↪ "sha256_v1"]][0]
17     prov["exact_alias_of"] = prov["output_feature_name"].map(alias_of)

```

Scales one stock OHLC by 2 and 4 and records max\_abs\_diff of each factor on that stock (Section 4.6; Appendix E.6).

### Listing J.9: Scale-invariance audit

```

1 def scale_ohlc(pv: pd.DataFrame, instrument: str, k: float) -> pd.DataFrame:
2     out = pv.copy()
3     inst = out.index.get_level_values("instrument").astype(str)
4     mask = inst == str(instrument)
5     for col in ["$open", "$high", "$low", "$close"]:
6         out.loc[mask, col] = out.loc[mask, col].to_numpy() * k
7     return out
8 ...
9 def scale_invariance_audit(out: Path) -> tuple[pd.DataFrame, list[str]]:
10     ...
11     base = compute_all_factors(pv0)
12     for k in (2.0, 4.0):
13         pv_k = scale_ohlc(pv0, target, k)
14         alt = compute_all_factors(pv_k)
15         ...
16         for col in list(LOOP9_LIBRARY8) + RD13:
17             ...
18             diff = (aa[mask] - bb[mask]).abs()
19             max_abs = float(diff.max())
20             ...
21             invariant = bool(max_abs < 1e-5 or (np.isfinite(med_rel) and
↪ med_rel < 1e-7))
22             if invariant is False:
23                 failing.append(col)
24             ...
25     ...

```

Qlib's \$factor field is written as 1/DlyCumFacPr in Listing J.4 (line 3), so its level depends on the end-of-sample cumulative adjustment (Section 4.6; Appendix E.5).

The original computed ATR on the absolute true range; the corrected version divides the true range by the previous close first (Section 4.6).

**Listing J.10:** Loop 2 factor as generated (excerpt)

```
1 def calculate_atr_20d_x_volume_ratio_20d():
2 ...
3     atr_20d = true_range.rolling(window=20, min_periods=20).mean()
4 ...
```

**Listing J.11:** Loop 2 factor after correction

```
1 def calculate_atr_20d_x_volume_ratio_20d():
2     ...
3     true_range = pd.DataFrame(
4         np.maximum(np.maximum(tr1.values, tr2.values), tr3.values),
5         index=tr1.index,
6         columns=tr1.columns,
7     )
8     rel_tr = true_range / prev_close
9     atr_20d = rel_tr.rolling(window=20, min_periods=20).mean()
10    ...
11    factor = atr_20d * vol_ratio_20d
12    ...
```

## J.4 Signal Fusion and Statistical Inference

These excerpts implement daily winsorisation and z-scoring, the frozen W1 (Fused-RD13) linear fusion, validation-only weight selection on the RD8 grid, and moving-block bootstrap inference (Chapters 5 and 6).

Daily 1%/99% winsorisation and z-scores, then the frozen W1 weights  $0.7 \cdot PV + 0.3 \cdot \text{FUND}$  (Fused-RD13) (Chapter 5).

**Listing J.12:** Cross-sectional winsorise, z-score and W1 fusion

```
1 def winsorize_series(s: pd.Series, lower: float = 0.01, upper: float = 0.99)
    ↪ -> pd.Series:
2     if s.notna().sum() < 5:
3         return s
```

```

4     lo, hi = s.quantile(lower), s.quantile(upper)
5     return s.clip(lo, hi)
6 def zscore_series(s: pd.Series) -> pd.Series:
7     std = s.std(ddof=0)
8     if std == 0 or np.isnan(std):
9         return s * 0.0
10    return (s - s.mean()) / std
11 ...
12 def preprocess_scores(df: pd.DataFrame, score_cols: list[str]) -> pd.
    ↪ DataFrame:
13     out = df.copy()
14     for col in score_cols:
15         zcol = f"{col}_z"
16         out[zcol] = np.nan
17         for dt, g in out.groupby("datetime", sort=False):
18             s = winsorize_series(g[col].astype(float))
19             out.loc[g.index, zcol] = zscore_series(s)
20     return out
21 ...
22 WEIGHT_SCHEMES: dict[str, dict[str, tuple[float, float]]] = {
23     "W1": {"bull": (0.7, 0.3), "neutral": (0.7, 0.3), "bear": (0.7, 0.3)},
24 }
25
26 def build_dynamic_score(
27     df: pd.DataFrame,
28     scheme: dict[str, tuple[float, float]],
29     state_col: str = "state_label",
30 ) -> pd.Series:
31     wq = df[state_col].map({k: v[0] for k, v in scheme.items()}).astype(float
    ↪ )
32     wf = df[state_col].map({k: v[1] for k, v in scheme.items()}).astype(float
    ↪ )
33     return wq * df["quant_score_z"] + wf * df["fundamental_score_z"]

```

On the RD8 trajectory,  $s$  is chosen on 2018–2019 validation Rank IC only (Chapter 5).

**Listing J.13:** Validation-only fusion weight selection (RD8 grid)

```

1 def select_w_star(valid_rank_ic_by_w: dict[float, float]) -> float:

```

```

2     items = sorted(valid_rank_ic_by_w.items(), key=lambda kv: kv[0])
3     best_ric = max(v for _, v in items)
4     candidates = [w for w, v in items if abs(v - best_ric) < TIE_EPS]
5     return min(candidates, key=lambda w: abs(w - 0.5))

```

Moving-block bootstrap with  $B = 2000$  and block length 20 for paired mean Rank IC differences (Section 3.8; Appendix D.4).

**Listing J.14:** Moving-block bootstrap of paired  $\Delta$ Rank IC

```

1 def moving_block_indices(n: int, block: int, rng: np.random.Generator) -> np.
    ↪ ndarray:
2     ...
3     idx = np.concatenate(pieces)[:n]
4     return idx.astype(int)
5 ...
6 def run_bootstrap(daily_by_w, *, pair, period, w_star):
7     ...
8     for _b in range(BOOT_B):
9         idx = moving_block_indices(n, BOOT_BLOCK, rng)
10        means = {w: float(np.nanmean(ric_mat[w][idx])) for w in W_GRID}
11        ...
12    for w in W_GRID:
13        arr = np.asarray(boot_means[w], dtype=float)
14        ci_rows.append({"ci95_low": float(np.nanpercentile(arr, 2.5)), "
    ↪ ci95_high": float(np.nanpercentile(arr, 97.5)), ...})
15    ...

```

## J.5 Portfolio Evaluation

These excerpts wrap TopK-Dropout backtests and run the 2026 random-portfolio placebo (Chapters 6 and 7).

Fixes close deals, 1 bp costs, SPY benchmark and risk\_degree before calling the shared backtest runner (Chapter 6).

**Listing J.15:** TopK-Dropout backtest wrapper

```

1 def load_port_cfg(topk: int, n_drop: int) -> dict[str, Any]:
2     cfg = f2p.load_port_config()
3     cfg["strategy"]["kwargs"]["topk"] = int(topk)

```

```

4     cfg["strategy"]["kwargs"]["n_drop"] = int(n_drop)
5     cfg["strategy"]["kwargs"]["hold_thresh"] = HOLD_THRESH
6     cfg["strategy"]["kwargs"]["risk_degree"] = RISK_DEGREE
7     for k, v in rpa.STRATEGY_DEFAULTS.items():
8         cfg["strategy"]["kwargs"].setdefault(k, v)
9     cfg["backtest"]["account"] = ACCOUNT
10    cfg["backtest"]["benchmark"] = BENCHMARK
11    cfg["backtest"]["exchange_kwargs"]["deal_price"] = "close"
12    cfg["backtest"]["exchange_kwargs"]["open_cost"] = OPEN_COST
13    cfg["backtest"]["exchange_kwargs"]["close_cost"] = CLOSE_COST
14    cfg["backtest"]["exchange_kwargs"]["min_cost"] = 0
15    cfg["backtest"]["exchange_kwargs"]["limit_threshold"] = None
16    return cfg
17 def run_bt(pred: pd.DataFrame, topk: int, n_drop: int, start: str, end: str)
    ↪ -> dict[str, Any]:
18     cfg = load_port_cfg(topk, n_drop)
19     cfg["backtest"]["start_time"] = start
20     cfg["backtest"]["end_time"] = end
21     dates = pred.index.get_level_values("datetime")
22     sub = pred.loc[(dates >= pd.Timestamp(start) - pd.Timedelta(days=10)) & (
    ↪ dates <= pd.Timestamp(end))]
23     return rpa.run_backtest(cfg, sub)

```

Draws daily random rankings 500 times under the same TopK-Dropout rules and returns the empirical percentile (Section 7.3; Appendix H.2).

#### Listing J.16: Random-portfolio placebo

```

1 def placebo_runs(
2     universe_by_date: dict[pd.Timestamp, list[str]],
3     dates: list[pd.Timestamp],
4     px: pd.DataFrame,
5     *,
6     topk: int,
7     n_drop: int,
8     eval_start: pd.Timestamp,
9     eval_end: pd.Timestamp,
10    n: int,
11    seed: int,

```

```

12 ) -> pd.DataFrame:
13     rng = np.random.default_rng(seed)
14     ...
15     for i in range(n):
16         ...
17         for dt in dates:
18             tickers = universe_by_date.get(dt, [])
19             ...
20             scores = rng.random(len(tickers))
21             ...
22             w = topk_dropout_weights(sc, dates, topk=topk, n_drop=n_drop)
23             daily = portfolio_from_weights(w, px, eval_start=eval_start, eval_end
↪ =eval_end)
24             ...
25             results.append({"run": i, "cum_net": float((1 + r).prod() - 1), ...})
26     return pd.DataFrame(results)
27 ...
28 def percentile_of(x: float, dist: np.ndarray) -> float:
29     ...
30     return float((dist <= x).mean() * 100.0)

```

## J.6 Broker Execution and Reconciliation

These excerpts implement fail-closed pre-trade checks and rotation from the actual broker book (Sections 3.9 and 7.4).

Stops when the account is not paper, NAV/FX/positions cannot be confirmed, or the `n_drop=1` rotation is ill-posed; sells and buys are taken from the broker book versus the target names (Sections 3.9 and 7.4; Appendix H.3).

**Listing J.17:** Fail-closed pre-trade checks and actual-book rotation

```

1 def main() -> int:
2     target = json.loads(TARGET_JSON.read_text(encoding="utf-8"))
3     wanted = {str(t).upper() for t in target["tickers"]}
4     ...
5     if not app.still_paper() or (app.managed_accounts[0] if app.
↪ managed_accounts else "") != EXPECTED_ACCOUNT:
6         ...

```

```
7         return 1
8     ...
9     held = set(by_sym)
10    sell_names = sorted(held - wanted)
11    buy_names = sorted(wanted - held)
12    ...
13    if len(buy_names) != 1:
14        fatal.append(f"n_drop=1 requires exactly one buy; sell={sell_names}
↪ buy={buy_names}")
15    if not sell_names:
16        fatal.append(f"nothing to sell; sell={sell_names} buy={buy_names}")
17    ...
```